Algorithmic Trading

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Notation List

$x \in E,\ x \notin E$	membership / non-membership
$E \subseteq F,\ \emptyset$	subset relation / empty set
$X \times Y,\ f : X \to Y$	Cartesian product / function from X to Y
$\Longrightarrow,\ \Longleftrightarrow$	implies / if and only if
$:=$	defined to be equal to
$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	standard number systems
$\mathbb{F}$	field of scalars, usually $\mathbb{R}$ or $\mathbb{C}$
a, b, c, λ	scalars
$\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{0}$	vectors and the zero vector
$\mathcal{U}, \mathcal{V}, \mathcal{W}$	vector spaces or subspaces, depending on context
$\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$	finite ordered list, often a basis
$\text{span}(E),\ \dim(\mathcal{V})$	span of E / dimension of $\mathcal{V}$
$\mathcal{U} + \mathcal{W},\ \mathcal{U} \oplus \mathcal{W},\ \mathcal{V}/\mathcal{U}$	sum, direct sum, quotient space
$T : \mathcal{V} \to \mathcal{W},\ \mathcal{L}(\mathcal{V}, \mathcal{W})$	linear map / space of linear maps
$\ker(T),\ \text{range}(T)$	kernel and range of T
$[\mathbf{v}]_\beta,\ [T]_{\gamma \leftarrow \beta}$	coordinate vector / matrix representation
$A, B, C,\ \text{Mat}(m, n, \mathbb{F})$	matrices / $m \times n$ matrices over $\mathbb{F}$
$A^{-1},\ A^T,\ A^*,\ \text{tr}(A), \det(A)$	standard matrix operations

1 Preface and Prerequisites

1.1 Brief Overview

Lorem Ipsum
To be completed once most of the book is done

1.2 Reading Roadmap

Lorem Ipsum. To be completed once most of the book is done

This content builds upon the foundational works of Rishi K. (Narang, 2013), Raja (Velu, 2020), and Marcos Lopez (Prado, 2018), among others, whose insights form the backbone of our discussion.

1.3 Time Value of Money

Definition 1.3.1. *Time Value of Money Formulas*

Concept	Formula	Notes
Expected Annual Rate	$\text{EAR} = (1 + r_p)^m - 1$ $\text{EAR} = e^r - 1$	r_p periodic rate, m periods/year Continuous
Future Value (Continuous)	$FV_N = PVe^{r_s N}$	r_s stated rate, N years
Present Value (Frequent)	$PV = FV_N(1 + \frac{r}{m})^{-mN}$	m periods/year
Ordinary Annuity	$FV_N = A\left[\frac{(1+r)^N - 1}{r}\right]$	First CF one period from now
Annuity Due	$FV_N = A\left[\frac{(1+r)^N - 1}{r}\right](1 + r)$	First CF today
Perpetuity	$PV = \frac{A}{r}$	Infinite cash flows

Definition 1.3.2. *Return Metrics*

Metric	Formula / Description	Notes
Internal Rate of Return	$\text{NPV} = 0$	PV inflows = PV outflows
Money-Weighted Return	IRR on portfolio	Timing and size of CFs
Time-Weighted Return	$r = \sqrt[1/n]{\prod_{i=1}^{n}(1 + HPR_i)} - 1$	Geometric mean
Annualised Return	$r = (1 + HPR)^{365/\text{Days}} - 1$	HPR scaled to annual
Continuously Compounded	$r_{CC} = \ln(1 + HPR)$	Log return
Gross Return	Before fees	Pre-fees
Net Return	After fees	Post-fees
Pre-Tax Nominal	Before taxes	Pre-tax
After-Tax Nominal	After taxes	Post-tax
Real Return	$(1 + r_{\text{Real}}) = \frac{(1+r_{\text{Nom}})(1+r_{\text{Risk}})}{(1+\pi)}$	Inflation-adjusted
Leveraged Return	$r_L = \frac{r(V_0+V_B) - r_B V_B}{V_0}$	V_B borrowed, r_B borrow cost

1.4 Overview of Systematic Investments

A schematic of a live 'production' trading strategy is shown below, but does not include everything else necessary to create the strategy (i.e., research tools).

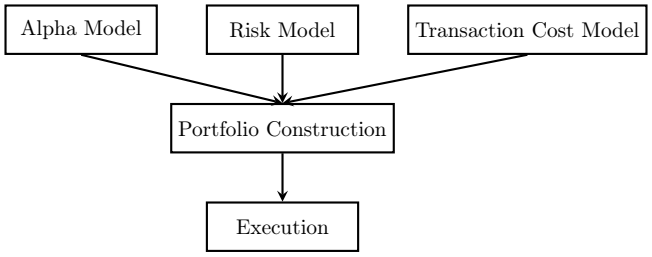


Figure 1: Live Production Trading Strategy Overview

At its core, the trading system is organised into three primary modules:
The trading system has three modules:

i. Alpha model: predicts the future of the instruments considered for trading, i.e. directional alpha

ii. Risk model: limits amount of exposure to factors that are unlikely to generate returns but could drive losses, i.e. directional exposure limit on an asset class

iii. Transaction cost model: determine if the cost of the trades needed to migrate from current portfolio to new portfolio is desirable to the portfolio construction model.

These models feed into a portfolio construction model that balances the tradeoffs of profit and risk to determine the best portfolio to hold. The model finds the differences in trades that need to be executed.
The execution model then takes the required trades, and using inputs such as urgency in which the trades need to be executed and dynamics of liquidity in the markets, executes the trades in an efficient and low cost manner.

Method 1.4.1. *Chains of Production for Alpha Signals*

i. Data Curation: for collecting, cleaning, indexing, storing, adjusting, and delivering all data to production chain. Requires experts in market microstructure and data protocols such as FIX.

ii. Feature Analysis: transform raw data into informative signals. Requires experts in information theory, signal extraction and processing, visualisation, labelling, weighting, classifiers, feature importance techniques. Feature analysts collect and catalogue libraries of findings.

iii. Strategists: informative features are transformed into actual investment algorithms. Strategists will parse libraries of features for ideas to develop an investment strategy. Require data scientists with deep knowledge of financial markets and economy. Features may be discovered by black box, but strategy is developed in a white box.

iv. Back-testers: assess profitability of investment strategy under various scenarios. Requires data scientists with deep understanding of empirical and experimental techniques. Good back-tested incorporates in analysis meta-information on how strategy was created.

v. Deployment Team: integrate strategy code into production line. Requires algorithm specialists and mathematical programmers. To ensure deployed solution is logically identical to prototype, and to optimise implementation sufficiently such that production latency is minimised.

vi. Portfolio Oversight: once strategy is deployed, follows lifecycle.

1. Embargo: initially, strategy is run on data observed after end date of backtest. If embargoed performance is consistent with backtest, strategy is promoted to next stage.
2. Paper Trading: strategy run on live, real-time feed. Performance accounts for data parsing latencies, calculation latencies, execution delays, and other time lapses between observation and positioning.
3. Graduation: strategy manages real position, whether in isolation of as part of ensemble. Performance evaluated precisely, including attributed risk, returns, and costs.
4. Re-allocation: based on production performance, allocation is re-assessed frequently and automatically. Strategy allocation follows a concave function, Initial allocation is small. As time passes and strategy performs as expected, allocation is increased. Over time, performance decays and allocations become gradually smaller.
5. Decommission: if strategy perform below expectations for sufficiently extended period of time, strategy is discontinued.

1.4.1 Alpha Models Overview

Theory-driven models tests theories of why markets behave in a manner, and see if they can be used to predict the future. Strategies utilising price-related data are trend and mean reversion; strategies utilising fundamental data are value/yield, growth and quality. Usually more than one model is used in combination.

Definition 1.4.2. *Theory Driven Models*

i. Trend Following: markets move in given direction long enough that the trend can be identified. As more data support the bull/bear thesis in an uncertain market, more market participants will adopt the same thesis and hence move the asset price to a new equilibrium.
Moving average crossover indicator strategy has less than one point of return for every point of downside risk taken, as market behaviour are unstable and episodic.

ii. Mean Reversion: markets move in opposite direction to the prevailing trend. Short-term imbalances between buyers and sellers due to liquidity forces prices to move abruptly in one direction, which increases probability of trend reversion as liquidity issue is resolved.
Statistical arbitrage bets on convergence of prices of similar stocks whose prices have diverged.
Longer-term trends can occur despite smaller oscillations around these trends occurring in the shorter term, hence both strategies may be used in conjunction.

iii. Value/Yield: value strategies uses ratios of fundamental factor against the price of the instrument, inverted to keep the ratio consistent. The higher the yield, the cheaper the instrument.
Buying undervalued security and selling overvalued security is a *carry trade*. The difference between yield received and yield paid is the *carry*.
Quant Long Short (QLS) ranks stocks by attractiveness based on various factors such as value, then buy the higher-ranked stocks while shorting the lower-ranked stocks.

iv. Growth: make predictions based on asset's expected or historically observed level of economic growth. Forward-looking growth expectations are typically used as a metric.
Growth is trending, and strongest growers are becoming more dominant relative to competitors. Macro growth factors may be used on foreign exchange, while micro growth factors may be used on companies.

v. Quality: All else being equal, it is better to long high quality and short low quality. Capital safety is important. Factors include earnings quality, equity-to-debt ratios etc.

Data-driven models are more difficult to understand, with more complicated mathematics. Relies on data mining, more technically challenging and far less widely practiced. Typically more used in high-frequency space, as they can discern how market behaves without caring about the economic theory or rational.

Method 1.4.3. *Strategy Parameters*
An implementation approach requires a forecast target, time horizon, bet structure, investment universe, model specification, and run frequency.

i. Forecast Target: models may forecast direction, magnitude, duration of move, and may include probability into the forecast. Signal strength is of importance, defined by a larger expected return and/or higher likelihood of return. A higher level of signal strength results in a bigger bet taken on the position.

ii. Time Horizon: models may have forecast horizons ranging from microseconds to years. There are more variability between short-term and long-term strategies, as short-term strategies are making very large number of trades compared to long-term strategies.

iii. Bet Structure: models can be made to forecast an instrument relative in itself or to others. For relative forecasts, smaller clusters (pairs) or larger clusters (sectors) may be used. For pairs, few assets can be compared precisely and directly. Large cluster grouping may eliminate impact of general movement of the sector and hence focus on the relative movement of stocks within the sector, allowing for clearer distinction between group behaviour and idiosyncratic behaviour. Clusters may be created either via statistical methods or using heuristics (i.e., fundamentally defined industry groups).
Statistical methods may be fooled by data, leading to bad grouping. Heuristic grouping may be imprecise for conglomerates, and may be too rigid. Relative alpha strategies tend to exhibit smoother returns during normal times than intrinsic alpha strategies, but may face incorrect groupings during stressful periods. This may be mitigated by utilising several grouping techniques in concert.

iv. Investment Universe: choices made on geography, asset class, instrument class, and exclusions. Liquidity is preferred so estimations of transaction costs are reliable. Large quantities of high quality data is required, which is found in highly liquid and developed markets. Instruments with consistent behaviour is preferred, hence biotech stocks are excluded due to sudden, violent price changes. Hence, the most common asset classes and instruments modelled are common stocks, futures (on bonds and equity indices) and forex.

v. Model Specification: focuses on definition of the strategy mathematically, and may be the source of alpha. Specification details in terms of machine learning or data mining techniques are also defined, to assist in fitting models to the data and setting parameter values. Refitting frequency is also defined to refresh the model and make it adapt to current market conditions; may lead to greater risk of overfitting.

vi. Run Frequency: defined from monthly to real time frequency. Increasing frequency of runs lead to greater number of transactions and hence higher transaction costs, and risk of moving portfolio based on noisy data. Less frequency of runs lead to smaller number of larger-sized trades, hence may move the market with block trades; may also miss opportunities to trade at more favourable prices.

Method 1.4.4. *Blending of Models*
Most common approaches are linear models, nonlinear models, and machine learning models. If models are not combined, then several portfolios are constructed based on output from each model, then combined using portfolio construction techniques. The best method depends on the model.

i. Linear Models: require independence of factors, and each factor to be additive. To determine the weight of each alpha factor, multiple regression techniques may be used.

ii. Nonlinear Models: used when factors are not independent, or the relationship changes over time. Conditional models base the weight of one factor on the reading of another factor. Rotational models assign weights of factors that fluctuate over time based on updated calculations of the various signal's weights, giving higher weights to factors with better performance recently.

iii. Machine Learning Models: developing machine learning strategies takes as much effort to produce one true investment strategy as to produce a hundred. The complexities include data curation and processing, HPC infrastructure, software development, feature analysis, execution simulations, backtesting etc.
Decades ago, macroscopic alpha based on simple tools like econometrics are common, but this is quickly diminishing. Microscopic alpha however, becomes more abundant, but requires heavy ML tools.

1.4.2 Risk Models

Risk model concerns the intentional selection and sizing of exposures to improve the quality and consistency of returns. By pursing an alpha, we want to be invested in the movement of the exposure to profit in the long run.

Method 1.4.5. *Factor-Based Models*
Factor-based models decompose asset returns into contributions from systematic factors and idiosyncratic components. The most common factors include:

i. Market Factor: Captures the overall movement of the market.

ii. Size Factor: Reflects the differential risk associated with companies of varying market capitalizations.

iii. Value Factor: Accounts for risk due to discrepancies between market prices and fundamental valuations.

iv. Momentum Factor: Measures the tendency of asset prices to continue in their current trajectory.

This allows traders to understand which elements drive portfolio risk and adjust exposures accordingly.

Method 1.4.6. *Statistical Models*
Statistical risk models leverage historical data and probabilistic techniques to quantify risk parameters.

i. Historical Simulation: Directly computing risk metrics from past return distributions.

ii. Monte Carlo Simulation: Generating a large number of potential future return scenarios to estimate risk under diverse conditions.

iii. Parametric Methods: Employing analytical formulas based on assumed return distributions to calculate key risk measures.

These are useful for dynamically updating risk assessments as new market data become available.

Method 1.4.7. *Limiting Size of Risk*
The quantitative risk models that limit the size of risk varies by the manner in which size is limited, how risk is measured, and what is having its size limited.
Size limits can be limited by hard constraints and penalties. A hard limit may be arbitrary, hence penalty functions may be built to allow a position to increase beyond the limit level, only if the alpha model expects a significantly larger return. The levels of limits and penalties may be determined from either theory or data.
To measure risk, there are two methodologies. The first is longitudinal, and measures risk through the volatility of an instrument. The second is to measure the correlation or covariance between assets (dispersion).
Size limiting may be applied to single positions and groups of positions (sectors, asset classes). It may also be applied to various types of risks and the amount of portfolio leverage.

Method 1.4.8. *Limiting the Types of Risk*
To eliminate unintentional exposure as there is no expectation of being compensated sufficiently for accepting them. This can be achieved through theoretical or empirical risk models.

i. Theory-Driven Risk Models: focuses on systematic risk factors, derived from economic theory. Systematic risks cannot be diversified away. Equity may have market risk, sector risk, market capitalisation risk etc. Fixed income may have interest rate risk.

ii. Empirical Risk Models: uses historical data to determine the unnamed systematic risks that should be measured and mitigated. Uses principal component analysis (PCA) to discern unnamed systematic risks that may correspond to named risk factors. Used by statistical arbitrage traders who are betting on exactly the component of an asset's return not explained by systematic risks.

1.4.3 Transaction Cost Models

Trade is made only if it increases the odds or magnitude of return (from alpha model), or if it decreases the odds or magnitudes of loss (from risk model). However, this improvement should be higher than cost of trading. The transaction cost model is not designed to minimise cost of trading, only to inform portfolio construction engine the cost of making any given trade.

Remark 1.4.9. *Transaction Cost Components*

i. Commissions and Fees: paid to brokerages (access to other market participants), exchanges (improved transaction security) and regulators (operational infrastructure) for the services provided. The bank's infrastructure is used by quants, where the brokerage commissions are rather small on a per-trade basis. Brokers also collect clearing and settlement fees. Clearing is the activity involving regulatory reporting and monitoring, tax handling, and handling failure, taken place in advance of settlement. Settlement is the delivery of securities in exchange for payment in full.

ii. Slippage: the change in price between the time the quant system decides to transact and the time when the order is at the exchange for execution. Trend-following strategies suffer most from slippage as assets are already moving in desired direction; mean-reverting strategies suffer the least from slippage. The lower the latency to market, the smaller the slippage. The more volatile an asset, the bigger the slippage.

iii. Market Impact: measures how much an order moves the market by its demand for liquidity. The impact of the trade on the market is unknown until the trade has already been completed. There may also be interaction between slippage and market impact (i.e., selling when a stock is trending upwards).

Definition 1.4.10. *Types of Transaction Cost Models*

i. Flat Model: cost of trading is the same, regardless of size of order. Model is reasonable if size traded is nearly always about the same, and liquidity remains sufficiently constant.

ii. Linear Model: cost of trading increases at a constant rate relative to size of order. Better estimate than flat transaction cost model.

iii. Piece-Wise Linear Model: using piece-wise linear functions to model costs. Balance between simplicity and accuracy; better accuracy than flat or linear models.

iv. Quadratic Model: most computationally intensive, but also most accurate.

1.4.4 Portfolio Construction Models

Comes in two major forms: rule-based, optimisers. Rule-based models are based on heuristics, can be exceedingly simple or rather complex, and derived from human experience (trial and error). Optimisers comprises of an objective function and uses algorithms to reach the end goal.

Definition 1.4.11. *Rule-Based Models*

i. Equal Position Weighting: used if portfolio manager believes that if a position is good enough to own, no other information is needed in determining its size. Strength of signal is not used as input in weighting. Model assumes that there is sufficient statistical strength and power to predict not only direction but also magnitude relative to other forecasts in the portfolio. Portfolio takes few large bets on 'best' forecast, many smaller bets on less dramatic forecasts; may take excess risk in an idiosyncratic event on a seemingly attractive position, resulting in adverse selection bias.

ii. Equal Risk Weighting: adjust position sizes inversely to volatilities or a measure of risk. More volatile positions given smaller allocations, less volatile positions given larger allocations.
When unit of risk is equalised, it is almost always a backward-looking measurement such as volatility. If volatility changes with time, then model will be misled.

iii. Alpha-Driven Weighting: position size based primarily on alpha model. Alpha signal determines size of position, but usually with size limits. Constraints used also includes limits on size of total bet on a group. May also have a function that relates the magnitude of forecast to size of position.
If model used in futures trend following, might suffer sharp drawdowns. Reliance on accuracy of alpha.

iv. Decision-Tree Weighting: decision path to arrive at the allocation for given instrument, depending on type of alpha model and type of instrument. Constraints may include percentage limits for allocation.
Model size grows dramatically if more alpha models or mode types of positions are included.

Remark 1.4.12. *Optimisers Models Parameters*
Harry Markowitz's mean variance optimisation (MVO) as the pioneer model. Models are based on principles of modern portfolio theory (MPT). Inputs include asset expected return (mean), asset variance, expected correlation matrix. Other inputs include size of portfolio in currency terms, desired risk level (volatility or expected drawdown), and other constraints such as liquidity, universe limits.
Model uses an objective function and an algorithm to seek the goal, usually maximising return of portfolio relative to volatility of portfolio returns.

i. Expected Return: alpha models as basis of expected return, which also includes expected direction.

ii. Expected Volatility: stochastic volatility forecasting methods is commonly used, as volatility may have high and low periods, with occasional jumps. GARCH model is most used.

iii. Expected Correlation: as instrument correlations are not stable over time, it is more appropriate to group assets together before computing correlation within the group.

Method 1.4.13. *Optimisation Techniques*

i. Unconstrained Optimisation: most basic form with no constraints. Might provide a single-instrument portfolio, where all money will be invested in instrument with highest risk-adjusted return.

ii. Constrained optimisation: constraints include position limits, limits on various groupings of instruments. Might result in constraints driving the portfolio construction more than the optimiser.

iii. Black-Litterman Optimisation: blends investor expectations with a degree of confidence about those expectations, and these with historical precedent evident in the data. Adjusts historically observed correlation levels by utilising investor's forecast of return for the various instruments.

iv. Grinold and Kahn's Approach: builds a portfolio of signals, instead of sizing positions. To build factor portfolios, each of which are usually rule-based portfolios based on a single type of alpha forecast. Each portfolio backtested, then series of returns are then treated as instruments of a portfolio by the optimiser. Number of factor portfolios is more manageable, usually not more than 20. What is optimised is then a handful of factor portfolios. The model allows for inclusion of risk model, transaction cost model, portfolio size, and risk targets as inputs.

v. Resampled Efficiency: to improve the inputs to optimisation by addressing oversensitivity to estimation error. To resample data using Monte Carlo simulation to reduce estimation error in inputs to the optimiser.

vi. Data-Mining Approaches: machine learning techniques such as supervised learning or genetic algorithms used, as MVO involves searching many possible portfolios to find the best.

1.4.5 Execution Model

Two basic ways to execute trade: through electronic, or through human intermediary. For electronic execution, achieved through direct market access (DMA), which allows traders to utilise the infrastructure and exchange connectivity of brokerage firms to trade directly on electronic markets.
Execution algorithms can be acquired through building, using broker's, or a third-party software vendors.
Brokerages offer portfolio bidding, where the 'blind' portfolio for transaction is described by characteristics such as valuation ratios of longs and shorts, sector breakdown, market capitalisation etc. Broker then quote a fee in basis points in terms of the gross market value of portfolio traded. Hence, certainty is provided by the broker to the trader. Once agreement reached, broker receives fee and assumes risk of trading out the portfolio at future market prices, which may be better or worse than prices guaranteed.

Remark 1.4.14. *Order Execution Algorithm Parameters*

i. Aggressive vs Passive: algorithm make decision of passive vs aggressive order, depending on how immediately the trader wants to do the trade. Market orders are considered aggressive. Limit order at current best order is fairly aggressive, while limit order below current bid is passive.
Many exchanges pay providers of liquidity for placing passive orders, charging traders for using liquidity provided. Orders that cross the spread are using liquidity by using a passive order placed by another

trader, reducing liquidity available. Paying for liquidity sweetens deal for passive order, only if order is actually executed; passive trader gets better transaction price and a commission rebate from the exchange. Momentum strategies uses more aggressive orders; mean reversion uses more passive orders. A stronger, more certain signal will be executed with greater aggressiveness than a weaker or less certain signal. A middle ground will be to put limit orders between best current bid and offer.

ii. Large vs Small Order: a large order may be broken into many smaller orders over a window of time, but risk price moving in adverse direction. Size of chunk depends on transaction cost model estimate, and analysis of correct level of aggressiveness.

iii. Hidden vs Visible Order: a queue as a visible order gives away a bit of information. Hidden order will provide no information to the market, staving off imbalances, but reduces priority of trade in the queue. Algorithmic trading utilising hidden order is 'iceberging', which is taking a single larger order and chopping it into many smaller chunks, most posted to order book as hidden orders.

iv. Order Routing: if there are several pools of liquidity for the same instrument, smart order routing will be used, which determines which pool of liquidity is most suitable for sending a given order. Depth of liquidity on various ECNs and connectivity speeds are also considered in smart order routing.

v. Cancelling and Replacing Orders: traders may place larger number of orders with no intention of execution, then rapidly cancelling them and replacing them with other orders. This allows gaining of information on how market responds to the changing depth of the book, providing information on how to profit from the pattern of reaction. If trader wants to buy a large number of shares, he may enter a large number of small orders to sell the shares further away from market and cancel, improving market perception.

Definition 1.4.15. *High Frequency Trading*
Alpha driving strategies on extremely near-term bets (seconds or less) are *microstructure alphas*, focusing on liquidity patterns in order book. Larger quants may also use this to guide execution models, improving costs of entering trades. Small differences over a single trade add up significantly in the long run.To trade microstructure alpha as independent high frequency strategies, large investments in infrastructure and research must be done. Machine learning techniques may also be used to discern patterns in execution of other player orders. The more inferior the execution models, the easier it is to discern the pattern, allowing the ML strategy to profit from these patterns in the future. Patterns in the shorter timescale are somewhat stable.

Definition 1.4.16. *HFT Shark Strategy*
Designed to detect large orders that are iceberged, by sending series of very small trades; if each of these small orders get filled quickly, this may be a sign of a large and iceberged order. The shark simply front-run this large, hidden order by placing visible trades in front of the iceberged order. The iceberg strategy must then push prices up to execute trades. When the iceberged order is complete, prices will be pushed up favourably for the shark, which can then exit the position with a quick and relatively riskless profit.

Remark 1.4.17. *HFT Trading Infrastructure*
Using a broker that act as trading agent allows the infrastructure requirements to be handled by the broker, instead of dealing with the regulatory and other constraints.
High frequency strategies may use colocation or sponsored access. Colocation setup is where trader attempts to place trading servers as physically close to the exchange as possible.
Financial Information eXchange (FIX) protocol is the choice of real-time electronic communication among users. The software that implements the FIX protocol is free and open source (FIX engine). High frequency traders will likely build their own FIX engines to ensure optimal speeds.

1.4.6 Research

Definition 1.4.18. *Scientific Method*

1. Researcher observe a phenomenon in the market and construct a theory.
2. Researcher seeks out information to test the theory.
3. Researcher tests the theory, and with enough confidence, risk some capital on the validity of the theory.

Remark 1.4.19. *Sources of Alpha Idea Generation*

1. Observing the market, using the scientific method to test the theory
2. Academic literature, requiring significant time to read academic journals, working papers, and conference presentations for ideas. Literature from other fields such as astronomy, physics, or psychology, may provide ideas relevant to quant finance problems.
3. Migration of a researcher or portfolio manager from one quant shop to another.

4. Lessons from activities of discretionary traders

Remark 1.4.20. *Model Quality Assessment*

i. Cumulative profit graph: if profit profile is not smooth, with long periods of inactivity, sharp losses and gains, then the model may have issues

ii. Average annual rate of return: indicates how well the strategy made on historical data

iii. Variability of returns: the less variable the level of returns, the better the strategy. May look at lumpiness of returns, which is the portion of strategy's total returns that comes from periods that are significantly above average (measures consistency of returns).

iv. Worse Peak-to-Valley Drawdowns: measures maximum decline from any cumulative peak in profit curve. The lower the drawdown the better the strategy. Also, to measure recovery period after drawdowns; the shorter the recovery period the better the strategy.

v. Predictive Power: R-squared statistic may be used, which shows how much of the variability of the predicted asset have been accounted for. A exceedingly high R^2 in would be 0.05 out of sample. Instrument returns may be bucketed by deciles; a model with reliable predictive power is one that appropriately buckets the instruments correctly.

vi. Percentage Winning Trades, Winning Time Periods: whether the strategy tends to make profits from a small portion of trades that do very well, or from a large number of trades.

vii. Ratios of Returns vs Risk: Statistics such as risk-adjusted return, Sharpe ratio, information ratio, Sterling ratio, Calmer ratio, Omega ratio.

viii. Relationship with Other Strategies: value-add of new strategy compared with results of existing strategy with and without the new idea.

ix. Time decay: understand strategy returns if trades are initiated on lagged basis after receiving a trading signal. Determine strategy sensitivity to timeliness with information received, and crowdedness of strategy.

x. Sensitivity to specific parameters: high quality strategy has small changes in outcomes from slight changes in parameters. Or else this may be a sign that model may be overfitted.

xi. Overfitting: plot a graph of parameter value vs function outcome; a good model has a flatter curve with no jumps. Models that are parsimonious (less parameters) uses less assumptions, hence less overfitting.

Remark 1.4.21. *Other Considerations in Model Testing*
Overestimation of trading costs may cause portfolio to hold positions for longer than optimal, and underestimation may result in high portfolio turnover and bleed from trading costs. Assumptions on availability of short positions must also be made; hard-to-borrow lists must be taken into consideration.

1.4.7 Risk Assessment

Definition 1.4.22. *Model Risks*
Quant models has model risk, the risk that the model does not accurately describe, match, or predict the real-world phenomenon. Each component of the quant model may all have model risk.

i. Inapplicability of Modelling: occurs when quant model is mistakenly applied to a problem. May also occur with misapplication of a technique to a given problem.

ii. Model Misspecification: occurs when the model doesn't fit the real world. Model may work fine most of the time, but fail when an extreme event occurs.

iii. Implementation Errors: erors in programming or architecting systems. Architectural error may also occur when models are loaded in a wrong sequence.

Definition 1.4.23. *Regime Change Risk*
Quant models are based on relationships prevalent in historical data. If there is a regime change, the historical relationships and behaviour may be altered, hence the model may lose effectiveness.

Definition 1.4.24. *Exogenous Shock Risk*
Risks driven by information that is not internal to the market, i.e., terrorist attacks, start of wars, bank bailouts, change in regulation such as in shorting rules. May require discretionary overrides.

Definition 1.4.25. *Contagion Risk*
Happens when other investors hold the same strategies. First part of risk factor relates to how crowded the quant strategy is. Second part relates to what else is held by other investors that could force them to exit the quant strategy in a panic (ATM effect).

Quant liquidation criss may be driven by size and popularity of quantitative strategies, subpar returns from operators leading up to the crisis, the practice of funds cross-collateralising many strategies against each other, and risk targeting (risk managers target a specific level of volatility for their funds or strategies).

Method 1.4.26. *Risk Monitoring Methods*

i. Exposure Monitoring Tools: with current positions held, the positions are grouped for the various exposures (i.e., valuation, momentum level, volatility) to monitor gross and net exposure to various sectors and industries, buckets of market capitalisation, various style factors.

ii. Profit and Loss Monitors: with current portfolio, compare that with previous day closing price. Intraday performance charts are used. May also look at source of profit, hit rate (percentage of time strategy makes money on a given position).

iii. Execution Monitors: shows progress of executions, i.e., which orders are currently being worked on, which ones are completed, with transaction size and prices. Fill rates for limit orders are used for more passive execution strategies. Slippage and market impact are also monitored.

iv. System Performance Monitors: checks for software and infrastructure errors. Checks performance of CPUs, speeds of various stages of automated processes, latency in communication of messages.

1.5 Exploratory Data Analysis

1.5.1 Data Taxonomy

A brief overview of the types of data used in systematic trading.

Four essential types of financial data

Fundamental Data	Market Data	Analytics	Alternative Data
Assets	Price/Yield/IV	Analyst Recommendation	Satellite/CCTV
Liabilities	Volume	Credit Ratings	Google Searches
Sales	Dividend/Coupons	Earnings Expectations	Twitter/Chats
Costs/Earnings	Open Interest	News Sentiment	Metadata
Macro Variables	Quotes/Cancellations	...	...
...	Aggressor Side		
	...		

Remark 1.5.1. *Fundamental Data Characteristics*

i. Data published is indexed by last date included in report, which precedes date of release.

ii. Data is often backfilled or re-instated, and data vendor may overwrite initial values with corrections.

iii. Data is extremely regularised and low frequency.

Remark 1.5.2. *Market Data Characteristics*

i. Raw feed contains unstructured information, such as FIX messages (allow full construction of trading book), or full collection of BWIC (bids wanted in competition) responses.

ii. FIX data is not trivial to process, $\sim$ 10TB generated on daily basis

Remark 1.5.3. *Analytics Data Characteristics*

i. Derivative data as processed based on original source. Signal already extracted from the original source.

ii. Costly, methodology used in production may be biased or opaque.

Remark 1.5.4. *Alternative Data Characteristics*

i. Produced by individuals, business processes, and sensors.

ii. Primary information that has not made it to other sources.

iii. Cost and privacy concerns. May be useful if it annoys data infrastructure team.

Definition 1.5.5. *Reference Data*

i. Trading Universe: evolving daily to incorporate new listings, de-listings etc. Knowing when a particular stock no longer trades is important to avoid survivor bias.

ii. Symbology Mapping: ISIN, SEDOL, RIC, Bloomberg Tickers etc. Data is not static, symbols may change, complicating historical data merges. Mapping needs to persist as point-in-time data and allow for historical 'as-of-date' usage, require implementation of bi-temporal data structure.

iii. Ticker Changes: for reasons described in symbology mapping. To maintain historical table of ticker changes to seamlessly go up and down time series data.

iv. Corporate Actions Calendars: contain stock and cash dividends (announcement, execution date), stock splits, reverse splits, rights offer, mergers and acquisitions, spin off, free float or shares outstanding adjustments, quotation suspensions etc.
For dividends, announcements may coincide with more volatility, jumps in price time series. Allow building of strategies that look to benefit from the added volatility.
For stock splits, reverse splits, rights offers, all historical data need to be adjusted backward to reflect the split (both volume and price).
For M&A, spin-offs, to account for changes in valuation, hence used in Merger Arbitrage strategies.
Suspensions result in gaps in data, may impact backtesting.

v. Static Data: country, sector, primary exchange, currency, and quote factor. May be used to group instruments based on fundamental similarities (hence for pairs trading). Maintaining a table of quotation currency per instrument necessary to aggregate positions at portfolio level.

vi. Exchange Specific Data: individual exchanges have variety of differences to be accounted for when designing trading strategies. First group of information concerns the hours and dates of operations:

1. Holiday Calendar: Strategies trading simultaneously in several markets and leveraging correlation may not perform as well if one market is closed and another is open.
2. Exchange Sessions Hours: Different sessions (Pre-Market, Continuous Core, After-Hour etc.); auction times and respective cutoff times for oder submission; lunch break restricting intraday trading and auctions before/after lunch; settlement times for futures market. Daylight Saving Time (DST) adjustments; length of trading hours during course of the year; different trading hours by venue.
3. Disrupted Days: Exchange outages or trading disruptions, market data issues. To be recorded so they can be filtered out when building or testing strategies.

Second group of information governing the mechanics of trading:

1. Tick Size: Minimum eligible price increment; may vary by instrument and as a function of price.
2. Trade and Quote Lots: Minimum size increment for quotes or trades.
3. Limit-Up and Limit-Down Constraints: Maximum daily fluctuations of securities, and whether trading is paused or can only be traded at better prices than the threshold.
4. Short Sell Restrictions: Restrict shot sells not to trade at price worse than last price, or not to create a new quote that will be lower than the lowest prevailing quote. Impact ability to source liquidity.

vii. Market Data Condition Codes: vary per exchange and asset class, and each market event may be attributed to several codes at once. To build mapping table of condition codes and what they mean (i.e., auction trade, lit or dark trade, cancelled or corrected trade, regular trade, off-exchange trade reporting, block-size trade, trade originating from multi-leg order such as option spread trade etc.). To access liquidity for trading algorithm, trades published for reporting purposes must be excluded and not be used to update some of the aggregated daily data used in construction of trading strategies.

viii. Special Day Calendars: days wth distinct liquidity characteristics to be accounted for in both execution strategies and in alpha generation process. These (non-exhaustive) irregular events may be:

1. Half trading days preceding Christmas and following thanksgiving in US
2. Ramandan even in Turkey
3. Taiwan market opening on weekend to make up for lost trading days during holiday periods
4. Korean market changing trading hours on day of nationwide university entrance exam
5. Brazilian market opening late on day following the Carnival
6. Last trading days of months and quarters (investors rebalance portfolios)
7. Index rebalancing dates, where intraday volume distribution is significantly skewed toward EODs
8. Options and futures expiry dates (quarterly/monthly expiry, Triple Witching in US, Special Quotations in Japan) where excess trading volume and different intraday patterns result from hedging activity and portfolio adjustments.

Model normal days first. Special days are modelled either independently, or using normal days as baseline.

ix. Futures-Specific Reference Data: to know which contract was live at any point of time by using expiry calendar, and the most liquid contract. Equity index futures are most liquid for first contract available (front month), energy futures such as oil are more liquid for second contract. Hence to know which contract carry the most significant price formation characteristics, and what is true liquidity available.
Note there is no real standardised expiry frequency that applies across markets. When computing rolling-window metrics, to account for potential roll dates (due to investors rolling forward positions) that may have happened during the time span. May blend volume time series prior to roll date and after roll date. Futures market also have different market phases during the day with significantly different liquidity characteristics. Various market data metrics (volume profile, average spread, average bid-ask sizes etc) should be computed separately for each market phase by maintaining a table of start and end times of each session for each contract.

x. Options-Specific Reference Data (Options Chain): expiry date and strike price combination (option chain). Map of equity tickers to option tickers with strike and expiry dates allow for design for more complex investment and hedging strategies (i.e., distance to strike, change in open interest of puts and calls).

xi. Market-Moving News Releases: macroeconomic announcements. To maintain calendar of dates and times of their occurrences to assess their impact on strategies. Central bank announcements or meeting minutes releases about major economies (FED/FOMC, ECB, BOE, BOJ, SNB), Non-Farm Payrolls, Purchasing Managers' Index, Manufacturing Index, Crude Oil Inventories etc.
Stock-specific releases such as earnings calendars, specialised sector events (for healthcare, biotech etc).

xii. Related Tickers: tickers that are related to each other as they fundamentally represent the same underlying asset. Allows efficient opportunity exploitation. Primary tickers to composite tickers mapping (for markets with fragmented liquidity), dual listed/fungible securities in US and Canada, ADR or GDR, local and foreign boards in Thailand etc.

xiii. Composite Assets: ETFs, Indexes, Mutual Funds etc. May be used to achieve desired exposures, or as cheap hedging instruments, and provide arbitrage opportunities when they deviate from NAV. To maintain information such as time series of their constituents and value of any cash component, divisor used to translate NAV into quoted price, constituent weights.

xiv. Latency tables: for higher frequency trading strategies. Contains distribution of latency between different data centres for more efficient order routing, and reordering data that are recorded in different locations.

Definition 1.5.6. *Market Data Feed*

i. Level I Data (Trade and BBO Quotes): trades and top of book quotes. Enough to reconstruct Best Bid and Offer (BBO). Also contains information in form of trade status (cancelled, reported late etc), trade and quote qualifiers (odd lot, normal trade, auction trade, Intermarket Sweep, average price reporting, on which exchange etc). May be used to analyse sequence of events and decide if a given print should be used to update the last price and total volume traded at a point in time.

ii. Level II Data (Market Depth): addition of quote depth data, displays all lit limit order book updates (price changes, addition or removal of shares quoted) at any level in the book, for all of the lit venues in fragmented markets.

iii. Level III Data (Full Order View): message data. Each order arriving is attributed a unique ID for tracking over time, and is precisely identified when it is executed, cancelled, or amended. Possible to build a full (with national depth) book at any moment intraday. Example from US market:

1. Timestamp: number of milliseconds after midnight
2. Ticker: equity symbol (up to 8 char)
3. Order: Unique order ID
4. T: message type. 'B' is add buy order; 'S' is add sell order'; 'E' is execute outstanding order in part; 'C' is cancel outstanding order in part; 'F' is execute outstanding order in full; 'D' is delete outstanding order in full; 'X' is bulk volume for cross event; 'T' is execute non-displayed order
5. Shares: order quantity for 'B', 'S', 'E', 'X', 'C', 'T' messages. Zero for 'F', 'D' messages
6. Price: order price for 'B', 'S', 'X', 'T' messages. Zero for cancellation and executions. Last 4 digits are decimal, padded to right with zeroes. Divide by 1000 to convert to currency value.
7. MPID: Market Participant ID associated with transaction (4 char)
8. MCID: Market Centre Code for originating exchange (1 char)

A few special types of orders worth mentioning are:

1. Order subject to price sliding: execution price may be one cent worse than display price at NASDAQ; ranked at locking price as hidden order, displayed at the price one minimum price variation inferior to locking price. New order ID will be used if order is replaced as a display order.
2. Pegged order: based on NBBO, not routable, new timestamp given upon repricing; display rule vary over exchanges
3. Mid-point peg order: non-displayed, may result in half-penny execution
4. Reserve order: displayed size is ranked as displayed limit order; reserve size is behind non-displayed orders and pegged orders in priority.
   Minimum display quantity is 100, amount replenished from reserve size when it falls below 100 shares; New timestamp created, displayed size re-ranked upon replenishment.
5. Discretionary order: displayed at one price while passively trading a more aggressive discretionary price. Order becomes active when shares are available within discretionary price range. Order ranked last in priority. Execution price may be worse than display price.
6. Intermarket sweep order: can be executed without need for checking prevailing NBBO.

Using these data, we may model: the pattern of inter-arrival times of various events; arrival and cancellation rates as a function of distance from nearest touch price; arrival and cancellation rates as a function of other information, such as in the queue on either side of the book, order book imbalance etc.
Once modelled, we may analyse: the impact of market order on limit order book; chances for limit order to move up the queue from given entry position; probability of earning the spread; expected direction of price movement over a short horizon.

Definition 1.5.7. *Binned Data*

i. Open, High, Low, Close (OHLC) and Previous Close Price: indication on trading activity and intraday volatility. Distance traveled between lowest and highest points is indication of market sentiment. Previous close has to be adjusted for corporate actions and dividends.

ii. Last Trade before Close (Price/Size/Time): how much the close price may have jumped in final moments of trading; how stable it is as a reference value for next day.

iii. Volume: trading activity indicator, especially when level jumps from long term average. Collect volume breakdown between lit and dark venues for execution strategies.

iv. Auctions Volume and Price: price discovery event when significant volume prints occur.

v. VWAP: indication of trading activity on the day. Easier to algorithmically execute large orders with VWAP than a single print.

vi. Short Interest/Days-to-Cover/Utilisation: good proxy for investor position. Short pressure an indication of upcoming short term moves: large short interest is bearish view from institutional investors. Utilisation level of available securities to borrow gives indication of how much room is left for future shorting. Days-to-Cover to assess magnitude of potential short squeeze (if sellers unwind position, fraction of available daily liquidity needed); larger value indicates larger potential of sudden upswing on heavily shorted securities.

vii. Futures Data: insight into activity or large investors through open interest data. Offer arbitrage opportunities if their basis exhibits mis-pricing compared to one's dividend estimates.

viii. Index-Level Data: source of relative measures for instrument specific features (index OHLC, volatility). Normalised features identify individual instruments deviating from their benchmarks.

ix. Options Data: information on position of traders through open interest and Greeks.

x. Asset Class Specific: yield/benchmark rates (repo, 2y, 10y, 30y), CDS spreads, US Dollar Index

Definition 1.5.8. *Granular Intraday Microstructure Activity*

i. Number and Frequency of Trades: proxy for activity level, and how continuous it is. Low number of trades mean harder execution, and may be more volatile

ii. Number and Frequency of Quote Updates: similar proxy for activity level

iii. Top of Book Size; proxy for liquidity of instruments (larger top of book size makes it possible to trade larger order quasi immediately)

iv. Depth of Book (price and size): similar proxy for liquidity

v. Spread Size (average, median, time weighted average): proxy for cost of trading. Parametrised distribution used to identify opportunities if they are cheap or expensive

vi. Trade size (average, median): to identify intraday liquidity opportunities when examining volume available in the order book.

vii. Ticking time (average, median): representation of how often one should expect changes in the order book first level. For execution algorithms for which the frequency of updates (adding/cancelling child orders, re-evaluating decisions etc.) should commensurate with characteristics of the traded instrument.

Daily distribution can be used as start of day estimates and updated intraday with online Bayesian updates. Last group of daily data is derived from previous two groups but stored pre-computed to save time during research phase, or to be used as normalising values:

i. X-day Average Daily Volume (ADV) / Average Auction Volume

ii. X-day Volatility (close-to-close, open-to-close etc)

iii. Beta with respect to index or sector (plain beta, or asymmetric up-days/down-days beta)

iv. Correlation matrix

When binning data, this may be grouped into bins ranging from a few seconds to 30 minutes. Minute bar data are used for volume and spread profiles to prevent introducing excess noise due to market friction.

Definition 1.5.9. *Fundamental Data and Other Data*

i. Key Ratios: Earnings Per Share (EPS), Price-to-Earning (P/E), Price-to-Book (P/B), etc.

ii. Analyst Recommendations: aggregated values given consensus valuation

iii. Earnings data: estimations by research analysts provide quarterly earning estimates which can be used as indication of performance of stock before actual value is published

iv. Holders: sudden changes in ownership indicate changes in sentiment by sophisticated investor

v. Insiders Purchase/Sale: indicator of future stock price moves from group of people who have access to best possible information about the company

vi. Credit Ratings: credit downgrades resulting in higher funding costs have negative impact on equity prices

1.5.2 Financial Data Structures

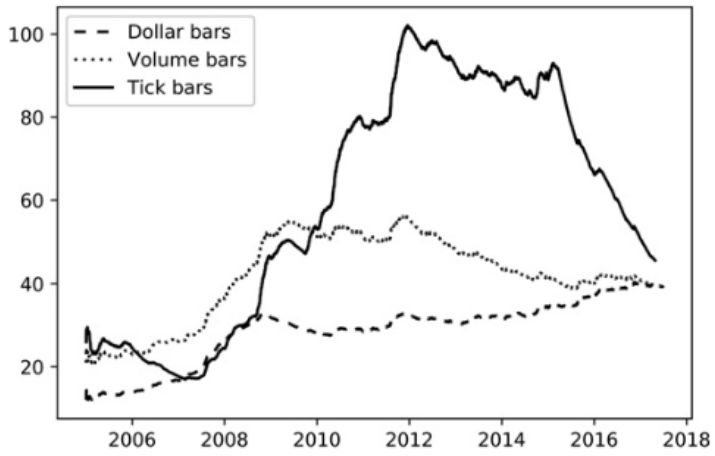


Figure 2: Average daily frequency of tick, volume, and dollar bars

Remark 1.5.10. *Standard BARS*
Method to transform a series of observations arriving at irregular frequency into a homogeneous series derived from regular sampling.

i. Time Bars: obtained by sampling information at fixed intervals. Information collected includes timestamp, volume-weighted average price (VWAP), open price, close price, high price, low price, volume etc.
To be avoided as markets do not process information at constant time interval. Time bars oversample information in low-activity periods and under-sample information in high-activity periods. Time bars exhibit poor statistical properties, i.e., serial correlation, heteroscedasticity, non-normality of returns.

ii. Tick Bars: sample variables extracted each time a pre-defined number of transactions take place. Allows synchronisation of sampling with a proxy of information arrival.
Sampling as a function of trading activity creates returns closer to IID Normal ((Thierry and Helyette, 2000)). When constructing tick bars, to be aware of outliers, as many exchanges carry out auction at open and at close; order book accumulates bids and offers without matching. Order fragmentation introduces some arbitrariness in number of ticks. Matching engine protocols may split one fill into multiple artificial partial fills as a matter of operational convenience.

iii. Volume Bars: samples every time a pre-defined amount of security's units that have been exchanged.
Achieves better statistical properties than sampling tick bars.
Convenient artefact for studying market microstructure theories.

iv. Dollar Bars: samples an observation every time a pre-defined market value is exchanged. Used when the analysis involves significant price fluctuations. Robust against corporate actions such as splits, reverse splits, issuance of new shares, buying back existing shares.
Bar size could be dynamically adjusted as a function of free-floating market cap of a company or outstanding amount of issued debt.

Remark 1.5.11. *Information-Driven Bars*
Method to sample more frequently when new (micro-structural) information arrives to the market.

i. Tick Imbalance Bars: sample bars whenever tick imbalance exceeds expectations. To determine tick index T such that accumulation of signed ticks exceeds a given threshold.
Let $\{(p_t, v_t)\}_{t=1,\dots,T}$ be sequence of ticks where p_t and v_t is the price and volume associated with tick t.
Let tick rule define a sequence $\{b_t\}_{t=1,\dots,T}$ where

$$b_t = \begin{cases} b_{t-1} & \text{if } \Delta p_t = 0 \\ \frac{|\Delta p_t|}{\Delta p_t} & \text{if } \Delta p_t \neq 0 \end{cases}$$

The tick imbalance at time T is defined as

$$\theta_T = \sum_{t=1}^{T} b_t$$

Compute expected value of θ_T at beginning of the bar,

$$E_0[\theta_T] = E_0[T](P[b_t = 1] - P[b_t = -1]) = E_0[T](2P[b_t = 1] - 1)$$

where $E_0[T]$ is expected size of tick bar, $P[b_t = 1]$ and $P[b_t = -1]$ is unconditional probability that a tick is classified as a buy and sell. In practice, $E_0[T]$ and $(2P[b_t = 1]-1)$ may be estimated as an exponentially weighted moving average of T and b_t values from prior bars.
Define the tick imbalance bar (TIB) as a T^* contiguous subset of ticks such that

$$T^* = \arg\min_T\{|\theta_T| \geq E_0[T] \mid 2P[b_t = 1] - 1\}$$

where the size of expected imbalance is implied by $|2P[b_t = 1] - 1|$.
When θ_T is more imbalanced than expected, a low T will satisfy the conditions.
TIBs are produced more frequently under presence of informed trading (asymmetric information that triggers one-side trading). TIBs are buckets of trades containing equal amounts of information.

ii. Volume/Dollar Imbalance Bars: sample bars when volume or dollar imbalances diverge from expectations. First, define imbalance at time T as

$$\theta_T = \sum_{t=1}^{T} b_t v_t$$

where v_t may represent ether number of securities traded (VIB) or dollar amount traded (DIB).
The expected value of θ_T at the beginning of the bar is then computed as

$$\begin{aligned} E_0[\theta_T] &= E_0\left[\sum_{t|b_t=1}^{T} v_t\right] - E_0\left[\sum_{t|b_t=-1}^{T} v_t\right] \\ &= E_0[T](P[b_t = 1]E_0[v_t|b_t = 1] - P[b_t = -1]E_0[v_t|b_t = -1]) \\ &= E_0[T](v^+ - v^-) \end{aligned}$$

where the initial expectation of v_t is decomposed into component contributed by buys and sells. Then

$$E_0[\theta_T] = E_0[T](2v^+ - E_0[v_t])$$

In practice, $E_0[T]$ and $(2v^+ - E_0[v_t])$ may be estimated as exponentially weighted moving average of T and $b_t v_t$ values from prior bars. Next, define VIB or DIB as a T^*-contiguous subset of ticks such that

$$T^* = \arg\min_T\{|\theta_T| \geq E_0[T] \mid 2v^+ - E_0[v_t]\}$$

where the size of expected imbalance is implied by $|2v^+ - E_0[v_t]|$.
When θ_T is more imbalanced then expected, a low T will satisfy the conditions.
VIB and DIB addresses concerns on tick fragmentation and outliers, and also addresses the issues of corporate actions, as the bar size is adjusted dynamically.

iii. Tick Runs Bars: sample bars when the sequence of buys in overall volume diverges from expectations. For the case when large traders sweep order book, use iceberg orders, or slice parent orders into multiple children, all leaving a trace of runs in the $\{b_t\}_{t=1,\ldots,T}$ sequence. Define length of current run as

$$\theta_T = \max\left\{\sum_{t|b_t=1}^{T} b_t - \sum_{t|b_t=-1}^{T} b_t\right\}$$

The expected value of θ_T at beginning of bar is computed as

$$E_0[\theta_T] = E_0[T]\max\{P[b_t = 1], 1 - P[b_t = 1]\}$$

In practice, $E_0[T]$ and $P[b_t = 1]$ may be estimated as exponentially weighted moving average of T and

proportion of buy ticks from prior bars. Next, define TRB as T^*-contiguous subset of ticks such that

$$T^* = \arg\min_T \{\theta_T \geq E_0[T] \max\{P[b_t = 1], 1 - P[b_t = 1]\}\}$$

where the expected count of ticks from runs is implied by $\max\{P[b_t = 1], 1 - P[b_t = 1]\}$.
When θ_T exhibits more runs than expected, a low T will satisfy these conditions.
Instead of measuring length of longest sequence, count number of ticks of each side without offsetting.

iv. Volume/Dollar Runs Bars: sample bars when volume or dollars traded by one side exceed expectation for a bar. First, define volume or dollars associated with a run as

$$\theta_T = \max\left\{\sum_{t|b_t=1}^{T} b_t v_t - \sum_{t|b_t=-1}^{T} b_t v_t\right\}$$

where v_t may either represent volume (VRB) or dollar amount exchanged (DRB). The expected value of θ_T at beginning of the bar is then

$$E_0[\theta_T] = E_0[T] \max\{P[b_t = 1]E_0[v_t|b_t = 1], (1 - P[b_t = 1])E_0[v_t|b_t = -1]\}$$

In practice, $E_0[T], P[b_t = 1], E_0[v_t|b_t = 1], E_0[v_t|b_t = -1]$ may be estimated as exponentially weighted moving average of T, proportion of buy ticks, buy volumes, and sell volumes from prior bars. Next, define a volume runs bar (VR) as T^*-contiguous subset of ticks such that

$$T^* = \arg\min_T \{\theta_T \geq E_0[T] \max\{P[b_t = 1]E_0[v_t|b_t = 1], (1 - P[b_t = 1])E_0[v_t|b_t = -1]\}\}$$

expected volume from runs is implied by $\max\{P[b_t = 1]E_0[v_t|b_t = 1], (1 - P[b_t = 1])E_0[v_t|b_t = -1]\}\}$.
When θ_T exhibits more runs than expected, volume from runs is greater than expected, a low T will satisfy these conditions.

Definition 1.5.12. *Multi-Product Series: ETF Trick*
To model a basket of securities as if it was a single cash product. To transform any complex multi-product dataset into a single dataset that resembles a total return ETF.

Method 1.5.13. *ETF Trick*
Produce a time series that reflects the value of \$1 invested. Changes in the series will reflect changes in PnL, series will be strictly positive, and implementation shortfall will be taken into account. The bars contain:

i. Raw open price of instrument $i = 1, \ldots, I$ at bar $t = 1, \ldots, T$: $o_{i,t}$
ii. Raw close price of instrument $i = 1, \ldots, I$ at bar $t = 1, \ldots, T$: $p_{i,t}$
iii. USD value of one point of instrument $i = 1, \ldots, I$ at bar $t = 1, \ldots, T$: $\varphi_{i,t}$. This includes forex rate.
iv. Volume of instrument $i = 1, \ldots, I$ at bar $t = 1, \ldots, T$: $v_{i,t}$
v. Carry, dividend, or coupon paid by instrument i at bar t: $d_{i,t}$. Variable can also be used to charge margin costs or costs of funding.

All instruments $i = 1, \ldots, I$ were tradable at bar $t = 1, \ldots, T$. Even if some instruments were not tradable over entirety of time interval $[t - 1, t]$, at least they were tradable at times $t - 1$ and t.
For basket of securities with allocation vector ω_t rebalanced (or rolled) on bars $B \subseteq \{1, \ldots, T\}$, the \$1 investment value $\{K_t\}$ is derived as

$$h_{i,t} = \begin{cases} \frac{\omega_{i,t} K_t}{o_{i,i+1}\varphi_{i,t} \sum_{i=1}^{I} |\omega_{i,t}|} & \text{if } t \in B \\ h_{i,t-1} & \text{otherwise} \end{cases}$$

$$\delta_{i,t} = \begin{cases} p_{i,t} - o_{i,t} & \text{if } (t-1) \in B \\ \Delta p_{i,t} & \text{otherwise} \end{cases}$$

$$K_t = K_{t-1} + \sum_{i=1}^{I} h_{i,t-1}\varphi_{i,t}(\delta_{i,t} + d_{i,t})$$

where $K_0 = 1$ is the initial AUM. Variable $h_{i,t}$ is the holdings of instrument i at time t, $\delta_{i,t}$ is change of market value between $t - 1$ and t for instrument i. Note profits or losses are being reinvested whenever $t \in B$, hence preventing negative prices. Dividends $d_{i,t}$ are already embedded in K_t.

The purpose of $\omega_{i,t}\left(\sum_{i=1}^{I}|\omega_{i,t}|\right)^{-1}$ is to de-lever the allocations.
Let τ_i be transaction cost associated with trading \$1 of the instrument. Three additional variables that the strategy needs to know for every observed bar t are:

i. Rebalance Costs: variable cost $\{c_t\}$ associated with allocation rebalance is

$$c_t = \sum_{i=1}^{I}(|h_{i,t-1}|\, p_{i,t} + |h_{i,t}|\, o_{i,t+1})\varphi_{i,t}\tau_i \quad \forall t \in B$$

Note c_t is not embedded in K_t, as shorting will generate fictitious proceeds when allocation is rebalanced. In code, $\{c_t\}$ is treated as a (negative) dividend.

ii. Bid-Ask Spread: the cost $\{\tilde{c}_t\}$ of buying or selling one unit of this ETF,

$$ildec_t = \sum_{i=1}^{I}|h_{i,t-1}|\, p_{i,t}\varphi_{i,t}\tau_i$$

When a unit is bought or sold, strategy must charge this cost $\tilde{c}_t$.

iii. Volume: volume traded $\{v_t\}$ is determined by least active member in the basket. Let $v_{i,t}$ be volume traded by instrument i over bar t. The number of tradable basket units is

$$v_t = \min_i\left\{\frac{v_{i,t}}{|h_{i,t-1}|}\right\}$$

Transaction costs functions may not be linear, and can be simulated by the strategy.

Method 1.5.14. *ETF Trick: Computation of Allocation Vector with PCA*
Consider an IID multivariate Gaussian process with means vector μ of size $N \times 1$, and covariance matrix V of size $N \times N$. First, perform spectral decomposition $VW = W\Lambda$, where columns in W are reordered so that elements of Λ diagonal are sorted in descending order. Second, given allocations vector ω, portfolio risk is

$$\sigma^2 = \omega' V\omega = \omega' W\Lambda W'\omega = \beta'\Lambda\beta = (\Lambda^{1/2}\beta)'(\Lambda^{1/2}\beta)'$$

where β is projection of ω on orthogonal basis. Third, Λ is a diagonal matrix, thus

$$\sigma^2 = \sum_{n=1}^{N}\beta_n^2\Lambda_{n,n}$$

The risk attributed to the nth component is

$$R_n = \beta_n^2\Lambda_{n,n}\sigma^{-2} = [W'n]_n^2\Lambda_{n,n}\sigma^{-2}$$

with $R'1_N = 1$, and 1_N is a vector of N ones.
Note $\{R_n\}_{n=1,\ldots,N}$ is distribution of risks across orthogonal components.
Next, compute vector ω which delivers user-defined risk distribution R. Note from earlier,

$$\beta = \left\{\sigma\sqrt{\frac{R_n}{\Lambda_{n,n}}}\right\}_{n=1,\ldots,N}$$

which represents allocation in new (orthogonal basis).
The allocation in old basis is $\omega = W\beta$. Rescaling ω re-scales σ, hence keeping risk distribution constant.

Method 1.5.15. *ETF Trick: Single Futures Roll*
To work with non-negative rolled series, derive price series of \$1 investment as follows:

i. Compute time series of rolled futures prices

ii. Compute return r as rolled price change divided by previous roll price

iii. Form a price series using these returns

These methods allow us to produce a continuous, homogeneous, and structured dataset from collection of unstructured financial data. Note however, that several ML algorithms do not scale well with sample size. ML algorithms achieve higher accuracy when they attempt to learn from relevant examples.

Method 1.5.16. *Sampling for Reduction*
To reduce the amount of data used to fit ML algorithm, downsampling could be used.

i. Sequential sampling at constant step size (linspace sampling)

ii. Sampling randomly using uniform distribution (uniform sampling)

Note both samples do not necessarily contain subset of most relevant observations.

Method 1.5.17. *Event-Based Sampling: CUMSUM Filter*
Bets are often placed after some event takes place, hence to let ML algorithm learn whether there is an accurate prediction function under these circumstances, CUSUM filter could be used.
This is a quality-control method, to detect shift in mean value of measured quantity away from a target value. Let $\{y_t\}_{t=1,\dots,T}$ be IID observations arising from a locally stationary process. The cumulative sums are

$$S_t = \max\{0, S_{t-1} + y_t - E_{t-1}[y_t]\}, \quad S_0 = 0$$

An action will be recommended at the first t satisfying $S_t \geq h$ for some threshold h (filter size).
Note $S_t = 0$ whenever $y_t = E_{t-1}[y_t] - S_{t-1}$, The zero floor means some downward deviations will be skipped.
The filter is set up to identify a sequence of upside divergences from any reset level zero.
The threshold is activated when

$$S_t \geq h \Leftrightarrow \exists \tau \in [1, t] \mid \sum_{i=\tau}^{t} (y_i - E_{i-1}[y_t]) \geq h$$

This concept of run-ups can be extended to include run-downs, giving symmetric CUSUM filter.

$$\begin{aligned} S_t^+ &= \max\{0, S_{t-1}^+ + y_t - E_{t-1}[y_t]\}, \quad S_0^+ = 0 \\ S_t^- &= \min\{0, S_{t-1}^- + y_t - E_{t-1}[y_t]\}, \quad S_0^- = 0 \\ S_t &= \max\{S_t^+, -S_t^-\} \end{aligned}$$

1.5.3 Data Labelling Techniques

Method 1.5.18. *Labelling with Fixed-Time Horizon Method*
Given features matrix X with I rows, $\{X_i\}_{i=1,\dots,I}$ drawn from some bards with index $t = 1, \dots, T$, where $I \leq T$, let an observation X_i be assigned a label $y_i \in \{-1, 0, 1\}$,

$$y_i = \begin{cases} -1 & \text{if } r_{t_{i,0}, t_{i,0}+h} < -\tau \\ 0 & \text{if } \left| r_{t_{i,0}, t_{i,0}+h} \right| \leq \tau \\ 1 & \text{if } r_{t_{i,0}, t_{i,0}+h} > \tau \end{cases}$$

$$r_{t_{i,0}, t_{i,0}+h} = \frac{p_{t_{i,0}+h}}{p_{t_{i,0}}} - 1$$

where τ is a pre-defined constant threshold, $t_{i,0}$ is index of bar immediately after X_i takes place, $t_{i,0} + h$ is index of h-th bar after $t_{i,0}$, and $r_{t_{i,0}, t_{i,0}+h}$ is price return over bar horizon h.

Remark 1.5.19. *Limitations of Fixed-Time Horizon Method*

i. Time bars do not exhibit good statistical properties (as seen earlier)

ii. The same threshold τ is applied regardless of observed volatility.
Compute daily volatility at intraday estimation points, applying span of n days t an exponentially weighted moving standard deviation.

Method 1.5.20. *Labelling with Triple-Barrier Method*
Labels an observation according to first barrier touched out of three barriers.

i. Set two horizontal barriers and one vertical barrier. Horizontal barriers are defined by profit-taking and stop-loss limits, which are a dynamic function of estimated volatility (realised or implied). Third barrier is the number of bars elapsed since the position was taken (expiration limit).

ii. If upper barrier is touched first, label observation as 1. If lower barrier is touched first, label observation as -1. If vertical barrier is touched first, either label by sign of the return or with 0.

Note that the method is path-dependent. To label an observation, need to account for entire path spanning $[t_{i,0}, t_{i,0} + h]$ where h defines the vertical barrier (expiration limit). Let $t_{i,1}$ be the time of first barrier touch with return as $r_{t_{i,0}, t_{i,1}}$. The horizontal barriers may not be symmetric.

Remark 1.5.21. *Triple-Barrier Method Configurations*
Denote a barrier configuration by triplet $[pt, sl, t1]$ which are the upper barrier, lower barrier, physical barrier. Set value as 0 if barrier is inactive, and 1 if barrier is active.
The three useful configurations are:

i. $[1, 1, 1]$: to realise profit, but have set a maximum tolerance for losses and a holding period.
ii. $[0, 1, 1]$: to exit after a number of bars, unless stopped-out.
iii. $[1, 1, 0]$: take profit as long as not stopped-out.

The three less realistic configurations are:

i. $[0, 0, 1]$: equivalent to fixed-time horizon method.
ii. $[1, 0, 1]$: position held until a profit is made or maximum holding period is exceeded, without regard for immediate unrealised losses
iii. $[1, 0, 0]$: position is held until a profit is made. Could lock in loose position for years.

The two illogical configurations are:

i. $[0, 1, 0]$: aimless. Hold position until stopped-out.
ii. $[0, 0, 0]$: no barriers. Position locked forever, no label generated.

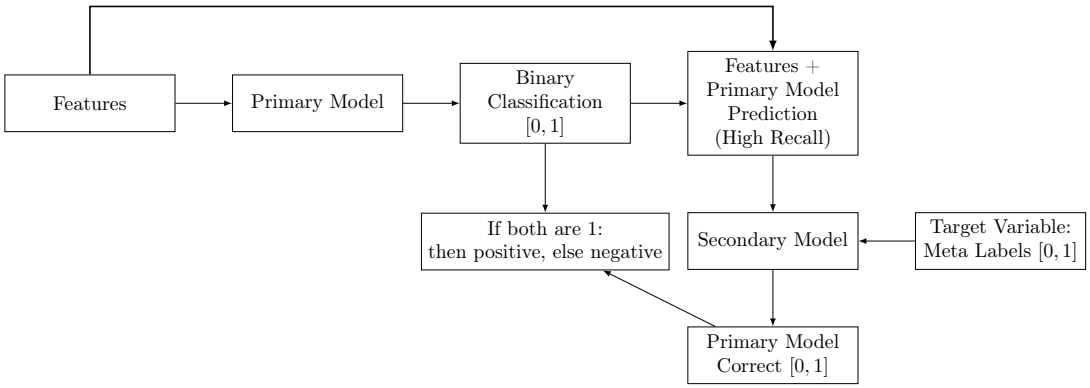


Figure 3: Meta-Labelling Process

Method 1.5.22. *Meta-Labelling*
The technique is particularly helpful to achieve higher F1-scores.
First, build a model that achieves high recall, even if precision is not particularly high. Second, correct for low precision by applying meta-labelling to positives predicted by primary model.
Meta-labelling will filter out false positives, where majority of positives have been identified by primary model. The second model's purpose is to determine if the positive from primary model is true or false.

i. Train a primary model (binary classification)
ii. A threshold level is determined at which the primary model has a high recall, ROC curves could be used to help determine a good level.
iii. Typical features of second model are as follows:
    i. Primary model features concatenated with predictions from first model.
    ii. Market state
    iii. Features indicative of false positives
    iv. Distribution related
    v. Recent model performance

    Meta Labels are used as target variable in second model. Fit the second model
iv. Prediction from the secondary model is combined with the prediction from the primary model and only where both are true, is your final prediction true.

Remark 1.5.23. *Limitations of Meta-Labelling*

i. If model has overfit the data, meta-labelling will not add much value

ii. If every trade is not treated as an independent observation, the meta-model is forced to determine day-to-day exposures, which is the wrong way to apply the technique

iii. Technique trades recall for precision. Require a large number of trades to train on, while being happy with reduction in trade frequency

1.5.4 Data Sample Weights

Note that most of ML literature is based on IID assumption, and ML applications usually fail in finance as these assumptions are unrealistic in the case of financial time series.

Remark 1.5.24. *Overlapping Outcomes*
Let label y_i be assigned to an observed feature X_i, where $y_i = f([t_{i,0}, t_{i,1}])$ is a function over the interval. When $t_{i,1} > t_{j,0}$ and $i < j$, then y_j will depend on common return $r_{t_{j,0},\min\{t_{i,1},t_{j,1}\}}$ (over interval $[t_{j,0}, \min\{t_{i,1}, t_{j,1}\}]$). The series of labels $\{y_i\}_{i-1,\ldots,J}$ are not IID whenever there is overlap between any two consecutive outcomes, i.e., $\exists i \mid t_{i,1} > t_{i+1,0}$. If this is resolved by restricting bet horizon to $t_{i,1} \leq t_{i+1,0}$, there is no overlap, but this will lead to coarse models where features sampling frequency is limited by horizon used to determine outcome. To investigate outcomes that lasted a different duration, samples have to be resampled with different frequency. In addition, if path-dependent labelling technique is to be applied, the sampling frequency will be subordinated to first barrier's touch. Hence, to use $t_{i,1} > t_{i+1,0}$, leading to overlapping outcomes.

Method 1.5.25. *Estimating Uniqueness of Label*
Let two labels y_i and y_j be concurrent at time t, both a function of at least one common return $r_{t-1,t} = \frac{p_t}{p_{t-1}} = 1$. To compute the number of labels that are a function of given return $r_{t-1,t}$:

i. For each $t = 1, \ldots, T$, form a binary array $\{1_{t,i}\}_{i=1,\ldots,I}$ where $1_{t,i} \in \{0, 1\}$.
Variable $1_{t,i} = 1$ if and only if $[t_{i,0}, t_{i,1}]$ overlaps with $[t-1, t]$ and $1_{t,i} = 0$ otherwise.

ii. Compute the number of labels concurrent at t, $c_t = \sum_{i=1}^{I} 1_{t,i}$

Method 1.5.26. *Average Uniqueness of Label*
To estimate label's uniqueness (non-overlap) across its lifespan.

i. Uniqueness of label i at time t is $u_{t,i} = 1_{t,i} c_t^{-1}$.

ii. Average uniqueness of label i is average $u_{t,i}$ over label's lifespan, $\overline{u}_i = (\sum_{t=1}^{T} u_{t,i})(\sum_{t=1}^{T} 1_{t,i})^{-1}$.

Note that $\{\overline{u}_i\}_{i=1,\ldots,I}$ are not used for forecasting the label, hence there is no information leakage.

Remark 1.5.27. *IID and Oversampling*
Probability of not selecting item i after I draws with replacement on set of I items is $(1 - I^{-1})^I$. As $I \to \infty$, note that $(1 - I^{-1})^I \to e^{-1}$. Number of unique observations drawn to be expected is $(1 - e^{-1}) \approx \frac{2}{3}$.
If maximum number of overlapping outcomes is $K \leq I$, probability of not selecting a particular item i after I draws with replacement on set of I items is $(1 - K^{-1})^I$. As sample size increase, probability can be approximated as $(1 - I^{-1})^{I\frac{K}{I}} \approx e^{-\frac{K}{I}}$. Implication is that incorrectly assuming IID draws lead to oversampling.

Method 1.5.28. *Sampling with Bootstrap, Redundancy*
Sampling with bootstrapping on observations where $I^{-1} \sum_{i=1}^{I} \overline{u}_i \ll 1$, in-bag observations will increasingly be redundant to each other, and very similar to out-of-bag observations. Two solutions may be:

i. Drop overlapping outcomes before performing bootstrap.
As overlaps are not perfect, dropping an observation due to overlap will lead to extreme loss in information.

ii. Utilise the average uniqueness $I^{-1} \sum_{i=1}^{I} \overline{u}_i$ to reduce undue influence of outcomes that contain redundant information. Ensure in-bag observations are not sampled at frequency much higher than uniqueness.

Method 1.5.29. *Sequential Bootstrap*
Draws made according to changing probability that controls for redundancy.

i. Observation X_i is drawn from uniform distribution, $i \sim U[1, I]$.
Probability of drawing any value i is $\delta_i^{(1)} = I^{-1}$.

ii. Second draw, to reduce probability of drawing observation X_j with highly overlapping outcome.
Let φ be sequence of draws (may include repetitions), where $\{\varphi^{(1)}\} = \{i\}$.
Uniqueness of j at time t is $u_{t,j}^{(2)} = 1_{t,j}(1 + \sum_{k \in \varphi^{(1)}} 1_{t,k})^{-1}$, which is the uniqueness from adding alternative j's to existing sequence of draws $\varphi^{(1)}$.

Average uniqueness of j is average $u_{t,j}^{(2)}$ over j's lifespan, $\overline{u}_j^{(2)} = (\sum_{t=1}^{T} u_{t,j})(\sum_{t=1}^{T} 1_{t,j})^{-1}$.
A second draw can be made based on updated probabilities $\{\delta_j^{(2)}\}_{j=1,\dots,I}$:

$$\delta_j^{(2)} = \overline{u}_j^{(2)} \left(\sum_{k=1}^{I} \overline{u}_j^{(2)} \right)^{-1}$$

where $\sum_{j=1}^{I} \delta_j^{(2)} = 1$. Do a second draw, update $\varphi^{(2)}$, and re-evaluate $\{\delta_j^{(3)}\}_{j=1,\dots,I}$.

iii. Process is repeated until I draws have taken place.

Process draws samples much close to IID, verified by increase in $I^{-1} \sum_{i=1}^{I} \overline{u}_i$.

Method 1.5.30. *Weighting Observations by Uniqueness and Absolute Return*
Let labels be a function for return sign ($\{-1, 1\}$ for standard label, $\{0, 1\}$ for meta-label). The sample weights can be defined in terms of sum of attributed returns over event's life-span, $[t_{i,0}, t_{i,1}]$,

$$\tilde{w}_i = \left| \sum_{t=t_{i,0}}^{t_{i,1}} \frac{r_{t-1,t}}{c_t} \right|, \quad w_t = \tilde{w}_i \left(\sum_{j=1}^{I} \tilde{w}_j \right)^{-1}$$

where $\sum_{i=1}^{I} w_i = I$. The method weigh an observation as a function of absolute log returns that can be attributed uniquely to it. Lower returns should be assigned higher weights.

Method 1.5.31. *Time Decay Weighting*
To let sample weights decay as new observations arrive.
Let $d[x] \geq 0 \; \forall x \in [0, \sum_{i=0}^{I} \overline{u}_i]$ be time-decay factors multiplying sample weights from earlier.
The final weight has no decay, $d[\sum_{i=1}^{I} \overline{u}_i] = 1$, and all other weights will adjust relative to that.
Let $c \in (-1, 1]$ be user-defined parameters that determines decay function as follows:

i. If $c \in [0, 1]$, then $d[1] = c$ with linear decay

ii. If $c \in (-1, 0)$, then $d[-c \sum_{i=1}^{I} \overline{u}_i] = 0$, with linear decay between $[-c \sum_{i=1}^{I} \overline{u}_i, \sum_{i=1}^{I} \overline{u}_i]$, and $d[x] \; \forall x \leq -c \sum_{i=1}^{I} \overline{u}_i$.

If given linear piecewise function $d = \max\{0, a + bx\}$, requirements are met by following boundary conditions:

i. $d = a + b \sum_{i=1}^{I} \overline{u}_i = 1 \Rightarrow a = 1 - b \sum_{i=1}^{I} \overline{u}_i$

ii. Contingent on c:

1. $d = a + b0 = c \Rightarrow b = (1 - c)(\sum_{i=1}^{I} \overline{u}_i)^{-1} \quad \forall c \in [0, 1]$
2. $d = a - bc \sum_{i=1}^{I} \overline{u}_i = 0 \Rightarrow b = [(c + 1) \sum_{i=1}^{I} \overline{u}_i]^{-1} \; \forall c \in (-1, 0)$

In the implementation, decay takes place according to cumulative uniqueness. Note that

i. $c = 1$ means there is no time decay

ii. $0 < c < 1$ means weights decay linearly over time, but every observation still receives strictly positive weight, regardless of age

iii. $c = 0$ means weights converge linearly to zero over time

iv. $c < 0$ means oldest portion cT of observations receive zero weight (erased from memory)

Method 1.5.32. *Class Weighting*
Weights for underrepresented labels. Critical in classification problems where the most important classes have rare occurrences. To assign higher weights to samples associated with those rare labels.

1.5.5 Fractionally Differentiated Features

Standard stationarity transformations (i.e. integer differentiation) reduce signal by removing memory. Although stationarity is necessary for inferential purposes, it is rarely the case that we want all memory to be erased. Fractionally differentiated processes exhibit long-term persistence and anti-persistence, hence enhancing the forecasting power compared to standard ARIMA approach.

Definition 1.5.33. *BackShift Operator*
Let B be the backshift operator applied to a matrix of real-valued features $\{X_t\}$, where $B^k X_t = X_{t-k}$ for any integer $k \geq 0$. By binomial expansion, we then have

$$(1-B)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-B)^k = \sum_{k=0}^{\infty} \prod_{i=0}^{k-1} (d-i) \frac{(-B)^k}{k!} = \sum_{k=0}^{\infty} (-B)^k \prod_{i=0}^{k-1} \frac{d-i}{k-i}$$
$$= 1 - dB + \frac{d(d-1)}{2!} B^2 - \frac{d(d-1)(d-2)}{3!} B^3 + \cdots$$

Remark 1.5.34. *Properties of Fractionally Differentiated Features*
Let d be a real (non-integer) positive number. The arithmetic series consists of dot product

$$\tilde{X}_t = \sum_{k=0}^{\infty} \omega_k X_{t-k}$$
$$\omega = \left\{1, -d, \frac{d(d-1)}{2!}, -\frac{d(d-1)(d-2)}{3!}, \ldots, (-1)^k \prod_{i=0}^{k-1} \frac{d-1}{k!}, \ldots\right\}$$
$$X = \{X_t, X_{t-1}, \ldots, X_{t-k}, \ldots\}$$

where ω are the weights, X are the values. Properties of these features are:

i. Long memory: if d is a positive integer number, then

$$\prod_{i=0}^{k-1} \frac{d-i}{k!} = 0 \quad \forall > d$$

and memory beyond that point is cancelled.

ii. Iterative weight generation: given sequence of weights ω, for $k = 0, \ldots, \infty$, the weights are

$$\omega_k = -\omega_{k-1} \frac{d-k+1}{k}, \quad \omega_0 = 1$$

iii. Convergence: For $k > d$, if $\omega_{k-1} \neq 0$, then

$$\left|\frac{\omega_k}{\omega_{k-1}}\right| = \left|\frac{d-k+1}{k}\right| < 1$$

and $\omega_k = 0$ otherwise. Hence weights converge asymptotically to zero.
For positive d and $k < d+1$, then $\frac{d-k+1}{k} \geq 0$, which makes initial weights alternate in sign.
For non-integer d, once $k \geq d+1$, ω_k will be negative if int$[d]$ is even, and positive otherwise.
In summary, $\lim_{k\to\infty} = 0^-$ when int$[d]$ is even, and $\lim_{k\to\infty} = 0^+$ when int$[d]$ is odd.
In special case $d \in (0,1)$, that $-1 < \omega_k < 0 \; \forall k > 0$. Alternate weight signs makes $\{\tilde{X}_t\}_{t=1,\ldots,T}$ stationary, as memory wanes or is offset over the long run.

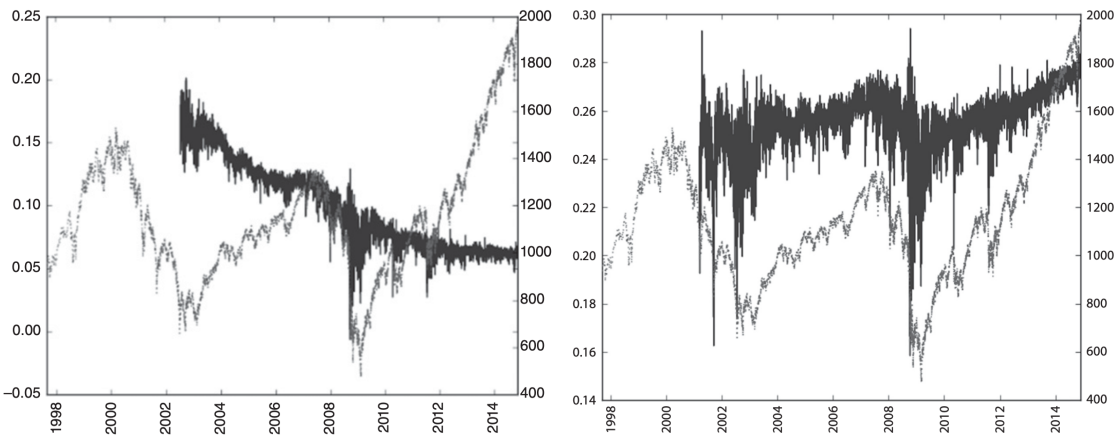


Figure 4: Fractional differentiation controlling for weight loss with expanding and fixed-width window

Method 1.5.35. *Expanding Window*
Given time series T with real observations $\{X_t\}_{t=1,\dots,T}$, for each l, the relative weight loss is defined as

$$\lambda_l = \sum_{j=T-l}^{T} |\omega_j| \Big/ \sum_{i=0}^{T-1} |\omega_i|$$

Given tolerance level $\tau \in [0,1]$, determine value l^* such that $\lambda_{l^*} \leq \tau$ and $\lambda_{l^*+1} > \tau$. This value l^* corresponds to the first results $\{\tilde{X}_t\}_{t=1,\dots,l^*}$, where weight-loss is beyond acceptable threshold $\lambda_t > \tau$.
From Remark 1.5.34, it is clear λ_{l^*} depends on convergence speed of $\{\omega_k\}$, which in turn depends on $d \in [0,1]$. For $d = 1, \omega_k = 0\ \forall k > 1$, and $\lambda_l = 0\ \forall l > 1$, hence it suffices to drop $\tilde{X}_1$.
As $d \to 0^+$, l^* increases, and larger portion of initial $\{\tilde{X}_t\}_{t=1,\dots,l^*}$ needs to be dropped to keep the weight loss $\lambda_{l^*} < \tau$. Note that there will be negative drift caused by negative weights added to initial observations as window is expanded. By controlling for weight loss, negative drift is still substantial as $\{\tilde{X}_t\}_{t=l^*+1,\dots,T}$ are computed on an expanding window.

Method 1.5.36. *Fixed-Width Window*
Drop weights after their modulus $|\omega_k|$ decreases below a given threshold τ. This is equivalent to finding the first l^* such that $|\omega_{l^*}| \geq \tau$ and $|\omega_{l^*+1}| \leq \tau$, setting a new variable $\tilde{\omega}_k$:

$$\tilde{\omega_k} = \begin{cases} \omega_k & \text{if } k \leq l^* \\ 0 & \text{if } k > l^* \end{cases}, \qquad \tilde{X}_t = \sum_{k=0}^{l^*} \tilde{\omega}_k X_{t-k} \quad \text{for } t = T - l^* + 1, \dots, T$$

Note that the same vector of weights is used across all estimates of $\{\tilde{X}_t\}_{t=l^*,\dots,T}$, hence avoiding negative drift caused by expanding window's added weights.
Distribution has skewness and excess kurtosis from memory, but it is stationary.

2 Macro and Global Market Analysis

Global macro is the strategy of identifying economic regime shifts early and expressing them through the most liquid instruments. The primary objective is surviving and profiting from crises, rather than maximising returns in stable periods ((Gliner, 2014)).

2.1 Regime and World State

2.1.1 Survey of Global Macro Landscape

Definition 2.1.1. *Global Macro*
Global macro is a top-down investment strategy that trades economic regimes, not securities.
The core idea is simple but powerful: macro forces dominate asset prices when they shift, and those forces operate across countries, asset classes, and policy domains. Global macro traders

i. Form views on growth, inflation, monetary policy, and geopolitics
ii. Express those views via equities, rates, FX, and commodities
iii. Are unconstrained by geography, asset class, or direction (long/short)

Global macro is often considered the most flexible and opportunistic hedge fund strategy, due to the scope of traded products and the number of markets it covers. Its aim is to preserve capital, using stringent risk management to limit drawdowns.

Remark 2.1.2. *Existence of Global Macro as a Strategy*
Macro exists because economic regimes are unstable, but markets price them slowly and asymmetrically.
Three structural reasons:

i. Policy is discrete, markets are continuous: Central banks and governments move in jumps (rate decisions, QE, capital controls). Markets must reprice entire distributions after the fact.
ii. Capital is sticky: Large allocators (pensions, insurers, sovereign funds) cannot reposition quickly. Macro traders exploit this inertia.
iii. Cross-asset linkages are under-arbitraged: Many investors specialise in a single asset class. Macro traders operate between silos.

Hence macro is a strategy of structural inefficiency, not informational advantage.

Remark 2.1.3. *Types of Global Macro Strategies*
The four broad approaches are:

i. Discretionary Macro
Relies on trader experience, intelligence, and knowledge to take subjective bets on global markets. Requires serious organization and data processing skills. Uses top-down analysis of risks and opportunities across industries, sectors, countries, and the macroeconomic situation.
Discretionary traders can execute *directional trades* (betting on asset direction) or *relative value trades* (pairing assets to capture value differentials). Strong in crises and regime breaks where human judgment outperforms models.
ii. Systematic Macro
Model-driven, employing quantitative factors to produce trading positions that remove human emotion. Prides itself on stringent process, strong back-tests, and ability to operate solely on quantitative analysis. Over long periods, systematic funds produce more consistent returns than discretionary strategies; however, in periods of high volatility, they tend to underperform (e.g., 2008).
The ability to trade multiple liquid asset classes allows these funds to scale to very large AUM.
iii. High-Frequency Macro
Uses sophisticated technology to trade very short-term (millisecond to hours) dislocations around macro events. Processing speed is paramount. Limited scalability compared to discretionary and systematic.
iv. CTAs (Commodity Trading Advisors)
Primarily futures-based, momentum-driven trend-following strategies. Position sizing methodology originated with the Turtle Traders. Perform well over longer periods but subject to large drawdowns.

It is wise to allocate to both discretionary and systematic macro in a balanced manner. Discretionary is negatively correlated during stress, while systematic provides predictable allocation with back-tested drawdowns.

Remark 2.1.4. *Trade Types in Macro*
Discretionary macro traders express views through two primary trade structures:

i. Directional Trades: Betting on an asset moving in a particular direction. Example: going long copper on a bullish commodities view.

ii. Relative Value Trades: Pairing or grouping assets to capture the relative value differential between them. Example: during European crisis, short Italian 5-year BTPs vs. long German Bobls, betting Italy will see yields rise relative to Germany.

iii. Thematic Trades: Longer-term positions based on structural macro views.
Examples: "fiat money debasement", hence long gold; "Japan debt unsustainability", hence short JGBs.
Thematic trades should be sized smaller than short-term trades because they can take years to play out and may incur negative carry.

Remark 2.1.5. *Return Profile Across Cycles*
Global macro's defining feature is return asymmetry across market regimes. Empirical characteristics:

i. Lower average volatility than equities

ii. Comparable or higher long-run returns

iii. Low or negative correlation to equities during crises

iv. Higher Sharpe ratio than most hedge fund strategies

Macro tends to underperform in bull markets (when staying long works), but outperform during dislocations, recessions, and policy shifts. This asymmetry exists because macro traders can go short, can hold cash, can trade volatility, rates, and FX directly (flexibility that equity-focused strategies lack).

Remark 2.1.6. *Behaviour in Economic Crisis*
In stress regimes:

i. Equity managers are trapped long (mandates prevent shorting or holding cash)

ii. Credit liquidity vanishes (bids disappear, spreads gap wider)

iii. Correlations converge toward 1 (diversification fails)

Macro funds, by contrast, can:

i. Reduce gross exposure rapidly

ii. Short risk assets directly

iii. Express views via rates, FX, or volatility instead of equities

iv. Profit from the dislocation rather than merely survive it

Remark 2.1.7. *Global Macro Correlation Properties*
Global macro's real value is conditional correlation:

i. In normal times: low correlation to equities

ii. In crises: negative correlation to equities

This makes macro an effective portfolio diversifier and tail-risk hedge. Combined with high liquidity (trading the most liquid markets globally), macro is the most popular hedge fund allocation by pension funds.

Remark 2.1.8. *Liquidity Advantage*
One of most liquid hedge fund strategies because it trades the most liquid underlying markets:

i. G10 FX and major EM currencies

ii. Government bond futures (Treasuries, Bunds, JGBs, Gilts)

iii. Equity index futures (S&P 500, Euro Stoxx, Nikkei, etc.)

iv. Major commodity futures (crude oil, gold, copper, etc.)

This liquidity allows macro funds to:

i. Scale to very large AUM without market impact

ii. Enter and exit positions quickly during stress

iii. Offer better redemption terms than illiquid strategies

2.1.2 Role of Central Banks in Global Macro

Remark 2.1.9. *Monetary Policy Regimes*
Central banks define the macro regime more than any other actor. Markets do not trade rates, they trade policy intent under constraints. Regimes fall into following types:

i. Conventional Policy Regime: Policy rate is the main tool. Yield curve responds normally. Transmission is through rates to credit, to growth, then to inflation.
ii. Unconventional Policy Regime: Policy rate loses effectiveness (zero lower bound). Balance sheet becomes the primary tool. Transmission shifts to asset prices, expectations, and FX.
iii. Crisis/Emergency Regime: Stability is more important than inflation targeting. Rules are suspended. Liquidity provision dominates all other objectives.

Remark 2.1.10. *Objectives of Central Banks*
Central banks typically have mandates that include:

i. Price Stability: Most central banks target inflation around 2%
ii. Full Employment: Explicit in Fed's dual mandate; implicit elsewhere
iii. Financial System Stability: Lender of last resort function; systemic risk monitoring
iv. Currency Stability: Explicit for some (SNB, PBOC); implicit for others

Under stress, these objectives may conflict. Central banks must choose which objective to sacrifice. Understanding these trade-offs is critical for macro trading.

Remark 2.1.11. *Conventional Central Bank Tools*

i. Policy Rate: Set by changing overnight rate at which banks lend and borrow to meet reserve requirements. The primary tool in normal times.
ii. Reserve Ratio: Raising reserve ratio decreases money multiplier and contracts money supply.
iii. Open Market Operations (OMO): Buying and selling government bonds in local currency. Repos are short-term agreements where the central bank agrees to repurchase securities within a specified time.
iv. Currency Intervention:
    1. Sterilised intervention: Trading own currency via FX reserves without changing monetary base
    2. Unsterilised intervention: Allows for changes in monetary base

Remark 2.1.12. *Taylor's Rule*
Maps macroeconomic deviations to policy stance:

$$i_t = r^* + \pi_t + \alpha(\pi_t - \pi^*) + \beta(y_t - y_t^*)$$

where r^* is the neutral real rate (unobservable, typically estimated 2–3%), π_t is current inflation, π^* is the inflation target, $(y_t - y_t^*)$ is the output gap, and α, β are response coefficients (typically 0.5).
The rule states that the nominal policy rate must rise more than one-to-one with inflation (Taylor Principle); otherwise real rates fall, inflation becomes self-reinforcing, and expectations de-anchor.
Central banks treat Taylor's rule as a benchmark, communication tool, and political shield.

Remark 2.1.13. *The Impossible Trinity*
A country cannot simultaneously maintain:

i. Fixed exchange rate
ii. Free capital flows
iii. Independent monetary policy

This constraint explains currency crises (ERM 1992, Asian crisis 1997), forced rate hikes or cuts, FX pegs breaking, and why EM central banks behave "irrationally" under stress. Countries must choose two of three.

Remark 2.1.14. *Central Bank Balance Sheet*

i. Monetary Base: Sum of currency in circulation and reserves held at the central bank. Can only be controlled via open market operations.

ii. Money Supply: Changes in money supply are leading predictors of GDP growth and inflation. The Quantity Theory of Money:

$$M \times V = P \times Y$$

where M is money supply, V is velocity of money (turnover rate), P is price level, Y is real output.

Post-2008, central bank balance sheets expanded dramatically. The Fed releases weekly H.4.1 report showing sources and uses of reserves.

Remark 2.1.15. *Types of Money Supply*

i. $M0$: Physical currency in circulation (notes and coins)
ii. $M1$: $M0$ + demand deposits, checking accounts, traveler's checks
iii. $M2$: $M1$ + savings deposits, money market accounts, small time deposits, retail money market funds
iv. $M3$: $M2$ + large time deposits, institutional money market funds, repos, eurodollars

Central banks primarily control $M0$; broader measures depend on banking system credit creation.

Remark 2.1.16. *Zero Lower Bound and the Liquidity Trap*
When central bank rate is at or near zero:

i. Rate cuts lose marginal effectiveness
ii. Expectations dominate outcomes
iii. Forward guidance becomes critical
iv. Fiscal policy starts to matter more

Yield curves flatten naturally; carry trades distort; volatility is suppressed artificially.
Some central banks have gone to negative rates, testing the effective lower bound.

Remark 2.1.17. *Quantitative Easing (QE)*
Central banks purchase financial assets (government bonds, MBS, corporate bonds), creating electronic reserves to expand the balance sheet. QE works through multiple channels:

i. Portfolio rebalancing (investors pushed into riskier assets)
ii. Risk premia compression (term premium, credit spreads)
iii. Signalling effects (commitment to low rates)
iv. Exchange rate channel (currency weakening)

Bank lending does not mechanically follow reserve creation. Transmission depends on credit demand and bank capital. Effectiveness diminishes with successive rounds.
Fed research shows QE1 lowered 10-year yields by 100bp; QE2 by only 14bp.

Remark 2.1.18. *Hawks and Doves*
Central bankers have individual biases that shape their policy views:

i. Hawks: Wary of inflation, favour tighter policy, defend currency value
ii. Doves: Favour growth and employment, tolerate higher inflation, support looser policy

When a known hawk makes dovish statements (or vice versa), it signals a significant shift and moves markets more than statements consistent with known bias.
Tracking individual central banker positions is essential for anticipating policy shifts.

Remark 2.1.19. *Central Bank Communication*
Central bank communication is a policy tool in itself:

i. Statements: Released after each meeting; market moves on hawkish/dovish tone changes
ii. Minutes: Released with lag (2 to 3 weeks); reveal vote splits and internal debates
iii. Press Conferences: Chair/President Q&A; tone and body language matter
iv. Projections: Dot plots (Fed), staff forecasts; reveal expected rate path
v. Speeches: Individual members signalling; watch for coordinated messaging

Example: BOE June 2012 minutes revealed 5–4 vote against QE (vs 8–1 in May), correctly predicting QE expansion in July.

Remark 2.1.20. *Major Central Banks*

i. Federal Reserve (Fed): Dual mandate (price stability, full employment). FOMC meets 8 times/year with 12 voting members (7 governors, NY Fed, 4 rotating).
Most important central bank globally due to USD reserve status.

ii. European Central Bank (ECB): Price stability mandate (HICP $< 2\%$). Governing Council of 25 members. Key rates: Main Refinancing Operations (MRO), deposit facility rate. Tools include LTRO, TLTRO, OMT, APP.

iii. Bank of Japan (BOJ): Targets 2% inflation (chronically undershoots). Policy Board of 9 members. Pioneer of QE, yield curve control (YCC), negative rates.

iv. Bank of England (BOE): 2% inflation target. MPC of 9 members with individual accountability. First major central bank to publish inflation forecasts.

v. Swiss National Bank (SNB): Inflation target $< 2\%$. Notable for 2011–2015 EUR/CHF floor at 1.20 and subsequent abandonment. Massive FX reserve accumulation.

vi. People's Bank of China (PBOC): Multiple objectives including growth, stability, FX management. Manages CNY within trading band. Uses reserve requirements actively.

Remark 2.1.21. *ECB-Specific Tools*

i. LTRO/TLTRO: Long-Term Refinancing Operations provide multi-year funding to banks against collateral. 3-year LTROs in 2011 to 2012 injected over EUR1 trillion.

ii. OMT (Outright Monetary Transactions): Announced September 2012. Unlimited bond purchases for countries in EFSF/ESM programs. Never actually used but stabilised spreads.

iii. APP (Asset Purchase Programme): QE-equivalent. Purchases of government bonds, covered bonds, corporate bonds, ABS.

iv. TARGET2: Payment system showing intra-Eurozone imbalances. German TARGET2 claims rose to EUR700B+ during crisis, reflecting capital flight from periphery.

Remark 2.1.22. *Fed Crisis-Era Programmes*
During and after 2008, the Fed deployed unprecedented tools:

i. Central Bank Swap Lines: USD liquidity to foreign central banks to prevent LIBOR spikes

ii. TAF (Term Auction Facility): 28–84 day unsecured lending without discount window stigma

iii. TSLF (Term Securities Lending Facility): Treasury lending against lower-quality collateral

iv. CPFF (Commercial Paper Funding Facility): Commercial paper purchases to prevent market freeze

v. TALF (Term ABS Loan Facility): Lending against consumer ABS to support credit markets

vi. Agency MBS Purchases: $1.25 trillion in QE1 alone

Understanding these tools helps anticipate future crisis responses.

2.1.3 Economic Data Releases and Demographics

Remark 2.1.23. *Data Releases and Market Impact*
Data releases are central bank inputs, not just statistics. Markets price expectations before the release; the move comes from the surprise, not the level. Key principles:

i. Surprise Direction: Above/below consensus drives immediate price action. Magnitude of surprise matters more than absolute level.

ii. Revision Risk: Prior releases are frequently revised. Initial release moves markets; revisions move markets again if material.

iii. Hierarchy: Not all data is equal. NFP and CPI dominate in the US; PMIs dominate globally due to frequency and forward-looking nature.

iv. Regime Dependence: What data matters depends on the regime. In low-inflation regimes, growth dominates. In high-inflation regimes, inflation dominates everything.

Back-testing asset responses to past data releases (queries) can inform trade sizing and direction. However, false positives are common. If wrong, cut quickly.

Remark 2.1.24. *Growth Indicators*
Growth data sets the directional bias for risk assets, but markets trade changes, not levels.

i. GDP (Headline, QoQ, YoY): Measures total production in current or constant prices (nominal, real GDP). Released quarterly with preliminary, revised, and final prints. Backward-looking but sets narrative.

ii. Purchasing Managers' Indices (PMIs): Survey business managers on employment, new orders, inventories, production, supplier deliveries, prices, backlogs, imports, and exports on scale of better/same/worse vs prior month. Above 50 = expansion; below 50 = contraction. Most important leading indicator.

iii. Industrial Production: Measures growth in manufacturing, utilities, and mining. More cyclical than services. Released monthly. Leading indicator for GDP forecasting.

iv. Trade Volumes, Baltic Dry Index (BDI): Shipping costs for dry bulk cargo. Leading indicator on global trade activity since it cannot be speculated (no futures market).

PMIs and surveys move markets more than GDP because they are higher frequency and forward-looking.
Growth surprises affect equities (earnings expectations), commodities (demand), FX (via growth differentials).

Remark 2.1.25. *Inflation*
Inflation data is primarily a policy variable. It constrains or enables central bank action.

i. Consumer Price Index (CPI)/Core CPI: Prices paid by consumers for basket of goods and services. Core excludes food and energy (volatile). Central banks target core.

ii. Producer Price Index (PPI): Average change in selling prices received by domestic producers. Leads CPI by 1–3 months as input costs pass through.

iii. PCE (Personal Consumption Expenditures): Fed's preferred inflation measure. Differs from CPI in weights and methodology.

iv. Inflation Expectations: Market-based (breakevens from TIPS) or survey-based (Michigan, NY Fed). Central banks watch expectations closely. De-anchoring is a regime change.

Inflation surprises directly affect real yields, curve shape, FX via real rate differentials.
Inflation matters most when central banks are near constraints (ZLB, credibility questioned).

Remark 2.1.26. *Employment Data*
Labor data bridges growth, inflation, and central bank reaction function.

i. Non-Farm Payrolls (NFP): Number of workers in US excluding farms, private households, nonprofits, self-employed, and military. Released first Friday of each month. Among highest-volatility releases for rates, FX, equities.

ii. Unemployment Rate: Unemployed as percentage of labor force. Headline number; can be distorted by participation changes.

iii. Participation Rate: Labor force as percentage of civilian population. Structural decline (aging) vs cyclical decline (discouraged workers) matters.

iv. Wage Growth (Average Hourly Earnings): More important than job count for inflation outlook.

v. JOLTS (Job Openings and Labor Turnover Survey): Shows job openings, hires, quits. Quits rate indicates labor market tightness and worker confidence.

Wage growth drives services inflation, which is stickier than goods inflation.
Fed's dual mandate makes employment data directly policy-relevant.

Remark 2.1.27. *External Accounts*
External balances define structural currency pressure and financing needs.

i. Current Account: Net exports + net foreign income + transfer payments. Deficit countries need capital inflows or currency depreciation; surplus accumulate reserves and have structurally stronger currencies.

ii. Capital Account/Financial Account: Portfolio flows, FDI, banking flows. Can finance current account deficits but creates vulnerabilities (sudden stop risk).

iii. Trade Balance: Exports minus imports. Incremental changes in exports or imports can have big impact on trade balance as percentage of GDP.

iv. Terms of Trade (ToT): Export prices divided by import prices. Improves when export prices rise relative to imports. Currency depreciation typically worsens ToT.

v. FX Reserves: Central bank holdings of foreign currency and gold. Countries with large reserves can intervene; watching reserve changes indicates intervention.

Persistent current-account deficits require higher yields to attract capital or eventual currency adjustment.

Remark 2.1.28. *Government/Fiscal Indicators*
Fiscal data determines long-term sustainability, not short-term price action.

i. Budget Balance: Revenues minus expenditures. Deficit = borrowing required. Deficit/GDP ratio matters more than absolute level.
ii. Debt-to-GDP: Stock measure of accumulated borrowing. Matters more for EM and Euro periphery than for reserve currency issuers.
iii. Primary Balance: Budget balance excluding interest payments. Shows underlying fiscal stance; positive primary balance means debt/GDP can stabilise if growth exceeds interest rate.
iv. Debt Sustainability: Interest payments as % of revenue. Rising share means deteriorating sustainability.

Fiscal deterioration matters when markets doubt financing ability or central bank independence.
Sovereign risk appears first in bond spreads, CDS, FX (especially EM).

Remark 2.1.29. *Consumption Indicators*
Consumption drives developed-market growth (70%+ of US GDP).

i. Retail Sales: Consumer demand for finished goods. Durable vs non-durable breakdown matters.
ii. Consumer Confidence: Survey of expected financial situation. Conference Board and Michigan surveys in US. Leading indicator, confidence leads spending, but noisy.
iii. Personal Income and Spending: Income growth enables spending growth. Savings rate = income minus spending.
iv. Savings Rate: Higher savings = lower current consumption but more sustainable. US savings rate has declined structurally; Asian savings rates remain high.

Consumption data is most relevant for equity indices, cyclical sectors, domestic currencies.
US consumer accounts for $\sim 15\%$ of global GDP directly through consumption.

Remark 2.1.30. *Housing Indicators*
Housing is cyclically important and credit-sensitive.

i. Building Permits: Lead housing starts since permits are required before construction. Most forward-looking housing indicator.
ii. Housing Starts: Counted when construction begins. Gauge of housing demand and construction activity.
iii. Existing/New Home Sales: Volume of transactions. Prices vs volumes can diverge.
iv. Home Price Indices: Case-Shiller, FHFA. Lag actual market conditions.
v. Household Formations: Structural driver of housing demand. Post-crisis decline reduced demand.

Housing matters for rate sensitivity, consumer wealth effects, and financial stability.

Remark 2.1.31. *Business and Industry Indicators*
These indicators show where growth is coming from and where it's heading.

i. Manufacturing vs Services PMI: Manufacturing is more cyclical and globally linked. Services dominate GDP but move slower.
ii. New Orders: Most forward-looking PMI subcomponent. Rising new orders = future production.
iii. Inventories: High inventories relative to sales = future production cuts. Inventory cycles matter for commodities and industrial equities.
iv. Capacity Utilisation: Actual output relative to potential. Over 100% = overheating, inflation risk. Well under 100% = slack, deflation risk.
v. Durable Goods Orders: Capital expenditure intentions. Core capital goods orders (ex-aircraft, ex-defence) is key business investment indicator.

ISM New Orders minus Inventories spread is a reliable leading indicator of manufacturing cycles.

Remark 2.1.32. *Demographics*
Slowest-moving macro variable, but the hardest constraint. Demographics cannot be QE'd.

i. Working-Age Population (15–64): Drives potential growth. Many developed countries have peaked or will peak soon.

ii. Prime Saving Age (35–69): Peak saving years. Countries past this peak see declining savings and current accounts.

iii. Dependency Ratio: Non-working population (young + old) divided by working population. Higher ratio = more fiscal pressure, lower savings.

iv. Median Age: Summary statistic for demographic structure. Europe/Japan: 40s; US: 38; EM: 20s–30s.

v. Retirement Age vs Life Expectancy: Gap determines pension system sustainability. Life expectancy rising; retirement ages barely moving.

Demographics affect long-term growth, savings rates, current accounts, equilibrium real rates.

Remark 2.1.33. *Demographic Implications by Country*
Peak working-age populations determine structural trajectories:

i. Already peaked: Germany (2006), Japan (2016), Netherlands (1989), UK (2009), US (2007)

ii. Peaking soon: China (2032), Korea (2023), Russia (2025), Spain (2022)

iii. Long runway: India (2050), Indonesia (2050), Nigeria (2050), Philippines (2050)

Countries past peak will see declining current accounts, rising dependency ratios, pressure on fiscal positions. Jean-Claude Trichet: "Current account balance is an important summary indicator that signals losses of competitiveness and emerging imbalances."

Remark 2.1.34. *Building Queries for Data Events*
Queries back-test asset behaviour around past occurrences of specific events:

i. Identify the event (e.g., BOE QE announcement, Fed rate hike, NFP beat)

ii. Find historical dates with similar conditions

iii. Calculate asset returns +1 day, +2 days, +1 week, +1 month after event

iv. Compute hit rates (percentage of times asset moved in expected direction)

v. Assess risk/reward: average return vs standard deviation

Queries are powerful for short-term event trading and understanding portfolio exposure to upcoming releases. Caveat: False positives are common. If position moves against you immediately after event, cut quickly.

Remark 2.1.35. *Key Data Release Calendar (US)*
Monthly cadence of major releases:

i. First Friday: Employment Report (NFP, unemployment rate, wages)

ii. Mid-month: CPI, Retail Sales, Industrial Production

iii. Third week: Housing Starts, Building Permits

iv. Late month: GDP (quarterly), PCE, Durable Goods

v. First business day: ISM Manufacturing PMI

vi. Third business day: ISM Services PMI

FOMC meetings (8 per year) and ECB meetings (6 per year) often dominate all other releases.
Economic calendars (Bloomberg ECO, Reuters, Trading Economics) are essential tools.

2.2 Transmission and Cross-Asset Mapping

This section services as a primer for four basic asset classes and their usage in global macro trading.

2.2.1 Building Blocks of Global Macro

Remark 2.2.1. *Macro Strategy Product Groups*
The four core product groups in global macro are:

i. Fixed Income (FI) (Policy and Growth Sensor): encodes growth expectations, inflation expectations, central bank credibility. Yield curves summarise the entire macro narrative; steepening means growth/reflation; flattening or inversion means slowdown/recession. Fixed income tells what the market thinks will happen before it happens.

ii. Foreign Exchange (FX) (Adjustment Mechanism): FX reflects relative, not absolute, macro conditions. Sensitive to rate differentials, capital flows, balance of payments, risk-on/risk-off regimes. FX absorbs and redistributes macro shocks across countries.

iii. Equities (Growth Expression): equities are long growth, long liquidity, short uncertainty. Equity indices aggregate earnings expectations, discount rates, risk appetite. Volatility (VIX) is a second-order macro signal, not just an equity signal. Equities express the confidence (or fear) embedded in the macro outlook.

iv. Commodities (Inflation and Real-Economy Link): driven by physical supply and demand, inventories, geopolitics. Highly sensitive to growth cycles, inflation shocks, currency moves (especially USD). Commodities anchor macro views in real economic constraints.

These are the most liquid markets globally, which allows macro strategies to scale to large AUM while maintaining execution quality.

Remark 2.2.2. *Relationship Between Assets*
These relationships are regime-dependent, not permanent laws:

i. Growth Shocks: rates rise, equities rally, cyclicals outperform, commodities rise. FX favours growth-linked currencies (AUD, CAD, EM FX).

ii. Recession Shocks: rates fall, curves flatten/invert, equities sell off, USD strengthens (flight to safety), commodities weaken (except gold).

iii. Inflation Shocks: nominal rates up, real rates determine equity impact. Commodities outperform. FX punishes low-credibility central banks.

iv. Liquidity Shocks: USD strengthens sharply, credit spreads widen, correlations spike toward 1, volatility explodes. Cross-currency basis blows out.

A single macro thesis should have expressions across multiple asset classes, each with different risk profiles and convexity characteristics.

Remark 2.2.3. *Asset Correlation Dynamics*
Asset correlations change under stress. Diversification fails when needed most, where assets that appear uncorrelated in calm periods may collapse together in crises.
Macro traders profit when they anticipate correlation breakdowns and position for nonlinear regime shifts. This is why macro portfolios use convex instruments (options), size conservatively, and focus on liquidity.
Traders must know what assets they are trading in aggregate. If entire portfolio has assets with high correlation to one another, they can be stopped out of all positions more easily if the market turns against them.

Remark 2.2.4. *Position Sizing Framework*
Position sizing is arguably more important than knowing what to buy or sell. The first rule of investing is to preserve capital and avoid loss.
Position sizing aims to adjust each position for volatility. This normalises all positions such that losses are more predictable and have the same probability of occurring:

i. Calculate N = 20-day exponential moving average of True Range

ii. True Range = max(High − Low, High − Previous Close, Previous Close − Low)

iii. Use $2N$ for stop-loss calculation (two standard deviations) to avoid being stopped out by normal volatility

iv. With $2N$, there is approximately 2% chance of being stopped out on any given day (vs. 16% with $1N$)

This methodology, developed by the Turtle Traders, allows traders to take positions in a broad range of markets while monitoring risk consistently.

Remark 2.2.5. *Risk/Reward Discipline*
Given human error and transaction costs, traders can only hope to be right about 50% of the time. Having good risk/reward on each trade will make or break long-term success.
As a baseline, always strive for a 1:3 risk-to-reward ratio or higher:

i. For every trade risking 25 basis points, expect to make at least 75 basis points
ii. Set targets before the trade is executed to avoid biases of irrational human decision making
iii. Log reasons for each trade in a trade journal
iv. If original thesis no longer holds true, take trade off, but do not create excuses for taking trades off early

Remark 2.2.6. *Gap Risk*
Gap risk is one of the biggest risks when sizing positions. Gap risk occurs when prices change without any trading in between (e.g., overnight, over weekends). This risk is especially acute in:

i. EM currencies and assets (less liquid, more event-driven)
ii. Positions held over weekends or holidays
iii. Periods of elevated geopolitical risk
iv. Earnings/data release windows

Remark 2.2.7. *Risk Utilisation*
Risk-taking is nonlinear. Even with 50% conviction, should not run 50% of risk utilisation.

i. Base case: set a maximum VaR limit (e.g., 5% daily VaR at 95% confidence)
ii. If up 5% on the year, allow VaR limit to increase to 6%
iii. If down 2.5% on the year, decrease VaR limit to 4%
iv. At the start of each new year, reset to 0% return regardless of prior year performance

The intention is to create steady returns while preserving capital and avoiding blow-up risk.

Remark 2.2.8. *Stress Testing*
Stress testing aims to go beyond VaR to find potential worst-case scenarios:

i. Run historical back-test on P&L for previous 500 trading days ($\approx$ 2 years)
ii. Sort from best to worst days; examine the average of the 5 worst days (worst 1%)
iii. In periods of low volatility, also test against extreme historical events: October 1987 crash, 1998 LTCM/Russian default, 2008 financial crisis, 2011 Japan earthquake, 2015 CNY devaluation, 2020 COVID crash

Knowing maximum loss outside of VaR helps traders understand exposure to tail events.

Remark 2.2.9. *Performance Metrics*
Key metrics for evaluating macro trading performance:

i. Sharpe Ratio: $\frac{E[R]-R_f}{\sigma}$, risk-adjusted return; higher is better. Assumes normal distribution.
ii. Sortino Ratio: Sharpe with only downside volatility, differentiating between good and bad volatility.
iii. Maximum Drawdown: peak-to-trough decline; critical for understanding worst-case capital loss.
iv. Value at Risk (VaR): 95% or 99% daily VaR (expected loss exceeded on 1 in 20 or 100 days).

When evaluating strategies, test Sharpe ratios rather than raw returns. Break down P&L by asset class, duration, and conviction level to identify strengths and weaknesses.

Remark 2.2.10. *Macro View Thesis*
A macro view expressed in only one asset class is incomplete. A single macro thesis should have multiple expressions, each with different risk profiles:

i. Growth view: rates (duration), equities (beta), FX (pro-cyclical currencies), commodities (cyclicals)
ii. Inflation view: commodities (energy, agriculture), breakevens, FX (commodity currencies), TIPS
iii. Crisis/Deflation view: bonds (duration), USD/JPY/CHF (havens), volatility (VIX, swaptions), gold
iv. Policy divergence view: FX (rate differentials), yield curve spreads, relative equity indices

Having multiple expressions allows for confirmation across asset classes and provides natural hedging if one leg underperforms.

2.2.2 Foreign Exchange in Global Macro

FX plays three special roles simultaneously:

i. Macro shock absorber

ii. Macro transmission channel

iii. Macro expression instrument

Remark 2.2.11. *FX Market Structure*
The FX market is the world's largest and most liquid market, and operates 24 hours per day, 5.5 days per week across global financial centers (London, New York, Tokyo, Singapore, Hong Kong).
Key participants include: central banks (intervention, reserve management), commercial banks (market-making, proprietary trading), corporations (hedging trade flows), asset managers (currency overlay, alpha generation), hedge funds (macro speculation, carry trades), and retail traders.
The market is predominantly OTC, with spot, forwards, and swaps as the main instruments. FX futures trade on exchanges (CME) but represent a small fraction of total volume.

Remark 2.2.12. *Role of the US Dollar*
Global macro is de facto USD-centric. USD is the global reserve currency, most commodities are USD-priced, global funding markets clear in USD, and cross-border leverage is USD-denominated.
The USD is involved in approximately 85% of all FX transactions globally. Many "local" macro trades are actually USD liquidity trades in disguise.
Stress often appears first in US Dollar Index (DXY), cross-currency basis, EM FX before local assets.
The DXY Index measures USD against a basket of six major currencies: EUR (57.6%), JPY (13.6%), GBP (11.9%), CAD (9.1%), SEK (4.2%), CHF (3.6%). The DXY is not trade-weighted and over-represents EUR.

Remark 2.2.13. *Special Drawing Rights (SDRs)*
SDRs are international reserve assets created by the IMF to supplement member countries' official reserves. The SDR basket is reviewed every five years. SDRs serve as a unit of account for IMF transactions and provide insight into official reserve diversification trends.

Remark 2.2.14. *Currency Regimes*
Currency regimes matter more than models. FX does not behave uniformly. Regime dominates signal.

i. Free-floating: price discovery happens in FX

ii. Managed/pegged: pressure builds elsewhere such as in rates, reserves, capital controls

iii. Hard pegs/Currency boards: FX looks stable until it breaks violently

FX valuation only works within regime constraints. Ignore the regime and "cheap" can stay cheap forever.

Remark 2.2.15. *Central Bank Intervention*
Central banks intervene in FX markets for multiple objectives:

i. Defend a peg or managed float band

ii. Smooth excessive volatility

iii. Counter speculative attacks

iv. Accumulate or deploy reserves

Intervention effectiveness depends on: reserve adequacy, policy credibility, coordination with monetary policy, and market liquidity conditions. Intervention against fundamental pressures typically fails. Watch for reserve depletion as a leading indicator of forced adjustment.

Remark 2.2.16. *Valuation Techniques*

i. Market Implied and Risk-Appetite Signals

   1. Equity Index Price Performance: strong local equities (vs peers) attract inflows and support currency; sharp equity underperformance foreshadows FX weakness via risk-off deleveraging and foreign selling.
   2. Credit/Sovereign Risk: widening sovereign spreads/CDS signals default/policy credibility/funding stress, weakening FX or raising devaluation risk premia. Also a regime detector for EM FX.
   3. Sentiment/Positioning: call-put skew on three-month risk reversals. Acts as a multiplier. Can create overshoots/undershoots around fair value in crowded carry or crisis episodes.

ii. Trade and Competitiveness Fundamentals

1. Trade Balance/External Account: structurally strong trade position often supports FX; persistent deficits can make the currency reliant on financing.
2. Trade-Weighted Index (TWI): reflects currency's effective competitiveness against trading partners; the currency basket that matters for the real economy.
3. Economic Activity: strong activity can support FX via higher expected returns and inflows, but it can also worsen the trade balance.
4. Export Partner Growth: mapping who buys exports and whether those buyers are expanding/contracting. Export partner growth leads exporter's FX through expected trade receipts.

iii. Macro Sustainability Indicators

1. Debt-to-GDP: high public debt with FX liabilities or weak domestic savings can increase risk premia and currency vulnerability, as it tightens the policy constraint set (inflation vs austerity vs default).
2. Current Account Balance: persistent deficits require capital inflows; sudden stops trigger sharp FX adjustment.
3. FX Reserves: adequacy (months of import cover, short-term debt coverage) determines policy space.

iv. Long-Term Valuation Anchors

1. Purchasing Power Parity (PPP): the idea that exchange rates should adjust so that identical goods cost the same in different countries. In practice, PPP convergence can take years or decades.
2. GDP Per Capita: structural development/productivity proxy. Richer, higher productivity economies tend to sustain stronger real exchange rates over time (Balassa-Samuelson effect).

v. Carry Indicators

1. OIS Differential: compares respective policy rates of two countries. Carry is the interest rate differential between two currencies.
2. LIBOR/SOFR Differential: reflects unsecured interbank rates. Can be decomposed into risk-free rate + credit spread component. Using relative spreads gives a sense of direction; interest rate differential drives demand for a currency.
3. Risk-Adjusted Carry: adjusts carry for volatility. Outperforms standard carry trades and is useful in cross-currency analysis:

$$\text{Carry-to-Risk Ratio} = \frac{\text{3M Carry (Rate Differential)}}{\text{30D Realized Volatility}}$$

Within high-yielding currency pairs, one would ideally want the highest carry-to-risk ratio. Volatility and carry components are always in flux, so adjustments should be made frequently.

Remark 2.2.17. *Carry Trade Mechanics*
Carry strategies are typically executed by borrowing in lower-yielding currencies and buying higher-yielding currencies. The carry trade is a crowded trade; in large risk-off markets, the strategy suffers significant losses as positions unwind simultaneously.
Carry is the return earned on holding a currency, assuming the exchange rate is held constant. In practice, exchange rate moves often dominate carry returns. Historically, carry strategies exhibit positive returns on average but with significant negative skewness (small gains, occasional large losses).

Remark 2.2.18. *Carry Trade Risk Factors*
Key risks to carry trades include:

i. Crowding Risk: carry trades are popular, leading to violent unwinds during risk-off episodes.
ii. Gap Risk: EM currencies can gap significantly on weekends or during crises, bypassing stop-losses.
iii. Regime Shifts: central bank policy changes can rapidly alter rate differentials.
iv. Correlation Breakdown: in stress, all carry currencies tend to sell off together against funding currencies.
v. Liquidity Risk: EM FX can become illiquid precisely when exits are needed most.

Finding the carry-to-risk ratio is useful, as is comparing all currency pairs on a relative basis. Measuring the standard error of the sample and finding the most attractive pairs helps in mean-reverting carry strategies.

2.2.3 Equities in Global Macro

In macro, a trader should understand equity in terms of market behaviour in the country being traded, the sectors that make up those indices, and correlation risk.

Remark 2.2.19. *Top-Down Macro Approach to Equities*
Many macro traders take a top-down approach to trading equities:

i. Start with a global view
ii. Narrow to countries expected to outperform
iii. Select the best sectors within those countries
iv. Optionally, select best companies within sectors

If the investor has a knack for knowing where the next opportunity lies, the most reward typically comes from concentrated exposure rather than diversified index holdings.

Remark 2.2.20. *Major Equity Indices by Region*

i. United States: S&P 500, Nasdaq, DJIA, Russell 2000, Wilshire 5000
ii. Europe: Euro Stoxx 50, DAX, CAC 40, FTSE 100, FTSE MIB, IBEX 35, AEX
iii. Asia Ex-Japan: Shanghai Comp, Hang Seng, KOSPI, ASX 200, NZX 50
iv. Japan: Nikkei, Topix, JASDAQ
v. Emerging Markets: Bovespa, NIFTY, Micex/RTS, Mexico IPC, TOP 40/JALSH, BIST 30/100 etc.

Indices can be traded directionally or as relative value trades (long one index, short another). Different indices have different sector exposures: Bovespa is more commodity-sensitive than NIFTY; S&P 500 has more financial exposure than Nasdaq (technology-weighted).

Remark 2.2.21. *Index Construction Methods*
Index construction methodology affects interpretation:

i. Price-Weighted: considers only stock price, ignores market cap. A single large stock move can dominate.
ii. Market-Cap Weighted: weighted by market capitalisation. Large-cap stocks dominate performance.
iii. Equal-Weighted: each constituent has equal weight regardless of size.

Understanding weighting structure is essential for accurate interpretation of index moves.

Remark 2.2.22. *Sector Rotation Across Economic Cycles*
Different sectors outperform in different economic regimes:

i. Recession/Contraction: Consumer Staples, Utilities, Healthcare (defensive sectors)
ii. Recovery: Financials, Consumer Discretionary, Real Estate
iii. Expansion: Industrials, Materials, Technology
iv. Slowing Growth/Late Cycle: Energy, Materials (inflation hedges)

To best position an equity portfolio, it is critical to know whether the economy is in contraction, recovery, expansion, or slowing growth.

Remark 2.2.23. *Equity Derivatives Overview*
Key equity derivatives for macro trading:

i. ETFs: fast, cheap exposure to indices, sectors, countries, commodities, FX, FI. Trade like stocks.
ii. ADRs (American Depositary Receipts): foreign stocks listed on U.S. exchanges in USD. Subject to FX risk since underlying shares are held locally.
iii. Index Futures: liquid, leveraged exposure to equity indices.
iv. Options: provide asymmetric payoffs for hedging and directional views.

Remark 2.2.24. *The Volatility Index (VIX)*
The VIX measures implied volatility of S&P 500 options one month out. Key properties:

i. Negatively correlated with equity returns (volatility rises when equities fall)
ii. During large selloffs, VIX can move from mid-teens to 30+ in days

iii. VIX options provide asymmetric hedges for equity portfolios

iv. VIX futures typically trade in contango; during sharp selloffs, shift to backwardation

Volatility is measured relative to $\sqrt{252} \approx 16$. A VIX of 16 implies approximately 1% expected daily move; VIX of 24 implies 1.5% daily move ($24/16 = 1.5\%$).
V2X (Euro Stoxx), VXN (Nasdaq), VXD (DJIA), RVX (Russell 2000), GVZ (Gold), OVX (Crude Oil).

Remark 2.2.25. *Variance Swaps*
Variance swaps are OTC products providing pure variance exposure. Advantages over options:

i. Pure volatility exposure without delta hedging

ii. Priced with realised volatility (usually lower than implied)

iii. No interest rate or dividend risk

Payoff structure is convex and nonlinear. Long variance benefits disproportionately from large moves in either direction. Quoted by strike (reference realised volatility), vega notional, variance units, maturity.
Variance units $= \frac{\text{Vega Notional}}{2\times\text{Strike}}$, Payoff $= (\sigma^2_{\text{realised}} - K^2) \times$ Variance Units

Remark 2.2.26. *Dividend Swaps*
An OTC or exchange-traded product that allows one to take a view on dividends of an index to be higher, or lower, than a fixed amount.

Remark 2.2.27. *Equity Valuation Techniques for Macro*

i. Price-to-Book (P/B): P/B $> 2.5\times$ generally overbought; P/B $< 1.5\times$ oversold. Useful for detecting bubbles and value opportunities.

ii. Dividend Yield: dividend/price. Holder gets paid while waiting (long carry). Useful for relative value across markets.

iii. Price-to-Earnings (P/E): price per \$1 of earnings. Best for comparable analysis across countries, sectors, stocks. Forward P/E uses analyst EPS estimates.

iv. Free Cash Flow Yield: FCF/Price. Cash flow from operations minus capital expenditures. Graham and Dodd value investors use this as top screen.

v. Market Cap to GDP: useful for locating potential bubbles. Above 100% warrants caution; rapid rises to 150%+ indicate bubble conditions (Japan 1989).

Remark 2.2.28. *Leading Indicators for Equities*
Useful macro indicators that lead equity performance:

i. PMI (Purchasing Managers' Index): leading indicator scored 0-100. Sharp slope changes and deviations from 50 provide strong signals. ISM below 50 typically indicates U.S. recession.

ii. Baltic Dry Index: shipping prices for raw material dry bulk. Supply is fixed (ships take years to build), demand is inelastic. Leading indicator for raw material demand.

iii. CDX High Yield Index: CDS spread on high-yield corporates. Highly correlated with VIX; inversely related to equity prices. Rising spreads signal risk-off.

iv. Consumer Confidence: in the U.S., consumption accounts for $> 70\%$ of GDP. Higher confidence leads to higher spending and GDP growth.

v. Commodity Prices (YoY): rising commodity prices are inflationary, typically met by central bank tightening, which has bearish equity implications.

Remark 2.2.29. *AUD Volatility as Risk Barometer*
The Australian dollar is a main risk-on currency due to Australia's commodity exports and reliance on Asian economies. AUD moves in tandem with equity prices.
Three-month AUD/USD implied volatility is highly correlated with VIX and inversely correlated with S&P 500. In times of stress, the rate of change in AUD volatility shifts suddenly and aggressively.
Using VIX and AUD volatility combined on an absolute basis smooths out error as a more useful measure.

2.2.4 Fixed Income in Global Macro

Fixed income is the lifeblood of the global financial system. Bank credit and debt capital markets fund governments, corporations, and consumers. Understanding fixed income is essential for macro trading because yields encode growth expectations, inflation expectations, and central bank credibility.

Remark 2.2.30. *Fixed Income Universe*
The fixed income market can be decomposed into:

i. Money Markets: Short-term instruments (maturity < 1 year). Includes T-bills, commercial paper, repos, Fed Funds, SOFR.
ii. Government Bonds: Sovereign debt across the curve (2Y, 5Y, 10Y, 30Y). Risk-free rate benchmark.
iii. Inflation-Linked Bonds: TIPS in US, Linkers in UK, OATi in France. Index to CPI.
iv. Corporate Bonds: Investment grade (IG) and high yield (HY). Credit spread over risk-free rate.
v. Municipal Bonds: State and local government debt, often tax-exempt.
vi. Mortgage-Backed Securities (MBS): Agency and non-agency. Prepayment risk.
vii. Asset-Backed Securities (ABS): Auto loans, credit cards, student loans.

Fixed income is the largest of the four product groups by notional outstanding.

Remark 2.2.31. *Money Markets and Funding*
Money markets allow short-term capital flow between lenders and borrowers:

i. Treasury Bills (T-Bills): Short-term US government obligations (4-week, 13-week, 26-week, 52-week). Risk-free benchmark.
ii. Commercial Paper: Unsecured short-term corporate debt, typically 90–270 days. Lower rates than bank borrowing.
iii. Repurchase Agreements (Repos): Sale of securities with agreement to repurchase. Critical for short-term funding and leverage.
iv. Certificates of Deposit (CDs): Bank time deposits, 30 days to 5 years.
v. Banker's Acceptances: Promissory notes issued by firms to banks, typically for trade finance.

Money market stress (rising repo rates, LIBOR-OIS widening) often precedes broader financial stress.

Remark 2.2.32. *Key Reference Rates*
Reference rates anchor pricing across the fixed income universe:

i. Fed Funds Rate: Unsecured overnight lending rate between depository institutions.
ii. SOFR (Secured Overnight Financing Rate): Overnight Treasury repo rate as USD benchmark.
iii. LIBOR (Legacy): London Interbank Offered Rate. Historically the most important rate in global finance. Phased out post-2021 due to manipulation scandals.
iv. EONIA/EURSTR: Euro overnight rates. EURSTR replaced EONIA in 2022.
v. SONIA: Sterling overnight rate for GBP.

The transition from LIBOR to risk-free rates (SOFR, SONIA, EURSTR) is one of the largest market structure changes in decades.

Remark 2.2.33. *Overnight Index Swaps (OIS)*
OIS is a fixed-for-floating interest rate swap indexed against an overnight rate (Fed Funds effective in US, EONIA/EURSTR in Europe):

i. Excellent tool to hedge or speculate on central bank action
ii. Typically maturity less than two years
iii. Customisable dates for precise central bank meeting targeting
iv. Less volatile than LIBOR, especially in periods of stress

Example: If BOE rate is 0.50% and December MPC OIS trades at 0.625%, the market prices 50% chance of a 25bp hike.

Remark 2.2.34. *LIBOR-OIS Spread*
The LIBOR-OIS spread is a key gauge of funding stress:

i. LIBOR is unsecured interbank lending; OIS is effectively risk-free overnight rate
ii. Spread widens during credit stress as banks demand higher compensation for counterparty risk
iii. 2008 financial crisis: LIBOR-OIS spiked as institutions stopped trusting each other
iv. European crisis: Spread widened as European banks faced USD funding stress

A widening spread does not automatically indicate crisis. Serves as useful barometer for funding market pressure.

Remark 2.2.35. *Eurodollar Futures*
Eurodollar deposits are USD time deposits at banks outside Fed jurisdiction. Eurodollar futures are the most heavily traded futures contracts in the world:

i. Contract on 3-month LIBOR (transitioning to SOFR futures)
ii. Contract size: $\$1,000,000$ notional
iii. Price: $100 -$ 3-Month LIBOR Yield
iv. Tick value: 1 basis point = \$25
v. Months: March (H), June (M), September (U), December (Z)
vi. Colour codes by year: White (Y1), Red (Y2), Green (Y3), Blue (Y4), Gold (Y5)

Eurodollar futures used to hedge LIBOR exposure, manage weighted average maturity of short-term liabilities.

Remark 2.2.36. *Interest Rate Swaps*
Interest rate swaps allow exchanging one set of interest payments for another:

i. Fixed-for-Floating: Most common. Exchange fixed rate for floating rate (SOFR, EURSTR).
ii. Receiver: Receives fixed, pays floating. Profits if floating rates fall.
iii. Payer: Pays fixed, receives floating. Profits if floating rates rise.
iv. Swap curve driven by future rate expectations; increasingly exchange-traded post-Dodd-Frank.

Interest rate swaps are the largest derivatives market by notional. Used for hedging, speculation, and transforming asset/liability profiles.

Remark 2.2.37. *Forward Rate Agreements (FRAs)*
FRAs are OTC contracts to exchange a reference rate (SOFR, formerly LIBOR) for a fixed rate:

i. More customisable than Eurodollar futures (pick specific dates)
ii. Notation: "3×6" means 3-month forward, 3-month rate
iii. Fixing date, settle date, and maturity date are key terms
iv. Used for hedging or speculating on short-term rate movements

FRAs allow precise targeting of rate exposure around central bank meetings or data releases.

Remark 2.2.38. *US Treasury Futures*
Treasury futures are standardised contracts for trading various maturities:

i. 2-Year (TU): Contract size $\$200,000$, tick \$15.625
ii. 5-Year (FV): Contract size $\$100,000$, tick \$7.8125
iii. 10-Year (TY): Contract size $\$100,000$, tick \$15.625
iv. 30-Year (US): Contract size $\$100,000$, tick \$31.25
v. Ultra-Long (WN): 25+ years, contract size $\$100,000$, tick \$31.25

Treasury futures are physically settled (cheapest-to-deliver mechanics). Most traders roll positions before delivery. Futures curve typically in backwardation due to term premium.

Remark 2.2.39. *German Bund Futures*
German government bonds (Bunds) are the European benchmark:

i. Schatz (DU): 1.75–2.25 year maturity
ii. Bobl (OE): 4.5–5.5 year maturity
iii. Bund (RX): 8.5–10.5 year maturity
iv. Buxl (UB): 24–35 year maturity

Bunds trade as "risk-free" for the Eurozone. Spreads to Bunds (e.g., BTP-Bund spread for Italy) are key measures of peripheral sovereign risk.

Remark 2.2.40. *Yield Curve Trades*
Curve trades express views on the shape of the yield curve (done DV01 neutral):

i. Steepener: Long short-end, short long-end. Profits from curve steepening.
ii. Flattener: Short short-end, long long-end. Profits from curve flattening.
iii. Bull Flattener: Rates fall, long end falls more than short end.
iv. Bear Flattener: Rates rise, short end rises more than long end (often Fed hiking).
v. Bull Steepener: Rates fall, short end falls more than long end (often Fed cutting).
vi. Bear Steepener: Rates rise, long end rises more than short end (inflation fears).

The 2s10s spread (10Y yield minus 2Y yield) is a classic recession indicator. Inversion has preceded every US recession since 1970.

Remark 2.2.41. *Carry and Rolldown*
Fixed income positions have carry and rolldown components:

i. Carry: Coupon income − financing cost. + for steepeners, − for flatteners in normal yield curve.
ii. Rolldown: Return from bond "rolling down" a positively sloped yield curve as it approaches maturity.
iii. In DV01-neutral trades, carry differential between legs must be accounted for.
iv. Rolldown is deterministic if curve shape unchanged; carry is known at trade inception.

Total return = Price change + Carry + Rolldown. Carry and rolldown dominate in low-volatility environments.

Remark 2.2.42. *Treasury Inflation-Protected Securities (TIPS)*
TIPS are US government bonds indexed to inflation (CPI-U NSA):

i. Principal adjusts with CPI; coupon is fixed rate applied to adjusted principal
ii. 5Y, 10Y, and 30Y maturities issued quarterly
iii. Floor at par: At maturity, receive greater of par value or inflation-adjusted principal
iv. 3-month lag between CPI and index calculation

TIPS allow investors to take views on real rates or inflation. Short-end TIPS trade like food and energy.

Remark 2.2.43. *Breakeven Inflation*
Breakeven inflation rate is the implied inflation rate from TIPS:

$$\text{Breakevens} = \text{Nominal Treasury Yield} - \text{TIPS Real Yield}$$

For example, if 10Y Treasury yields 4.0% and 10Y TIPS yields 1.5%, 10Y breakeven is 2.5%.

i. Breakevens reflect market-implied inflation expectations
ii. Can be traded directly via breakeven swaps
iii. Rising breakevens = market expects higher inflation
iv. Fed monitors breakevens for inflation expectations anchoring

Negative real rates (TIPS yield < 0) indicate investors accepting guaranteed loss of purchasing power for safety.

Remark 2.2.44. *Sovereign Credit Risk*
Sovereign debt is issued by national governments. Key concepts:

i. Reference Entity: Legal name for CDS purposes (e.g., Hellenic Republic for Greece)
ii. Curve Inversion: When short-term yields exceed long-term yields, market prices default risk
iii. Credit Events: Failure to pay, restructuring, moratorium
iv. Recovery Rate: Typically assumed 40% for sovereigns; historical average 55% with high variance

Peripheral European spreads (Italy, Spain, Portugal, Greece) to German Bunds are key risk indicators.

Remark 2.2.45. *Credit Default Swaps (CDS)*
CDS provides insurance-like protection against credit events:

i. Protection Buyer: Pays spread (premium), receives par if credit event occurs
ii. Protection Seller: Receives spread, pays par minus recovery if credit event occurs
iii. Premium quoted as annual spread, paid quarterly (March, June, September, December)
iv. Can buy protection without owning underlying bond ("naked CDS")

CDS provides asymmetric payoff: pay small premium for potentially large payout. Attractive risk/reward for tail hedging. Counterparty risk is critical.

Remark 2.2.46. *CDS Credit Events*
Credit events trigger CDS settlement:

i. Failure to Pay: Sovereign/corporate misses interest or principal payment
ii. Restructuring: Postponement, reduction, or deferral of obligations
iii. Bankruptcy: Corporate only (sovereigns cannot file bankruptcy)

For sovereigns, restructuring is not automatic trigger. Both buyer and seller have right (not obligation) to trigger CDS. Greece 2012 required ISDA determination committee ruling.

Remark 2.2.47. *Historical Sovereign Defaults*
Recent sovereign defaults and restructurings:

i. Russia 1998: Devaluation and default on local currency obligations, $73B
ii. Argentina 2001: Largest sovereign default at time, $82B, 70% haircut
iii. Uruguay 2003: Contagion from Argentina, 5-year maturity extension
iv. Greece 2012: PSI restructuring, largest sovereign default in history, $200B+
v. Argentina 2020: Ninth default, $65B restructured

Recovery rates vary widely (20%–80%). "Orderly" restructurings tend to have higher recovery.

Remark 2.2.48. *Fixed Income ETFs*
Fixed income ETFs provide liquid, diversified exposure:

i. Treasury: SHY (1–3Y), IEF (7–10Y), TLT (20+Y), TBT (2x inverse 20+Y)
ii. TIPS: TIP (broad TIPS)
iii. Investment Grade: LQD (corporate IG), AGG (aggregate bond)
iv. High Yield: HYG, JNK (high yield corporate)
v. Emerging Markets: EMB, PCY (EM sovereign debt)
vi. Municipals: MUB (muni bonds)
vii. MBS: MBB (mortgage-backed)

ETFs provide quick and cheap exposure to markets otherwise requiring large minimums or dealer relationships.

Remark 2.2.49. *Duration and DV01*
Duration measures sensitivity to interest rate changes:

i. Modified Duration: Approximate percentage price change for 1% yield change
ii. DV01 (Dollar Value of 01): Dollar change for 1bp yield change $= \frac{\text{Duration} \times \text{Price}}{10000}$
iii. Longer maturity = higher duration = more rate sensitivity
iv. Curve trades are structured DV01-neutral to isolate curve shape from level

A 10Y bond with duration 8 loses approximately 8% if yields rise 100bp.

Remark 2.2.50. *Convexity*
Convexity measures how duration changes as yields change:

i. Positive convexity: Bond gains more when yields fall than it loses when yields rise (most bonds)
ii. Negative convexity: MBS, callable bonds. Gains are capped due to prepayment/call risk
iii. Convexity matters more for large yield moves
iv. Long convexity = long volatility; benefits from rate moves in either direction

Convexity adjustment:

$$\Delta P \approx -\text{Duration} \times \Delta y + \frac{1}{2} \times \text{Convexity} \times (\Delta y)^2$$

Remark 2.2.51. *Curve Inversion as Recession Indicator*
Yield curve inversion (short rates exceeding long rates) has predicted every US recession since 1970:

i. 2s10s inversion: 10Y yield below 2Y yield
ii. 3m10s inversion: 10Y yield below 3-month T-bill
iii. Typical lead time: 6–18 months before recession
iv. Mechanism: Markets expect Fed to cut rates in response to coming weakness

Curve inversion is necessary but not sufficient for recession. Duration and depth of inversion also matter.

Remark 2.2.52. *Fixed Income in Different Macro Regimes*

i. Growth Shock (positive): Rates rise, curve steepens, credit spreads tighten
ii. Recession: Rates fall sharply, curve bull steepens then flattens, credit spreads widen
iii. Inflation Shock: Nominal rates rise, real rates may fall initially, curve bear steepens, TIPS outperform nominals
iv. Deflation/Disinflation: Nominal rates fall, breakevens collapse, duration rallies
v. Financial Crisis: Flight to quality (Treasuries rally), credit spreads explode, funding markets seize

Fixed income tells what the market thinks will happen before it happens. The curve summarises the entire macro narrative.

2.2.5 Commodities in Global Macro

Commodities are driven by physical supply and demand, inventories, and geopolitics. They are highly sensitive to growth cycles, inflation shocks, and currency moves (especially USD). Commodities anchor macro views in real economic constraints.

Remark 2.2.53. *Contango and Backwardation*
The shape of the futures curve reveals market structure:

i. Contango: Near-term futures trade cheaper than further-dated contracts. Exists due to storage costs, financing costs, and opportunity cost of capital. Normal state for storable commodities like gold.
ii. Backwardation: Near-term futures trade at premium to further-dated contracts. Occurs during supply shortages or immediate demand spikes.
iii. Agricultural seasonality: Curves can exhibit both. Contango after harvest (abundant supply, storage costs), backwardation before harvest (depleted stocks, tight supply).

Contango creates negative roll yield; backwardation creates positive roll yield.

Remark 2.2.54. *Commodity Indices*
Major commodity indices for benchmarking and trading:

i. CRB (Commodity Research Bureau) Index: 22 commodities, broad representation of prices.
ii. S&P GSCI: Goldman Sachs Commodity Index. Production-weighted, heavily energy-exposed ($> 50\%$).
iii. Bloomberg Commodity Index (BCOM): Diversified, liquidity-weighted.
iv. ETFs: DBC (Deutsche Bank commodities, $> 50\%$ energy), DBA (agriculture only, includes corn, soybeans, sugar, cattle, coffee, cocoa).

Caution for ETFs: contango and roll costs can cause significant performance drag versus spot prices.

Remark 2.2.55. *Commodity Risk and Return Profile*
Commodities have distinct risk characteristics:

i. Higher volatility than FX, equities, and fixed income
ii. Historically attractive Sharpe ratio as a relative comparison
iii. Prior to 2008, exhibited negative correlation with equities (diversification benefit)
iv. Post-2008, correlations with risk assets increased significantly

v. Supply shocks can cause extreme price moves (geopolitics, weather, production disruptions)

Commodities are particularly useful as inflation hedges and for expressing views on global growth.

Remark 2.2.56. *Crude Oil Fundamentals*
Oil is the most important energy commodity, accounting for 33%of global energy demand:

i. Quality measures: API gravity (higher = lighter) and sulphur content (lower = sweeter)
ii. Light sweet crude: Most desirable (WTI, Brent). Heavy sour crude least desirable (Venezuela, Mexico)
iii. Products: Gasoline (cars), kerosene (diesel, jet fuel), heating oil, LPGs
iv. Global consumption approximately 90 million barrels per day

Remark 2.2.57. *Major Oil Benchmarks*
Key oil benchmarks and their characteristics:

i. WTI (West Texas Intermediate): Light sweet crude. Trades on NYMEX (CL). Delivery at Cushing, Oklahoma. Contract: 1,000 barrels. ETF: USO.
ii. Brent Crude: North Sea light sweet crude. Trades on ICE (CO). Prices more than half of world's internationally traded crude. Delivery at Sullom Voe, Scotland.
iii. WTI-Brent Spread: Historically WTI traded at premium (higher grade), but infrastructure constraints at Cushing and Middle East supply sensitivity pushed Brent to premium.

Remark 2.2.58. *OPEC and Oil Supply*
OPEC controls significant portion of global oil supply:

i. 12 member countries controlling $> 35\%$ of global production, 73% of reserves
ii. Saudi Arabia has largest spare capacity. "The Call on OPEC" typically means Saudi increases production
iii. Key NOCs: Saudi Aramco (largest company by value, 1/6 of global reserves), NIOC (Iran), PDVSA (Venezuela), INOC (Iraq), KOC (Kuwait)
iv. OPEC quota system attempts to coordinate production to stabilise prices

Political risk in OPEC members is critical for oil price forecasting.

Remark 2.2.59. *Oil Chokepoints and Infrastructure*
Critical infrastructure for global oil flows:

i. Strait of Hormuz: Most important chokepoint. 17 million bbl/d (20% of global traded oil). Between Iran and Oman. Geopolitical flashpoint.
ii. Strait of Malacca: 14 million bbl/d. Connects Indian and Pacific Oceans. Critical for Asian supply.
iii. Mandab Strait/Suez Canal: 3 million and 2 million bbl/d respectively. Piracy risk.
iv. Pipelines: Druzhba (Russia to Europe, 2,300 miles, 1.4 million bbl/d). Keystone (Canada to US).

Supply disruption at chokepoints causes immediate price spikes, especially in Brent.

Remark 2.2.60. *Crack Spread*
The crack spread measures refining margins:

i. Spread between crude oil input and refined product output (gasoline, heating oil)
ii. $3:2:1$ Crack Spread: 3 barrels crude vs. 2 barrels gasoline + 1 barrel heating oil (most common)
iii. High crack spread: refiners profitable, incentive to increase production
iv. Low/negative crack spread: margins squeezed, refiners slow production
v. Factors: Cushing supply, driving demand, refinery shutdowns, hurricanes

Conversion: gasoline/heating oil in cents/gallon, crude in dollars/barrel. 1 barrel = 42 gallons.

Remark 2.2.61. *Natural Gas*
Natural gas is the cleanest fossil fuel (30–50% less CO2 than gasoline/coal):

i. US and Russia account for 38% of global production
ii. Shale revolution: Horizontal drilling and hydraulic fracturing transformed US from importer to exporter
iii. Flaring: Burning unwanted natural gas, common in US due to regulatory requirements

iv. LNG: Liquefied at $-260°$F for transport, 600x less volume than gas state

v. Uses: Power generation, heating, fertiliser (ammonia), transportation

Natural gas futures (NG) trade on NYMEX. Contract: 10,000 MMBtu, delivery at Henry Hub, Louisiana. Quoted in \$/MMBtu. ETF: UNG (subject to contango drag).

Remark 2.2.62. *Gold*
Gold is both precious metal and monetary asset. Currency symbol: XAU.

i. Jewelry: 40%+ of demand. Investment: second largest demand driver

ii. Central banks hold 31,920 tonnes. US largest holder (8,134 tonnes, 73%of reserves)

iii. China overtook South Africa as largest producer

iv. India largest consumer (50% used for weddings)

v. Real rates: Gold inversely correlated with real interest rates. Negative real rates support gold.

Trading: GLD ETF (1/10 oz per share), gold futures (GC) on COMEX (100 troy oz contract).

Remark 2.2.63. *Silver*
Silver has both monetary and industrial properties. Currency symbol: XAG.

i. Highest electrical conductivity of all metals

ii. Industrial applications: 44% of demand (batteries, electronics, solar panels)

iii. Jewelry/silverware: 22% of demand

iv. Mexico and China account for 47% of global production

v. More volatile than gold; often trades as leveraged gold play

Trading: SLV ETF (1 oz per share), silver futures (SI) on COMEX (5,000 troy oz). "Triple 9s" (99.9% purity) is deliverable grade.

Remark 2.2.64. *Platinum*
Platinum is extremely rare with concentrated supply:

i. South Africa holds 80% of global reserves

ii. Autocatalyst (catalytic converters): 38% of demand

iii. Jewelry: 31% of demand

iv. Almost twice as heavy as gold (density 21 vs. 19)

v. Play on emerging market growth and automotive demand

Platinum futures (PL) trade on COMEX (50 troy oz). Platinum/gold ratio is a key relative value indicator.

Remark 2.2.65. *Copper*
Copper is considered a leading economic indicator:

i. Chile dominates production (27%); China dominates consumption (43%), stockpiles are closely watched

ii. Uses: Building construction (33%), infrastructure (15%), equipment manufacturing (52%)

iii. Major input for urbanising economies, plays into EM growth story

iv. Recycled copper ("secondary copper") represents significant market share

Trading: LME (LMCADS03, \$/ton, 25 tons/contract) or NYMEX (HG, cents/lb, 25,000 lbs/contract).

Remark 2.2.66. *Aluminium*
Aluminium is the most abundant metal in Earth's crust:

i. China accounts for 40%+ of both production and consumption

ii. Lightweight, durable, does not corrode

iii. Uses: Construction, transportation (less energy needed to move lighter vehicles), packaging

iv. One of few commodities where China is self-sufficient

v. Energy-intensive to produce (aluminium smelters locate near cheap power)

Trading: LME (LMAHDS03, \$/ton, 25 tons/contract).

Remark 2.2.67. *Agricultural Commodities Overview*
Key characteristics of agricultural trading:

i. Supply determined by weather, planting decisions, yields
ii. Demand relatively inelastic (food consumption stable)
iii. US is largest player in most agricultural products
iv. Best agriculture traders employ full-time meteorologists
v. Crop calendars define sensitive periods for each commodity
vi. Planting, flowering/silking, and harvest are critical stages

Remark 2.2.68. *Corn*
Corn is the largest cereal crop globally:

i. US accounts for 35%+ of production, 30%+ of consumption, largest exporter
ii. Western Corn Belt (Iowa, Illinois, Nebraska) produces 47% of US corn
iii. Uses: Livestock feed, ethanol (27% of US demand, growing), food
iv. Ethanol mandate: US Energy Policy Act 2005 requires ethanol blending, creating structural demand
v. China demand growth: feedstock demand from 0% (2007) to 5%+ of imports

Trading: CBOT (C), cents/bushel, 5,000 bushels/contract. 1 bushel = 56 lbs.
Delivery months: H, K, N, U, Z. Critical period: June–August (silking).

Remark 2.2.69. *Wheat*
Wheat has been cultivated for $10,000+$ years:

i. EU, China, India are largest producers; US is fourth largest but largest exporter
ii. Classifications: season (winter/spring), gluten content (hard/soft), colour (red/white/amber)
iii. Uses: Bread, flour, noodles, beer, feedstock
iv. Close substitute for corn. Prices move together
v. Russia export bans have caused supply shocks

Trading: CBOT (W), cents/bushel, $5,000$ bushels/contract. 1 bushel = 60 lbs.
Critical period: heading stage.

Remark 2.2.70. *Soybeans*
Soybeans are a critical oilseed with high protein content:

i. US largest producer (31%), Brazil second largest but largest exporter
ii. China largest consumer (crush) and importer (63% of global imports)
iii. Products: Soybean meal, vegetable oil, soy milk, tofu, edamame
iv. 90%+ of soybeans are genetically engineered (lower pesticide costs)
v. Brazil/Argentina weather critical, neighbouring countries, correlated weather risk

Trading: CBOT (S), cents/bushel, 5,000 bushels/contract. 1 bushel = 60 lbs. Most vulnerable during flowering.
Delivery months: F, H, K, N, Q, U, X.

Remark 2.2.71. *Cotton*
Cotton is the most important textile fiber:

i. China, India, US are largest producers (65% of production)
ii. China largest producer and consumer (textile industry)
iii. US third largest producer but largest exporter (30% of exports)
iv. Classifications: character (strength), grade (colour/purity), staple (fibre length)
v. More drought-resistant than other crops

Trading: ICE (CT), cents/lb, $50,000$ lbs/contract (≈ 104 bales). 1 bale = 480 lbs.
Delivery months: H, K, N, V, Z.

Remark 2.2.72. *Coffee*
Coffee has distinct supply characteristics:

i. Two types: Arabica (60% of production, higher grade, higher altitude) and Robusta (stronger taste, more caffeine, lower altitude)

ii. Brazil largest producer/exporter (37%). Vietnam second (mostly robusta)

iii. EU and US are largest importers (46% and 23% respectively)

iv. Biennial cycle: Coffee trees alternate high and low yield years

v. Tree life cycle approximately 20 years

Trading: ICE (KC), cents/lb of arabica, $37,500$ lbs/contract. Industry measures in bags (60 kg = 132 lbs). Critical period: blooming (water is critical).

Remark 2.2.73. *Ending Stocks and Supply/Demand Balance*
Ending stocks (inventory) are the most important driver of commodity prices:

i. Ending Stocks = Beginning Stocks + Production − Consumption

ii. Low stocks relative to consumption (stocks-to-use ratio) = price support

iii. High stocks = price pressure

iv. USDA WASDE reports provide monthly supply/demand estimates

v. Small changes in production or consumption can cause large price moves when stocks are tight

Remark 2.2.74. *Weather Risk in Agriculture*
Weather is the primary supply shock driver in agriculture:

i. Drought during critical growth stages (flowering, silking) is most damaging

ii. Livestock cascade: high grain prices → expensive feed → early livestock slaughter → short-term meat supply up, future supply down

iii. El Niño/La Niña cycles affect global weather patterns

iv. Monitor: NOAA drought monitors, crop condition reports, weather forecasts for key growing regions

Remark 2.2.75. *Commodities in Different Macro Regimes*

i. Growth Shock: Commodity demand rises, prices increase, especially energy and industrial metals

ii. Recession: Demand destruction, prices fall (except gold as safe haven)

iii. Inflation Shock: Commodities outperform as inflation hedge; energy and agriculture lead

iv. USD Weakness: Commodity prices rise (inverse correlation with dollar)

v. Geopolitical Crisis: Energy spikes on supply fears; gold rises on safe-haven demand

Commodities are the inflation and real-economy link in the macro framework.

2.3 Practitioner Frameworks

This section distills operational frameworks from leading global macro practitioners, drawn primarily from Steven Drobny's interviews ((Drobny, 2006)). Focus is on actionable principles rather than historical narratives.

2.3.1 Portfolio Construction and Risk Frameworks

Remark 2.3.1. *Systematic Risk Premia Allocation (Leitner Framework)*

i. Equal 20% risk allocation across: Equities (value screens), Fixed Income (5x leveraged short-duration), FX (carry + vol premia), Commodities (roll yield), Real Estate
ii. Additional 10% for 1–2 high-conviction opportunities annually
iii. Anchor to data first, then narratives; actively seek disconfirming evidence
iv. Not being invested = foregoing risk premia; short positions pay away premia

Remark 2.3.2. *Convexity and Optionality (Siva-Jothy/Anonymous Currency Manager)*

i. Maintain long gamma; never short gamma in macro portfolios
ii. 50–60% options (3–6 months); 75% of price moves occur 10–15% of time around data releases
iii. Systematically long options to capture regime shifts; leverage is function of correct macro analysis
iv. Never forced seller; liquidity-only mandate (G10 only)
v. Timing: buy options when consensus says inevitable and market at extreme

Remark 2.3.3. *Trade Construction Discipline (Drobny/Wadhwani)*

i. Views without actionable trades are useless; compare fundamentals to market perception
ii. Look for positioning extremes (2–5 year timeframes); CA deficit currency rising = longs building
iii. Minimum risk/reward $4:1$; after two stop-outs on same thesis, move on
iv. 3–4 major opportunities annually; overtrading as damaging as running losers
v. Positioning and sentiment dominate fundamentals short-term
vi. Stop-loss most critical; exit immediately when thesis invalidated

Remark 2.3.4. *Fixed Income Specific (Porter)*

i. Time horizon arbitrage: institutional short-termism creates medium-term opportunity
ii. Focus on one-year, one-year forwards (most volatile, greatest overshooting)
iii. Interest rate cycles 16–24 months; maximise income per VaR via front-end
iv. Curve steepener $\approx$ 90% bullish bonds; flattener $\approx$ bearish

2.3.2 Asset-Class Specific Frameworks

Remark 2.3.5. *Commodity Framework (Anderson)*

i. Pure microeconomics: supply/demand, cost curves, competitive positioning
ii. Fundamentals over price: ignore day-to-day moves, focus on economics
iii. Mean reversion: margins revert to mean, not prices
iv. Contrarian timing: front-page hysteria = reversal indicator (sell euphoria, buy panic)
v. Micro-economic inevitability: stress-test thesis against adverse macro scenarios. Long = assume low growth; short = assume high growth
vi. On-the-ground research: observe anecdotal evidence, see fundamental change before markets price it
vii. Only invest low-cost producers; high-cost assets can't be bought cheap enough
viii. Reject VaR for concentrated portfolios (assumes diversification); use volatility-adjusted sizing

Remark 2.3.6. *Equity Macro Framework (Bessent)*

i. Express macro views through equities: 50% of stock move = market, 30% = sector, 20% = stock-specific
ii. Bottom-up research for macro confirmation: micro insights reveal macro picture
iii. Entry points critical: good entry means rarely stopped out

iv. Concentrated portfolio: 8–14 large positions, not equally weighted
v. Paradigm shifts = major profits: identify when game changes, not how to play game better

Remark 2.3.7. *FX Framework (Anonymous Currency Manager)*

i. FX is tail of credit curve: Fed funds → short rates → govts → credit → equities → FX
ii. Currencies reflect relative sentiment/credit spreads between economies
iii. Three currency blocs: USD, EUR, Asia
iv. Directional concentration: 3–4 macro calls maximum (bonds, equities, USD vs blocs)
v. Five-month time horizon for thesis validation
vi. Basic law: bullish = long or flat, bearish = short or flat. Never relative value

Remark 2.3.8. *Emerging Markets Crisis Framework (Dimitrijević)*

i. Country implosion acts like company collapse: all instruments correlate
ii. Strict limits: 10% global macro fund, 15% EM fund per country
iii. Best EM trades after panics/disasters
iv. Clean sheet valuation: ignore yesterday's price. Ask: "would we invest at current price with clean sheet?"
v. Stop-loss mechanics: work in continuous markets, dangerous in distressed/value plays
vi. Internal stops only (counterparties invariably hit external stops)
vii. One-month cool-off after stop-out
viii. Hedge mismatch: short rich asset class as hedge to long attractive asset class within same country

2.3.3 Execution and Flow Analysis

Remark 2.3.9. *Flow and Cross-Market Linkages (Harris)*

i. Flow matrix: mental model tracking global money flows seeking highest risk-adjusted return
ii. Money craves stability; fear greater than greed drives capital flight
iii. Cross-market linkages: geopolitical events ripple through asset classes (e.g., Brazilian real devaluation → sell soybeans as farmers hedge USD revenues)
iv. Slower Fool Theory: exit before discontinuous leap into illiquidity
v. Risk management: 80% trades wrong, 20% profitable must exceed losses
vi. Small probes (reconnaissance) test thesis before scaling
vii. Never pray trades back to breakeven; clear books, start fresh
viii. Best at 3–4 concentrated positions; avoid overtrading

Remark 2.3.10. *Contrarian Indicators*

i. Concept exhaustion: magazine covers = fully discounted
ii. Gestalt shifts: volatility spikes when protection buying signals crowded positioning
iii. "Frozen ice theory": strong trades at edges, crowded in middle on proxies
iv. Central banks often counter-indicators in market timing

3 Market Microstructure

Market microstructure studies how trading rules, intermediaries, and order flow transform latent trading demands into transactions, prices, and allocations. The central departure from frictionless asset pricing is that the trading mechanism is not neutral: priority rules, transparency, inventory, information, and immediacy affect the prices at which trades occur.

3.1 Foundations of Trading and Market Organization

The main references for this section are (O'Hara, 1995), (Harris, 2002), (Madhavan, 2000), (Hasbrouck, 2007), and (Lehalle and Laruelle, 2018). Demsetz's treatment of immediacy and transaction costs is the canonical starting point for why trading mechanisms matter (Demsetz, 1968).

3.1.1 Roles of Financial Markets

Principle 3.1.1. *Core Market Functions*
A financial market is useful when it performs four linked functions:

i. *liquidity provision*: converts desired portfolio changes into executable trades;
ii. *price discovery*: aggregates dispersed beliefs and information into observable prices;
iii. *risk transfer*: reallocates exposures to agents more willing or able to bear them;
iv. *capital allocation*: directs savings toward issuers and projects with attractive risk-adjusted uses of capital.

Microstructure studies the frictions inside these functions rather than assuming they occur automatically.

Principle 3.1.2. *Immediacy Has a Price*
Asynchronous order arrival makes immediacy costly. The bid-ask spread compensates traders who absorb inventory, funding, processing, and adverse-selection risks (Demsetz, 1968).

Definition 3.1.3. *Liquidity Provision*
Liquidity is the ability to trade size quickly, cheaply, and with limited price impact. Supplied by limit orders, dealers, market makers, internalisers, and crossing systems, and consumed by marketable orders and urgent traders.

Mechanism 3.1.4. *Price Discovery*
Trades, quotes, cancellations, and depth transmit information about value, urgency, and risk-bearing capacity. Price discovery is strong when prices update quickly without being dominated by transient imbalances.

Mechanism 3.1.5. *Risk Transfer*
Markets separate cash-flow ownership from risk-bearing. Hedgers shed unwanted exposure; speculators, dealers, and market makers accept it for compensation.

Principle 3.1.6. *Capital Allocation*
Primary markets move capital from savers to issuers. Liquid and informative secondary markets provide exit prices, lower financing costs, and discipline poor uses of capital.

Principle 3.1.7. *Mechanism Relevance* (O'Hara, 1995; Madhavan, 2000)
In Walrasian abstraction, the mechanism disappears after supply and demand clear. In real markets, sequencing, asymmetric information, costly immediacy, and priority rules make the mechanism part of price formation.

Caveat 3.1.8. *Microstructure Trade-Off*
Market design rarely improves every function at once. Transparency, spread size, venue opacity, and priority rules trade off liquidity provision, execution cost, information leakage, and public price discovery.

3.1.2 Market Participants and Trading Motives

Model 3.1.9. *Trading Game View* (O'Hara, 1995; Harris, 2002)
A trading mechanism is a game in which participants submit orders under venue rules. The players include customers, brokers, dealers, market makers, and traders acting for their own account. Some demand liquidity, some supply liquidity, some carry information, and some process orders.
Intuition: a trader's best order depends on what others may know, how fast they can react, and how the venue ranks competing messages.

Notation 3.1.10. *Long and Short Positions*
A trader with a *long position* owns the asset and benefits when its price rises. A trader with a *short position* has sold an asset not owned, or has equivalent negative exposure, and benefits when its price falls. Closing a short position is called *covering*.

Taxonomy 3.1.11. *Buy-Side Motives*
Buy-side traders use markets to solve problems outside the market itself:

i. investors move wealth across time;
ii. borrowers move wealth from the future to the present;
iii. hedgers reduce business or portfolio risk;
iv. asset exchangers acquire assets they value more than what they tender;
v. gamblers trade for entertainment or payoff convexity rather than investment efficiency.

These traders buy liquidity services rather than sell them.

Taxonomy 3.1.12. *Information-Motivated Traders*
Informed traders trade because they believe prices do not yet reflect information. Value traders use slow information about fundamentals; news traders use recent information; arbitrageurs use relative mispricing across related instruments or venues. Their activity can improve prices, but only after other traders infer or absorb the information in their orders.

Definition 3.1.13. *Noise and Liquidity Traders*
Noise traders trade for reasons not directly tied to fundamental value: portfolio rebalancing, cash needs, index changes, behavioural demand, or mistaken beliefs. They are important because they make informed trading less obvious and give liquidity suppliers order flow that is not always toxic.

Mechanism 3.1.14. *Dealers and Market Makers*
Dealers trade for their own account. Market makers quote prices or otherwise stand ready to trade, often holding inventory only until an offsetting order arrives. Their compensation comes from spreads, rebates, information, or client relationships; their risks are inventory, adverse selection, funding, and operational failure. They are paid for immediacy: the client trades now, while the market maker manages the uncertainty of unwinding later.

Taxonomy 3.1.15. *Sell-Side Intermediaries*
Brokers arrange trades for clients without intending to own the position. Dealers trade as principal. Broker-dealers combine both functions, sometimes arranging agency trades and sometimes committing their own capital. The sell side exists because buy-side traders pay for search, immediacy, execution expertise, clearing access, and balance-sheet intermediation.

Practitioner Note 3.1.16. *Institutional Investors and HFTs*
Institutional investors often trade large parent orders that must be hidden, sliced, or negotiated to reduce market impact. High-frequency traders use speed, automation, and short holding periods; they may act as market makers, arbitrageurs, statistical traders, or order anticipators. Their role depends on strategy.

Taxonomy 3.1.17. *HFT Identification Markers*

i. automated order generation, routing, execution, and cancellation;
ii. low-latency infrastructure, co-location, and direct market-data feeds;
iii. short holding periods and low overnight inventory;
iv. high message counts relative to executed trades;
v. market making, cross-venue arbitrage, or very short-horizon statistical trading.

These markers describe technology and horizon, not whether the strategy is beneficial.

Mechanism 3.1.18. *HFT as Fragmentation Link*
Fast market makers connect fragmented venues by quoting related prices, arbitraging stale quotes, and hedging inventory across books. Their compensation is the spread, rebates, and short-lived information advantages; their cost is technology, inventory risk, and adverse selection. They act like moving bridges between books: useful when they keep prices aligned, fragile when those bridges are cancelled under stress.

Caveat 3.1.19. *Tighter Spread, Different Cost*
An HFT can improve the displayed spread by stepping inside an older quote, but investor execution cost may not fall by the same amount. Smaller trades, more prints, and higher routing complexity can shift fixed and reporting costs toward slower investors.

Caveat 3.1.20. *Investor Size Asymmetry*
The uncertainty created by meeting a fast counterparty is not borne equally. A small investor sees high variance in a small number of trades; a large institution diversifies this uncertainty across many child orders but still pays aggregate spread, impact, and signalling costs.

Caveat 3.1.21. *Blurring of Roles* (Lehalle and Laruelle, 2018)
Fragmentation blurs the distinction between liquidity takers and providers. The same agent may consume liquidity in one venue, supply it in another, and intermediate price formation through routing speed.

3.1.3 Trading Venues and Market Architecture

Principle 3.1.22. *Venue as Rule System*
A market is a rule system, not a place. Its rules govern access, order submission, visibility, execution, clearing, and settlement.

Definition 3.1.23. *Exchanges*
Exchanges centralise order interaction under membership, listing, trading, reporting, and clearing rules. They standardise access and make execution more reliable.

Definition 3.1.24. *OTC Markets*
OTC markets use bilateral or dealer-mediated negotiation rather than a central book. They fit bonds, swaps, FX, bespoke derivatives, and heterogeneous contracts.

Taxonomy 3.1.25. *Electronic Communication Networks (ECNs) and Alternative Trading Systems (ATSs)*
ECNs organise electronic matching outside traditional exchange structures. ATSs include dark pools, crossing systems, and specialised venues competing on anonymity, fees, block handling, or lower information leakage.

Taxonomy 3.1.26. *Dark Pools, Internalizers, and Crossing Networks*
Dark pools hide pre-trade interest; internalizers match against broker-dealer inventory or internal flow; crossing networks match natural buyers and sellers, often at lit-market reference prices. These venues reduce signalling but can weaken public price discovery.

Taxonomy 3.1.27. *Request for Quotation (RFQ) vs Central Limit Order Book (CLOB)*
RFQ markets request quotes from dealers, often bilaterally. A CLOB exposes standing interest multilaterally; RFQ protects relationships and discretion, while CLOBs improve comparability and public interaction.

Caveat 3.1.28. *Stale Quotes and Vanishing Liquidity*
RFQ quotes can become stale before acceptance; displayed CLOB liquidity can vanish before an order arrives. In fragmented electronic markets, observed liquidity is an estimate, not a guarantee.

Process 3.1.29. *Post-Trade Infrastructure*
Trading does not end at execution:

i. *clearing* matches buyer and seller records and validates trade terms;
ii. *settlement* exchanges cash and securities;
iii. *custody* holds assets on behalf of clients;
iv. *depositories* immobilise or dematerialise securities so ownership can transfer by book entry.

Clearinghouses may become buyer to every seller and seller to every buyer, reducing counterparty risk through margin and default management.

Practitioner Note 3.1.30. *Architecture Choice*
Venue choice depends on order size, urgency, information sensitivity, instrument standardisation, and counterparty risk. Small urgent orders favour visible and immediate venues; large or information-sensitive orders often favour dark, negotiated, or sliced execution.

Mechanism 3.1.31. *Liquidity Bridges*
Fragmented venues require routing, arbitrage, and market-making links between books. These links reduce price dispersion, but their compensation is paid through spreads, fees, latency races, or information leakage.

Practitioner Note 3.1.32. *Straight-Through Processing*
Straight-through processing sends trade details from execution to clearing, settlement, custody, and reporting with minimal manual intervention.

3.1.4 Auction and Dealer Market Structures

Taxonomy 3.1.33. *Quote-Driven vs Order-Driven*
Quote-driven markets trade through dealer quotes and balance sheets. Order-driven markets trade by matching public orders under priority rules.

Mechanism 3.1.34. *Continuous Auctions*
Orders execute whenever they arrive and match resting interest. Continuous trading provides immediacy but exposes prices to asynchronous order-flow imbalances.

Mechanism 3.1.35. *Call Auctions*
Orders accumulate, then execute at a single clearing price. Opening and closing calls concentrate liquidity when uncertainty and imbalance are high.

Mechanism 3.1.36. *Fixing Auctions*
Fixing auctions are call auctions used to create reference prices.
They are valuable when investors need one public price for index valuation, NAV computation, derivative settlement, opening risk transfer, or end-of-day inventory adjustment.

Principle 3.1.37. *Reference Price Monopoly*
Even when continuous trading fragments, primary exchanges often retain the auction flow that defines official reference prices. This makes auction volume structurally different from continuous-session volume and gives the primary market a separate source of liquidity concentration.

Taxonomy 3.1.38. *Auction Session Segmentation*

i. *opening call*: rebuilds the book after overnight information and sets the first tradable price;
ii. *intraday call*: concentrates liquidity around scheduled events or low-liquidity periods;
iii. *closing call*: sets the end-of-day reference price and absorbs benchmark-driven flow;
iv. *volatility call*: pauses continuous trading when price moves or imbalances breach venue rules.

Mechanism 3.1.39. *Dealer Markets*
Prices form through dealer quotes rather than a single public book. Dealers absorb timing gaps, manage inventory, and revise quotes as information and inventory change.

Taxonomy 3.1.40. *Hybrid Markets*
Hybrid markets combine order books, dealers, auctions, hidden orders, and off-book negotiation. They mix public order interaction with intermediary support.

Rule 3.1.41. *Priority and Access Rules* (O'Hara, 1995; Madhavan, 2000)
Priority, tick size, order type, display, halt, and access rules determine how trades are resolved. The same latent supply and demand can produce different prices, quantities, and costs under different rules.

Case Study 3.1.42. *Gold Fixing as Sequential Auction*
The London gold fixing resembles a call process: participants submit net interest, revise after tentative prices, and stop when net demand and supply balance. It produces a clearing price, not continuous immediacy.

Practitioner Note 3.1.43. *Why Large Orders Like Calls*
Calls reduce the need to reveal urgency sequentially. A large order can interact with accumulated opposite interest at one price, often with less signalling than chasing liquidity through the continuous book.

Caveat 3.1.44. *Auction Transparency Trade-Off*
Pre-auction imbalance messages help traders converge to a clearing price, but they can also invite probing, cancellation, and last-second repricing. More transparency improves price discovery only when participants cannot cheaply game the signal.

3.1.5 Market Quality and Efficiency

Definition 3.1.45. *Market Quality*
Market quality measures how well trading interest becomes low-cost execution and informative prices. Include liquidity, tightness, depth, resiliency, transparency, volatility control, and informational efficiency.

Definition 3.1.46. *Liquidity*
Liquidity is the ability to trade quickly, in size, at low cost, and with limited price impact. Spreads, depth, volume, turnover, fill rates, and price impact each capture only one face of it.

Taxonomy 3.1.47. *Four Liquidity Variables*

i. *traded volume*: realised liquidity, where traders actually agreed to trade;

ii. *quoted quantity*: offered liquidity at the best limits or across the visible book;

iii. *bid-ask spread*: price of immediacy for small trades;

iv. *volatility*: uncertainty faced by liquidity providers and liquidity takers.

Past volume is not enough: a market can trade actively and still have poor offered liquidity for the next order.

Definition 3.1.48. *Liquidity Risk*
Liquidity risk is uncertainty about the cost of turning a desired position into an executed trade. It includes spread risk, depth risk, market-impact risk, timing risk, and the risk that displayed liquidity disappears.

Taxonomy 3.1.49. *Tightness, Depth, and Resiliency*
Tightness is cost of an immediate small round trip, proxied by bid-ask spread. Depth is the quantity available at or near current prices. Resiliency is the speed with which liquidity returns after a shock or large trade.

Definition 3.1.50. *Book Size*
Book size is the quantity available at the best bid and ask, or within a specified number of price levels.
For small-tick instruments, several limits must often be summed; for large-tick instruments, the first limits may capture most executable size.

Caveat 3.1.51. *Book Size Is a Lower Bound*
Displayed book size understates tradable quantity when hidden, reserve, midpoint, or fast-replenishing liquidity exists. It also overstates certainty when posted quantities are short-lived.

Mechanism 3.1.52. *Market Impact and Decay*
Market impact is the price movement caused by trading pressure. Temporary impact decays when other traders disagree with the move; permanent impact remains when the order flow reveals information.

Principle 3.1.53. *Resiliency as Synchronisation*
Liquidity is resilient when opposing trading interests arrive before a price move becomes self-reinforcing. Poor synchronisation turns temporary imbalances into persistent price pressure (Lehalle and Laruelle, 2018).

Principle 3.1.54. *Informational Efficiency*
Informational efficiency means prices incorporate available information quickly and with limited noise. Visibility, update speed, and uninformed flow determine how much information enters prices.

Mechanism 3.1.55. *Bid-Ask Bounce*
Transaction prices alternate between bid and ask even when the efficient price is stable.
A simple bounce statistic compares alternations with continuations: if N^a is the number of price alternations and N^c the number of same-direction continuations, then $\eta = \frac{N^c}{2N^a}$.
Lower η means alternations dominate continuations, so more observed price movement is microstructure noise rather than information.

Caveat 3.1.56. *Transparency Paradox*
More shared information can improve price formation, but each trader prefers to reveal as little as possible before execution. Market design balances public price discovery against private information leakage.

Caveat 3.1.57. *Latency and Book State*
When order books update faster than an observer can receive data, decide, and route an order, the displayed book is stale. Available liquidity becomes probabilistic rather than deterministic.

Definition 3.1.58. *Nyquist-Shannon Latency Bound* (Lehalle and Laruelle, 2018)
For an order-book update frequency f, the latency bound is $\tau^\star = \frac{1}{2f}$. If round-trip observation and routing time exceeds this scale, deterministic use of the displayed book is no longer reliable.

Caveat 3.1.59. *Quality Trade-Offs*
Market quality criteria conflict. Displayed depth, tick size, dark trading, and transparency shift costs across liquidity provision, queue priority, information leakage, and price discovery.

Empirical Finding 3.1.60. *Intraday Liquidity Seasonality*
Volume, volatility, spreads, and book size are time-of-day conditional. Opens tend to have high volume, high volatility, and wide spreads; the middle of the day is closer to steady state; the close attracts benchmark flow, higher volume, and often larger best-limit size.

Principle 3.1.61. *Spread and Depth Substitute Imperfectly*
When the spread cannot narrow further because the tick binds, liquidity providers compete by increasing size at the best bid and ask. A larger book can then play a role similar to a smaller spread, but only for orders small enough to fit inside displayed depth.

Mechanism 3.1.62. *Spread-Volatility Link*
Market makers ask for wider spreads when volatility rises because adverse selection and inventory risk increase. The relation is cleaner after controlling for tick size, trading frequency, and volatility per trade.

Definition 3.1.63. *Flow Reactivity*
Flow reactivity measures how strongly trade counts rise with high-frequency volatility. A resilient venue keeps market makers active when uncertainty rises; a fragile venue loses displayed liquidity when it is most valuable.

Empirical Finding 3.1.64. *Volume Is Not Directly Comparable*
Reported volume depends on architecture. A direct match may count once, while dealer intermediation may count multiple legs, so cross-venue comparisons need the counting convention.

Caveat 3.1.65. *Visible Activity vs Economic Importance*
Visible markets are not always the largest exposures. Stocks attract attention, while real estate, bonds, currencies, swaps, and futures may dominate capital value, risk transfer, or notional activity.

3.1.6 Regulatory and Structural Evolution

Principle 3.1.66. *Regulation as Market Design* (Harris, 2002)
Regulation is market design by law. Good rules support communication, performance, investor protection, competition, and access; bad rules protect incumbents or raise trading costs.

Rule 3.1.67. *Decimalization*
Decimalization reduced U.S. equity tick sizes from fractions to cents. Smaller ticks narrowed quoted spreads but reduced displayed depth and queue stability.

Rule 3.1.68. *Reg NMS*
Reg NMS links U.S. equity venues through protected quotes and routing obligations. It protects displayed top-of-book prices, while depth, fees, latency, and hidden liquidity still differ across venues.

Rule 3.1.69. *MiFID and MiFID II* (Lehalle and Laruelle, 2018)
MiFID broke exchange monopolies by allowing trading outside historical exchanges; MiFID II tightened best-execution, transparency, and reporting duties. The combined effect was more competition, more venue types, and greater reliance on routing, reporting, and post-trade data quality.

Principle 3.1.70. *Best Execution After Fragmentation*
Best execution is not only the best displayed price. In fragmented markets it also depends on fees, likelihood of execution, latency, hidden liquidity, venue access, order size, and the investor's horizon.

Mechanism 3.1.71. *Fragmentation*
Fragmentation means the same instrument trades across multiple venues. It can lower fees and specialise execution, but splits liquidity and complicates best execution.

Taxonomy 3.1.72. *Levels of Fragmentation*

i. *operators*: new market operators compete with incumbents, then often merge as fixed costs bite;
ii. *venues*: one operator may run lit books, dark books, periodic auctions, and internalisation channels;
iii. *order books*: one venue may maintain several books with different access, transparency, or priority rules;
iv. *orders*: large parent orders are split through time and across compatible venues;
v. *technology*: protocols, market data feeds, and native gateways become part of venue competition.

Mechanism 3.1.73. *Structural Evolution of Fragmentation*
Post-MiFID fragmentation first reduced the monopoly share of historical exchanges, then consolidated around large pan-European operators. Complexity did not disappear: liquidity was segmented into lit books, dark pools, broker crossing systems, systematic internalisers, and specialised order books.

Taxonomy 3.1.74. *Post-MiFID Phases*

i. *entry*: new MTFs competed for order flow, often with bank sponsorship or imported technology;
ii. *convergence*: successful venues moved toward pan-European offers and common functionality;

iii. *dark segmentation*: broker crossing networks, dark MTFs grew around first-hand liquidity, lower signalling;

iv. *consolidation*: fixed technology, data, and connectivity costs pushed operators toward mergers.

Caveat 3.1.75. *Early MiFID Failures*
Turquoise and NASDAQ OMX Europe show that technology and sponsorship were not sufficient. Turquoise gained relevance only after its liquidity agreement ended and it became part of the LSE group; NASDAQ OMX's European MTF and router never reached durable scale.

Caveat 3.1.76. *Liquidity Segmentation*
Venue competition separates natural investor liquidity from opportunistic market-making liquidity. This can improve matching for specific order types, but the trader now pays for aggregation through routing logic, data, connectivity, and information leakage.

Taxonomy 3.1.77. *Technology Fragmentation*

i. *application protocols*: venue APIs and market-data formats such as ITCH, XDP, EMDI, or RWF;

ii. *network protocols*: multicast UDP, TCP, replay channels, and venue-specific recovery logic;

iii. *compression protocols*: FAST, ZLIB, or uncompressed feeds.

Even when trading rules converge, access remains fragmented through protocols, feeds, and operational detail.

Rule 3.1.78. *Transparency and Dark Trading*
Pre-trade transparency reveals resting interest; post-trade transparency reveals completed transactions. Dark trading reduces information leakage, so regulators often pair it with caps, waivers, or stronger reporting.

Mechanism 3.1.79. *Broker Crossing Networks*
Broker crossing networks internalise client flow before exposing it to lit markets.
They can protect large investors from signalling, but they also raise conflict questions when broker prop flow, market-making flow, and client flow meet in the same pool.

Rule 3.1.80. *MiFID II Dark Caps*
MiFID II constrained dark trading through double volume caps: a venue-level cap and an aggregate market-level cap. The intent is to preserve price formation on lit venues while still allowing limited dark execution.

Taxonomy 3.1.81. *MiFID Venue Statuses*
European transactions may occur on regulated markets, multilateral trading facilities, systematic internalisers, or OTC channels. Regulated markets and MTFs use multilateral rule books; systematic internalisers trade bilaterally against client flow; OTC trading keeps negotiation bilateral but still faces reporting obligations.

Rule 3.1.82. *Tick Size Reforms*
Tick size rules balance price competition against queue discipline. Large ticks widen spreads but support displayed queues; small ticks narrow spreads but encourage queue jumping, cancellation, smaller displayed size.

Caveat 3.1.83. *Tick Size as Market Share Tool*
In fragmented markets, venues can use smaller ticks to undercut incumbent quotes and attract flow. This may improve displayed prices in the short run, but can also start a race to economically meaningless price increments.

Rule 3.1.84. *Sub-Penny and Tick-Size Wars*
Sub-penny rules restrict displayed price improvement, but off-exchange venues can still internalise flow at tiny increments. In Europe, early post-MiFID venues used different tick tables to compete until harmonisation reduced tick-size competition.

Taxonomy 3.1.85. *Who Gains From Tick Changes*

i. *liquidity takers*: may gain from narrower spreads on small trades;

ii. *liquidity providers*: lose queue rents when stepping ahead becomes cheap;

iii. *large investors*: may face less displayed depth and higher search or slicing costs;

iv. *venues*: can gain share by offering finer price grids or specialised order types.

Principle 3.1.86. *Tick Policy Design*
Tick tables should vary with price and liquidity, stay stable enough for adaptation, and be revised using spreads, depth, order-to-trade ratios, and standard-size execution costs.

Rule 3.1.87. *Circuit Breakers and Structural Controls*
Halts, limit moves, short-sale constraints, margin rules, and position limits do not remove volatility. They change when and how traders express demand, supply, and leverage.

Taxonomy 3.1.88. *Circuit Breaker Scope*

i. *local*: halt one instrument or one venue;

ii. *related-instrument*: halt derivatives, ETFs, or baskets tied to the halted instrument;

iii. *market-wide*: halt a broad market after index-level stress;

iv. *dynamic bands*: reject or pause trades outside a price corridor;

v. *static thresholds*: trigger from a fixed reference price or previous close.

Principle 3.1.89. *Circuit Breaker Design*
Circuit breakers should stop disorderly liquidity demand, not ordinary price discovery. Useful triggers combine price movement with liquidity, volatility, correlation, or related-instrument stress.

Caveat 3.1.90. *Propagation Problem*
Halting one instrument can push pressure into substitutes. If ETFs, futures, options, or highly correlated names remain open, the halt may move price discovery elsewhere instead of calming it.

Rule 3.1.91. *Post-Flash-Crash Controls*
After the 2010 Flash Crash, U.S. market structure moved toward single-stock pauses, limit-up/limit-down bands, consolidated audit trail reporting, stub-quote restrictions, and tighter controls on naked market access.

Caveat 3.1.92. *Stub Quotes*
Stub quotes satisfy a formal two-sided quoting duty at prices far from the market. They are not real liquidity; executions against them reveal missing protection in the trading mechanism.

Principle 3.1.93. *Consolidated Tape as Resilience*
Helps fragmented markets retain a shared post-trade reference. Without it, stress or outages can leave participants unsure whether price discovery is happening in one market or several disconnected ones.

Taxonomy 3.1.94. *Regulatory Institutions*
Market oversight is shared across:

i. governmental regulators, such as securities and futures commissions;

ii. self-regulatory organisations, such as exchanges, clearinghouses, and trader associations;

iii. private standard setters, such as accounting, performance-reporting, and professional bodies;

iv. international organisations, such as IOSCO and the World Federation of Exchanges.

This division creates expertise, jurisdictional overlap, and regulatory competition.

Caveat 3.1.95. *Unintended Consequences*
Rules change incentives. Taxes may move liquidity abroad, bans may move risk into substitutes, protected quotes may increase routing complexity, and exchange rules may protect incumbents.

Mechanism 3.1.96. *Competition to Merger*
Opening trading to competition first produces innovative venues and lower explicit costs. Once practices converge, scale becomes decisive: mergers reduce fixed technology costs, expand connectivity, and preserve distinct order books under larger operating groups.

Practitioner Note 3.1.97. *Electronification Beyond Equities*
Electronification does not imply one architecture. Equities move to lit books and dark pools; fixed income and derivatives retain RFQ, dealer networks, and central clearing because contracts and counterparty risks differ.

3.2 Orders, Matching, and Limit Order Books

3.2.1 Order Types and Instructions

Definition 3.2.1. *Order* (Harris, 2002)
An order is a trade instruction. At minimum it identifies the instrument, side, and quantity; practical orders also specify price limits, activation rules, validity, display, routing, settlement, or broker discretion.

Taxonomy 3.2.2. *Order-Instruction Segmentation*

i. *economic terms*: instrument, side, quantity, and price constraint;
ii. *timing*: submission time, activation trigger, validity, and expiry;
iii. *execution constraint*: immediate execution, full execution, minimum size, or partial-fill permission;
iv. *display*: fully displayed, hidden, reserve, or iceberg quantity;
v. *handling*: held by strict rules, or not-held with broker discretion;
vi. *post-trade terms*: settlement, clearing, and account capacity.

The label of an order is therefore only a shorthand for a bundle of instructions.

Taxonomy 3.2.3. *Core Order-Type Segmentation*

i. *market*: pay immediacy and consume the best available liquidity;
ii. *limit*: specify a worst acceptable price and wait in the book if not marketable;
iii. *stop*: become active only after a trigger price is reached;
iv. *market-if-touched*: become active when price moves favourably to the touch price;
v. *IOC/FOK*: cancel any unfilled quantity immediately, or require full immediate execution;
vi. *pegged*: reference the midpoint, best bid, best offer, or another venue-derived price;
vii. *hidden/iceberg*: reduce displayed size while keeping some executable interest undisplayed.

Principle 3.2.4. *Order Choice Trades Certainty for Leakage*
Aggressive orders buy execution certainty but reveal urgency. Passive and hidden orders reduce signalling, but they introduce queue risk, missed fills, and adverse selection.

Mechanism 3.2.5. *Market Order Cost Segmentation*

i. *spread cost*: a small market buy normally pays the ask and a small market sell receives the bid;
ii. *price improvement*: negotiation, midpoint interaction, or discretionary handling may beat displayed quote;
iii. *market impact*: large orders walk the book and move prices to attract the other side;
iv. *execution-price uncertainty*: the quote can change between submission and execution.

The market order solves execution uncertainty by accepting price uncertainty.

Definition 3.2.6. *Marketable Limit Order*
A marketable limit order is priced aggressively enough to trade immediately if available liquidity is still present. It is like a market order with a price cap for buys or a price floor for sells; unfilled residual quantity either rests or cancels according to its other instructions.

Taxonomy 3.2.7. *Conditional Activation Segmentation*

i. *stop order*: activates after adverse price move; buy stops trigger after price rises, sell stops after price falls;
ii. *stop-limit order*: activates after the stop, but can execute only within the specified limit price;
iii. *market-if-touched order*: activates after a favourable price move, then behaves like a market order.

Stops tend to demand liquidity in the direction of the move; market-if-touched orders are more contrarian.

Caveat 3.2.8. *Stops Are Not Limits*
A stop price controls activation, not execution price. A stop-limit adds a limit price after activation, but then loses execution certainty when the market gaps through the limit.

Rule 3.2.9. *Tick-Sensitive Orders*
Tick-sensitive orders condition execution on the direction of the last price change. A buy downtick order waits for a downtick or zero downtick; a sell uptick order waits for an uptick or zero uptick. These instructions are mainly useful when tick sizes are large enough that one tick is economically meaningful.

Taxonomy 3.2.10. *Time-in-Force Segmentation*

i. *day*: valid until the session close;
ii. *good-till-cancel*: valid until the trader or broker cancels it;
iii. *good-till-date*: valid until a specified date or time;
iv. *IOC*: execute immediately and cancel the unfilled balance;
v. *FOK*: execute the full quantity immediately or cancel;
vi. *market-on-open/close*: execute in the opening or closing auction.

Validity instructions decide when an order can become stale enough to disappear.

Taxonomy 3.2.11. *Quantity and Handling Segmentation*

i. *all-or-none*: the whole quantity must execute;
ii. *minimum acceptable quantity*: each fill must meet a specified minimum size;
iii. *display size*: only part of the order is shown;
iv. *not-held*: the broker may use discretion over timing, price, and exposure.

These instructions trade fill probability against information leakage and accountability.

Taxonomy 3.2.12. *Special Instruction Segmentation*

i. *spread orders*: trade two related instruments as one package, usually with a limit on the price difference;
ii. *substitution orders*: let the broker choose among acceptable substitutes;
iii. *special settlement*: request nonstandard settlement, often around record dates or corporate actions.

Caveat 3.2.13. *Instruction Semantics Are Venue-Specific*
The same label can behave differently across venues. Minimum quantity, display refresh, priority loss after modification, pegging reference, and cancellation rules must be read as part of the market design.

3.2.2 Order Lifecycle

Process 3.2.14. *Order State Segmentation*

i. *pending*: submitted but not yet accepted by the broker or venue;
ii. *accepted*: valid for handling, but not necessarily executable yet;
iii. *working*: live in the market or with a broker seeking execution;
iv. *part-filled*: some quantity executed, with residual quantity still governed by the order instructions;
v. *terminal*: filled, cancelled, expired, rejected, or killed by a contingency.

The economic order exists before the public book sees it, but the market only reacts once it is working.

Process 3.2.15. *Limit-Order Lifecycle*

i. *insert*: add displayed or hidden liquidity at a chosen price and size;
ii. *modify*: change price, quantity, instructions, or expiry. Loses priority when price worsens or size increases;
iii. *cancel*: remove resting quantity before execution;
iv. *execute*: trade when an incoming marketable order crosses the resting quote.

Taxonomy 3.2.16. *Message-Channel Segmentation*

i. *private channel*: order entry, acknowledgements, rejects, cancels, and fills between member and venue;
ii. *public channel*: trades, best quotes, depth updates, status messages shared with market-data subscribers.

Private messages control the order; public messages update the common view of the book.

Principle 3.2.17. *Absent Trader Representation*
Limit, stop, validity, and quantity instructions encode future trading contingencies. They let an absent trader participate, but stale contingencies can trigger at bad times.

Caveat 3.2.18. *Lifecycle Tags Change Priority*
Time-in-force, display quantity, hidden flags, minimum quantity, and cancel-on-disconnect rules can change whether an order rests, executes, loses priority, or vanishes under stress.

3.2.3 Limit Order Book Mechanics

Definition 3.2.19. *Best Quotes and Spread*
The *best bid* is the highest displayed buy price and the *best ask* is the lowest displayed sell price. The *bid-ask spread* is $a_t - b_t$, and the quoted midpoint is $m_t = (a_t + b_t)/2$.

Principle 3.2.20. *Liquidity Supply and Demand*
Resting limit orders supply immediacy to other traders; marketable orders demand it.
A standing quote is therefore both a price and a service: it lets someone else trade now.

Taxonomy 3.2.21. *Limit Price Placement Segmentation*

i. *marketable*: crosses the opposite best quote and can execute immediately;
ii. *at the market*: joins the current best bid or ask;
iii. *inside the market*: improves the best bid or ask and narrows the spread;
iv. *behind the market*: rests away from the best quote;
v. *far from the market*: carries low execution probability but low immediacy cost.

Aggressiveness raises fill probability and lowers waiting time, but it gives up more price concession.

Proposition 3.2.22. *Standing Limit Orders Are Trading Options*
Let q be order size, L the limit price, τ the cancellation or expiry time, and V_t the contra-side trader's value at time t. While live, a sell limit order gives the market a call-like payoff $q(V_t - L)^+$; a buy limit order gives the market a put-like payoff $q(L - V_t)^+$, where $x^+ = \max\{x, 0\}$.

Proof. For $t \le \tau$, an incoming buyer exercises a sell limit only when buying at L gives non-negative surplus, so the exercise value is $\max\{q(V_t - L), 0\}$. The buy-limit case is symmetric, with exercise value $\max\{q(L - V_t), 0\}$. □

Caveat 3.2.23. *Limit-Order Regret*
Limit orders reduce price concession but introduce two risks. The order may not execute and the trader must chase the price later; or it may execute exactly because better-informed flow is trading through it.

Definition 3.2.24. *Tick Size*
Tick size is the minimum price increment for limit orders.
It discretises prices and can be uniform, price-banded, or liquidity-banded.

Mechanism 3.2.25. *Binding Tick Constraint*
When the tick is large relative to the latent spread, the quoted spread is floored at one tick. Once the tick no longer binds, further reductions mostly change queue behaviour.

Model 3.2.26. *Harris Tick Discretisation Model*
Let $Ask_{j,t}$ and $Bid_{j,t}$ be the best ask and bid for stock j on trade t of day d, and let $\Psi_{j,t} = (Ask_{j,t} - Bid_{j,t})/tick_{j,d}$ be the observed spread in ticks. The model assumes an unobserved continuous spread $\widetilde{\Psi}_{j,t}$, then rounds it through a discretisation map $\Psi_{j,t} = \Phi(\widetilde{\Psi}_{j,t})$.
The tick is therefore not just a unit of account; it censors small spreads into one-tick quotes.

Definition 3.2.27. *Continuous Spread Expectation*
With turnover proxy T_j, volatility $\sigma_{j,d}$, price $P_{j,d}$, and tick size $tick_{j,d}$,
a compact specification is $\mathbb{E}[\widetilde{\Psi}_{j,t} \mid T_j, \sigma_{j,d}] = g(T_j, \sigma_{j,d})\frac{P_{j,d}}{tick_{j,d}}$, $g(T_j, \sigma_{j,d}) = \alpha T_j^{\beta} \sigma_{j,d}^{\gamma}$.
Here g is the latent relative spread, and α, β, and γ are fitted spread-response parameters.
Multiplying by $P/tick$ converts it into tick units.

Mechanism 3.2.28. *Tick-Size Counterfactual*
To estimate the effect of a smaller tick, keep the latent spread distribution fixed and change only the rounding grid. If δ is the tick-reduction factor and the new tick is $tick/\delta$, an observed k-tick spread becomes roughly k/δ old-tick units before re-rounding.
The largest expected spread decrease occurs when current spreads are near one tick.

Taxonomy 3.2.29. *Harris Model Inputs*

i. *liquidity*: turnover T_j, usually lowering latent spread;
ii. *risk*: volatility $\sigma_{j,d}$, usually raising latent spread;
iii. *price grid*: $P_{j,d}/tick_{j,d}$, controlling how strongly ticks discretise spreads;
iv. *distribution*: a chosen positive density for latent continuous spreads;

v. *fit check*: compare observed relative spreads with fitted discrete-spread expectations.

Principle 3.2.30. *Tick Size and Queue Discipline*
The tick defines how expensive it is to step ahead of an existing limit order.
Large ticks protect time priority by making queue jumping costly; small ticks make one-tick improvement cheap and weaken the value of being first in queue.

Mechanism 3.2.31. *Quoted Depth Under Small Ticks*
When the price grid becomes finer, liquidity providers can spread orders across more price levels. Best-limit depth often falls, order sizes shrink, and displayed liquidity must be refreshed more frequently.

Caveat 3.2.32. *Order-to-Trade Ratio*
A smaller tick can increase cancellations and modifications relative to trades. More messages do not necessarily mean more economic liquidity; they may simply reflect shorter quote life and more frequent repricing.

Mechanism 3.2.33. *Cumulative Depth*
To compare book quality across tick regimes, measure the cost of executing a fixed monetary size by rebuilding cumulative depth around the best bid and offer. Top-of-book spread alone ignores hidden liquidity, resiliency, and depth beyond the first level.

Principle 3.2.34. *CLOB as Price-Formation Device*
A central limit order book stores unmatched buying and selling interest until compatible prices meet. Each transaction therefore both transfers inventory and updates the public state of price formation.

Mechanism 3.2.35. *LOB Event Types*
Order-book dynamics are driven by three event families: insertions add liquidity, cancellations remove liquidity without trading, and marketable orders remove liquidity through transactions. Price changes occur when the best queue is depleted or a better price is inserted.

Taxonomy 3.2.36. *Tick-Size Quality Proxies*

i. *binding probability*: share of time or trades with spread equal to one tick;

ii. *spread leeway*: spread in ticks minus one;

iii. *depth*: displayed and cumulative size near the midpoint;

iv. *message intensity*: order-to-trade ratio, cancellations, and modifications.

Caveat 3.2.37. *Value of a Quote*
As ticks and displayed sizes shrink, quotes can become shorter-lived and less informative. A consolidated tape of trades cannot fully replace quote data if only the best limit is visible and deeper book information is lost.

Caveat 3.2.38. *RFQ Can Mimic a Book*
In RFQ markets, dealers answer bilateral requests rather than post firm multilateral quotes.
Aggregating several responses can look like a book, but the liquidity is conditional on the request, relationship, size, and information embedded in the enquiry.

3.2.4 Matching Algorithms

Principle 3.2.39. *Matching Is Market Design*
Matching rules decide which orders trade, at what price, and in what priority. Small rule differences can change displayed liquidity, queue value, auction participation, and the usefulness of a venue for large orders.

Mechanism 3.2.40. *Continuous Matching*
In continuous trading, an incoming marketable order executes immediately against resting interest. The trade-off is speed versus noise: prices update continuously, but liquidity can be thin when order flow is asynchronous.

Mechanism 3.2.41. *Call Matching*
In a call, orders are batched until an uncrossing time and matched at one auction price. The batch creates a thicker crossing point, making calls useful at openings, closings, expiries, and volatility interruptions.

Mechanism 3.2.42. *Frequent Batch Auctions*
Frequent batch auctions group arrivals into short calls. Price priority remains, but within-batch time priority is muted, reducing pure latency value.

Process 3.2.43. *Call Auction Price Rules*

i. *maximum executable volume*: choose prices that match the largest quantity;

ii. *minimum surplus*: among those prices, minimise unmatched quantity;

iii. *market pressure*: if surplus is only on the buy side choose the higher candidate, and if only on the sell side choose the lower candidate;

iv. *reference price*: if still tied, choose the candidate closest to the venue reference price.

If buy and sell interest do not cross, no auction price can be formed.

Definition 3.2.44. *Auction Surplus*
Auction surplus is the quantity left unmatched at a candidate clearing price. It is not excess volume in the day; it is the residual buy or sell pressure at the auction price.

Rule 3.2.45. *Randomised Uncrossing*
Some venues randomise the exact end of the call after the theoretical auction time. This weakens last-second order placement and cancellation, but it also makes the final book state less deterministic for participants.

Taxonomy 3.2.46. *Priority Segmentation*

i. *price priority*: better-priced orders execute first;

ii. *FIFO time priority*: earlier orders at the same price execute first;

iii. *pro-rata priority*: fills at one price are allocated partly by displayed size;

iv. *size priority*: larger orders receive explicit rank or allocation advantages;

v. *venue-specific priority*: hidden, displayed, broker, or auction-only flags may change rank.

Definition 3.2.47. *Theoretical Matching Curve*
During a call, the theoretical matching curve tracks how close the current indicative matched volume is to the final auction matched volume. It reveals whether the book builds smoothly, jumps at information times, or overshoots before orders are cancelled.

Caveat 3.2.48. *Indicative Volume Is Strategic*
Pre-auction indicative prices, volumes, and imbalances are signals, not commitments. A book can look deeper than the final auction if traders test information by submitting orders they later cancel or modify.

3.2.5 Queue Dynamics and Latency

Principle 3.2.49. *Condition on the Queue*
Queue dynamics are less random once the state of the book is specified. A cancellation at the best bid is not comparable when that queue has one lot, ten lots, or a much larger opposite queue.

Definition 3.2.50. *Queue Imbalance*
Queue imbalance compares available quantity on the best bid and ask. Predictive because a small best queue is easier to deplete, while a large opposite queue can make the next price move less likely to reverse immediately.

Mechanism 3.2.51. *Arrival-Departure Ratio*
For a queue, the arrival-departure ratio compares insertion intensity with cancellation plus transaction intensity. Ratios above one imply queue growth; ratios below one imply depletion.

Caveat 3.2.52. *Adverse Queue Selection*
A limit order is most likely to fill when marketable flow reaches it. That event is not neutral: fills are often more valuable to the taker precisely when the resting quote is stale, underpriced, or exposed to informed flow.

Process 3.2.53. *Queue-Reactive Estimation*

i. choose a starting time and record the queue state;

ii. wait for the next insertion, cancellation, or transaction;

iii. store the waiting time by event type and queue-size bucket;

iv. estimate state-dependent intensities from the average waiting times.

Caveat 3.2.54. *Mechanical Mean Reversion*
Queue-reactive simulations can generate too much price mean reversion: after a best queue disappears, the newly promoted queue may be large enough to pull prices back. Models often need memory, Hawkes-style excitation, or state resets after price moves.

Principle 3.2.55. *Latency Race Boundary*
Latency matters most when a trader can change queue position before others react. It matters less when the tick is wide, the queue is deep, or the order is deliberately not competing for first priority.

3.2.6 Hidden Liquidity and Dark Trading

Definition 3.2.56. *Dark Pool*
A dark pool is a trading venue where available liquidity is not displayed before execution. It is mainly used to reduce signalling, market impact, and exposure to opportunistic short-horizon traders.

Principle 3.2.57. *Reference Price Trading*
Dark pools usually do not form prices independently. Equity dark trades often execute at the midpoint or another reference price from a reliable lit market, so dark liquidity depends on lit price formation.

Taxonomy 3.2.58. *Dark Liquidity Pools*

i. *crossing networks*: match buy and sell orders, often at the primary midpoint;
ii. *internalisation pools*: brokers cross client flow or trade against broker inventory;
iii. *electronic market-makers*: operators or selected liquidity providers accept or reject incoming flow;
iv. *exchange dark books*: hidden order types inside regulated venues;
v. *dark aggregators*: route one order across several dark pools;
vi. *block pools*: focus on large institutional orders and natural liquidity.

Mechanism 3.2.59. *Integrated vs Midpoint Books*
Large hidden orders may interact with lit and hidden liquidity in integrated books. Smaller dark orders are typically crossed at the midpoint with basic limit, MAQ, and IOC controls.

Mechanism 3.2.60. *Reserve and Iceberg Display*
A reserve order displays $d < q$ and refreshes after the displayed slice trades. It hides total size, but repeated refreshes can reveal the parent and may reset priority.

Caveat 3.2.61. *Discovering Hidden Size*
Hidden size is learned by trading into it, not by observing the public queue. Large IOC or minimum-quantity probes can reveal reserve liquidity, so display instructions reduce but do not remove information leakage.

Caveat 3.2.62. *Dark Pools and Price Discovery*
Dark pools can reduce market impact, but they rely on lit quotes for reference prices. If too much flow moves dark, public price discovery may weaken; if dark flow is selective and low-impact, lit market quality can improve.

Empirical Finding 3.2.63. *Dark Activity Drivers*
Dark activity is usually concentrated in liquid stocks, high-volume days, volatile periods, and high-spread names. It tends to be lower when order imbalances are large relative to volume.

Mechanism 3.2.64. *Why Investors Use Dark Pools*
The main incentives are secrecy and price improvement. A midpoint execution lets a liquidity taker avoid paying the full spread while hiding the order from traders who might infer and trade against the parent order.

Caveat 3.2.65. *Dark Liquidity Is Conditional*
Dark liquidity is hard to estimate: misses are often invisible, and fills can still be adversely selected.

Caveat 3.2.66. *Toxic Liquidity and Gaming*
Dark pools can be gamed through probing, leakage, and adverse selection. Protections include minimum sizes, access controls, and filtering toxic flow.

Taxonomy 3.2.67. *Bilateral vs Multilateral Liquidity*
Multilateral books let many participants provide liquidity under common rules. Bilateral trading preserves selection and relationship control, which can help avoid toxic counterparties but may reduce open price discovery.

3.2.7 Cross-Venue Routing and Fragmentation

Principle 3.2.68. *Routing After Fragmentation*
In a fragmented market, liquidity is no longer represented by one order book. The execution problem becomes a joint choice of price, size, time, venue, fees, and information exposure.

Definition 3.2.69. *Smart Order Routing*
A smart order router (SOR) maps a trading instruction to venues according to displayed quotes, depth, historical fill rates, queue position, fees, latency, and order-type constraints. The execution algorithm usually chooses the parent schedule; the router chooses the spatial allocation of each child order.

Principle 3.2.70. *Temporal Then Spatial Split*
Optimal execution first chooses how fast to trade, balancing market impact against timing risk. In a fragmented market, each child order then needs a spatial split across venues, where the SOR tries to obtain at least the limit price and finish the child order quickly.

Process 3.2.71. *SOR Feedback Loop*

i. observe venue quotes, depths, fees, and recent fills;
ii. decide the venue split under the execution policy;
iii. send orders through venue gateways;
iv. receive execution reports and market-data updates;
v. update the next routing decision.

The decision loop is bounded by CPU time, network latency, matching-engine latency, and feed latency.

Model 3.2.72. *Toy SOR Allocation Model*
For K venues, let r_k be the fraction routed to venue k, $V(\omega)$ the child size, and $D_k(\omega)$ the executable liquidity under liquidity state ω. Captured liquidity is $C(r,\omega) = \sum_{k=1}^{K} \min\{r_k V(\omega), D_k(\omega)\}$, $\sum_{k=1}^{K} r_k = 1$.
An SOR learns $r = (r_1, \ldots, r_K)$ by increasing allocation to venues that fill reliably and reducing allocation to venues that disappoint. The min term is the key intuition: a router cannot capture more than it sends to a venue or more than the venue actually has available.

Mechanism 3.2.73. *SOR Share Feedback*
If $p_k(t)$ is venue k's market share and a is the fraction of participants using SORs, a simple share update is $p_k(t+1) = (1-a)p_k(t) + a r_k(t)$. SORs therefore turn routing weights into market share; venues with reliable liquidity gain flow, while noisy venues lose allocation slowly. When a is small, historical share anchors the system; when a is large, routing quality changes share quickly.

Taxonomy 3.2.74. *SOR Fragmentation Drivers*

i. *initial share*: incumbent venues start with the observed flow prior $p(0)$;
ii. *liquidity quality*: stable depth is easier for routers to learn and trust;
iii. *liquidity variance*: noisy fills push routers toward diversification;
iv. *SOR adoption*: higher a makes market share follow routing weights faster;
v. *reassessment speed*: faster learning accelerates convergence or instability.

Taxonomy 3.2.75. *Venue Selection Segmentation*

i. *price and depth*: displayed best quotes, hidden liquidity, and available queue size;
ii. *execution probability*: fill rate, cancellation rate, queue priority, and adverse selection;
iii. *economics*: commissions, access fees, maker-taker rebates, and clearing costs;
iv. *access*: latency, connectivity, order types, and venue-specific restrictions;
v. *information*: pre-trade visibility, post-trade reporting, and leakage from multi-venue signalling.

Definition 3.2.76. *Market Share as Routing Prior*
Market share is a rough prior for where trading occurs. For stock k and venue n, value share over $[t_1, t_2]$ is

$$M_k(n) = \left(\sum_{t_1 \le \tau_\ell^k \le t_2} P_\ell^k V_\ell^k \mathbf{1}_n(\delta_\ell^k) \right) / \left(\sum_{t_1 \le \tau_\ell^k \le t_2} P_\ell^k V_\ell^k \right)$$

Here $M_k(n)$ is stock k's value share on venue n, ℓ indexes trades, τ_ℓ^k is trade time, P_ℓ^k is price, V_ℓ^k is size, δ_ℓ^k is venue, and $\mathbf{1}_n(\delta_\ell^k)$ equals 1 if trade ℓ is on venue n and 0 otherwise.
Trade-count share replaces traded value by $\mathbf{1}_n(\delta_\ell^k)$ and divides by the number of trades. Both measures should separate continuous trading from fixing auctions.

Model 3.2.77. *Venue-Share Estimator*
For asset k and venue i, let T_k^i be turnover on venue i inside a fixed segment.
The estimated venue share is $\hat{p}_k^i = \frac{T_k^i}{\sum_j T_k^j}$. The same ratio can be built from trade count, displayed depth, or executable volume, but each choice answers a different routing question.

Model 3.2.78. *Index-Level Venue Share*
For an index universe $\mathcal{K}$ with weights w_k, aggregate asset-level shares by $\hat{p}^i_{\mathcal{K}} = \sum_{k \in \mathcal{K}} w_k \hat{p}^i_k$.

Here $\mathcal{K}$ is the index constituent set, w_k is asset k's index weight, $\hat{p}^i_k$ is asset k's estimated share on venue i, and $\hat{p}^i_{\mathcal{K}}$ is the index-level share on venue i. Index weights measure benchmark exposure; turnover weights measure where actual trading activity is concentrated.

Practitioner Note 3.2.79. *Estimator Comparison*
To compare an asset with its universe, use $d^i_k = \hat{p}^i_k - \hat{p}^i_{\mathcal{K}}$. Here d^i_k is the excess venue share of asset k on venue i relative to the chosen universe. If asset k is included in the benchmark, the two estimators are mechanically correlated; a leave-one-out universe gives a cleaner comparison.

Caveat 3.2.80. *Metric Choice*
The correct fragmentation metric depends on the question. Use all-in market share for venue revenue; use continuous-session trade share for the probability that an order executes on a venue; use value share when the unit of interest is currency traded.

Taxonomy 3.2.81. *Market Share Segmentation*

i. *continuous ex-fixing*: best for venue choice and execution probability;
ii. *opening, closing, and intraday calls*: best for reference-price and auction-liquidity questions;
iii. *traded value*: best for economic exposure and venue revenue;
iv. *trade count*: best for fill-frequency and small-order routing questions;
v. *stock or sector buckets*: needed because liquidity and capitalisation affect fragmentation.

Combining these segments can make a primary venue look more competitive because it owns the fixing auctions.

Empirical Finding 3.2.82. *Intraday Venue-Share Shape*
Primary markets often regain share near the open and close because auctions, reference-price demand, and early order-book formation concentrate flow. MTF shares can be flatter, rising, or inverted depending on SOR adoption, fee agreements, and the presence of professional liquidity providers.

Caveat 3.2.83. *Market Share Is Conditional*
Intraday market share is not a stable venue attribute. It changes with time of day, expiry windows, benchmark prices, news, and whether investors must trade on the venue that defines the relevant reference price.

Definition 3.2.84. *Fragmentation Efficiency Index*
Let q_n be the share of trading or liquidity observed in venue n, with $\sum_{n=1}^{N} q_n = 1$.
The Fragmentation Efficiency Index (FEI) is normalised entropy: $\mathfrak{F}(q_1, \ldots, q_N) = \frac{-\sum_{n=1}^{N} q_n \log q_n}{\log N}$.
Here N is the number of venues and $\mathfrak{F}$ is the dispersion score for the share vector q.
$\mathfrak{F} = 0$ when all flow is in one venue and $\mathfrak{F} = 1$ when flow is evenly split across venues.

Taxonomy 3.2.85. *Fragmentation Measurement Segmentation*

i. *trade-count FEI*: probability that the next print occurs away from the primary venue;
ii. *value-share FEI*: economic dispersion of traded notional;
iii. *displayed-liquidity FEI*: dispersion of visible book quantity, usually by price level;
iv. *session FEI*: separate continuous trading from auctions and benchmark windows;
v. *universe FEI*: compare large caps, small caps, sectors, or index constituents separately.

The same formula answers different questions only after the underlying share definition is fixed.

Caveat 3.2.86. *FEI Venue Count*
FEI depends on the considered venue set. If t is the measurement date and N is the venue count, adding an empty venue mechanically lowers the index from $\mathfrak{F}(t)$ to $\frac{\log N}{\log(N+1)}\mathfrak{F}(t)$. Compare FEI through time only after controlling for venue openings, closures, and data coverage.

Caveat 3.2.87. *Fragmentation of What*
Market-share and entropy metrics describe how traded value or trades are dispersed. They do not by themselves measure protocol costs, connectivity costs, hidden liquidity, or the difficulty of accessing each pool.

Mechanism 3.2.88. *Aggregated Liquidity*
Consolidating visible depth across venues can reduce impact. A market order that would walk the book on one venue may execute at the same or better price when routed across several books. The benefit comes from replacing one steep book with several shallower local walks.

Caveat 3.2.89. *Intraday Non-Stationarity*
Venue share and traded volume are time-of-day conditional.
Opens, closes, lunch periods, linked-market openings, scheduled news, and expiry dates can create repeatable volume spikes, so a single daily average is a weak routing prior.

Taxonomy 3.2.90. *Volume-Curve Segmentation*

i. *session*: open, continuous core, lunch break, close, and after-hours;
ii. *event*: macro releases, earnings, index rebalances, and derivative expiries;
iii. *reference price*: auction close, VWAP window, snapshot median, or settlement formula;
iv. *calendar*: weekday, month-end, daylight-saving shifts, holidays, and local trading hours;
v. *venue*: primary auctions, lit continuous books, dark pools, and internalised flow.

Caveat 3.2.91. *Average Trade Size*
Falling average trade size does not mechanically mean smaller investor orders.
Fragmentation can split one aggressive order across several passive orders and venues, so trade prints reveal book mechanics as well as investor size.

Empirical Finding 3.2.92. *Liquidity and Capitalisation Segmentation*
Across stocks, fragmentation tends to increase with liquidity and market capitalisation.
For a given stock, high-volume or stressed days can reduce fragmentation as flow returns to the historical venue for depth, certainty, or familiar trading processes.

Mechanism 3.2.93. *Why Large Caps Fragment First*
HFTs and liquidity arbitrageurs warehouse large-cap risk more easily: stock borrow is deeper, index or futures hedges are cleaner, inventory can be offset in correlated instruments. Small caps therefore catch up more slowly.

Rule 3.2.94. *Maker-Taker Pricing*
Maker-taker venues charge liquidity takers and pay rebates to liquidity makers. The displayed price can therefore differ from the economic price after fees and rebates.

Caveat 3.2.95. *Smart Routers vs Smart Fee Savers*
Routing cannot ignore fee schedules. Clearing and settlement may be netted by member inventory, while venue fees can depend on monthly turnover, cross-venue volume, or maker-taker status. The marginal cost of one child order is therefore path-dependent.

Mechanism 3.2.96. *Rebate Effects*
Rebates attract passive flow and can deepen displayed queues, but they also change routing incentives. A broker or router must compare net execution quality, not only the visible quote or the nominal likelihood of a fill. The economic best venue is the one with the best expected all-in fill, not necessarily the best displayed price.

Caveat 3.2.97. *Subsidised Share*
Maker agreements and fee holidays can create market share before a venue has durable natural flow. When the subsidy ends, share can fall quickly unless the venue has become part of traders' routing logic.

Caveat 3.2.98. *Post-Trade Barriers*
Fragmentation also depends on clearing, settlement, membership, and registration rules. A venue may fail to gain market share if post-trade access makes trading operationally expensive.

Case Study 3.2.99. *Spain as Fragmentation Control Case*
Spain's delayed MTF entry after MiFID makes it useful for separating fragmentation effects from broader European market stress. When alternative venues could not clear and settle efficiently, primary-market dominance persisted even while fragmentation advanced elsewhere.

Mechanism 3.2.100. *Event-Driven Volume Clustering*
Volume clusters when many traders need to act at the same reference time.
Linked market openings import information; scheduled macro releases create common reassessment points; derivative expiries force cash-future convergence and settlement trades.

Caveat 3.2.101. *Reference Rules Create Volume*
If a benchmark is computed from a close, auction price, or short snapshot window, traders with benchmark risk are pulled toward that window. Volume then reflects the rule as much as investor preference.

Caveat 3.2.102. *Duplicate Liquidity*
The same HFT market maker may quote on several venues and cancel remaining quotes once one venue trades. Aggregated displayed depth may overstate executable liquidity unless router discounts correlated cancellations.

Mechanism 3.2.103. *Spread-Share Feedback*
Best-execution and SOR logic move flow toward venues with better net spreads and reliable fills.
More flow attracts market makers, which can tighten spreads further. The same loop can run backward when spreads deteriorate or fills become volatile. This is why venue quality can become self-reinforcing after a venue crosses a minimum flow threshold.

Taxonomy 3.2.104. *Venue Quality Segmentation*

i. *tightness*: quoted and effective bid-ask spread;

ii. *depth*: displayed size and replenishment near the best limits;

iii. *resilience*: continuity of liquidity during volatile intervals;

iv. *coverage*: how evenly liquidity is supplied across index constituents;

v. *access*: members, connectivity, latency, order types, and data quality.

Definition 3.2.105. *Coverage Liquidity Metric*
Coverage asks how many stocks are needed to explain a target share of index turnover, commonly 80%.
Low coverage means liquidity is concentrated in a few names; high coverage means liquidity is more balanced across constituents.

Definition 3.2.106. *Gini Liquidity Imbalance*
A turnover Gini index summarises inequality in trading activity across stocks. A value near zero means turnover is evenly distributed; a value near one means a small subset of stocks dominates index liquidity.

Model 3.2.107. *Lorenz-Curve Gini*
Let $L(u)$ be the Lorenz curve for venue share, turnover, or actor activity after observations are ordered from smallest to largest contribution. The Gini coefficient is $G = 1 - 2\int_0^1 L(u)\,du$.
Here G is the Gini coefficient, u is the cumulative fraction of observations, and $L(u)$ is the cumulative fraction of the measured quantity. Equal contribution gives $G = 0$; complete concentration gives G close to 1.

Taxonomy 3.2.108. *Gini Measurement Segmentation*

i. *venue share*: concentration of trading across venues for one asset or universe;

ii. *asset turnover*: concentration of liquidity across index constituents;

iii. *participant activity*: concentration of messages, trades, or liquidity supply across actors;

iv. *time segment*: separate open, continuous session, close, and event windows.

The same Gini value is meaningful only after fixing the unit being ranked.

Model 3.2.109. *Kolmogorov-Smirnov Distribution Test*
For observations $X_1, \ldots, X_n$, the empirical distribution is $F_n(x) = \frac{1}{n}\sum_{k=1}^{n} \mathbf{1}_{\{X_k \le x\}}$.
For two samples with empirical distributions F_n^X and F_m^Y, two-sample statistic is $D_{n,m} = \sup_x \left|F_n^X(x) - F_m^Y(x)\right|$.
Here X_k is observation k, n and m are sample sizes, F_n^X and F_m^Y are empirical distribution functions, and $D_{n,m}$ is largest vertical gap. Reject equal distributions at α when $\sqrt{\frac{nm}{n+m}}D_{n,m} >$ Kolmogorov critical value.

Practitioner Note 3.2.110. *Venue-Share Distribution Comparison*
Use the KS test to compare cross-sectional venue-share distributions across venues or dates. A high Gini says trading is concentrated; a rejected KS test says the whole distribution differs.

Mechanism 3.2.111. *HFT Perimeter Extension*
Fast market makers usually enter liquid large caps first because hedging, borrowing, and inventory recycling are easier. As competition compresses returns, they extend toward less liquid names, improving coverage until hedging cost and inventory risk become uneconomic.

Caveat 3.2.112. *Primary Outage Dependency*
Many alternative venues peg orders, route decisions, or price controls to primary-market quotes. In a primary outage, those may pause alternative trading or create separate prices on venues that continue to trade.

Principle 3.2.113. *Fragmentation Needs Shared References*
Fragmented markets work best when participants share timely quotes, trades, and reference prices. Competition adds liquidity only when the information layer keeps venues synchronised.

Caveat 3.2.114. *Transparency Is Not Certainty*
Pre-trade transparency reveals quotes, not guaranteed liquidity.
Post-trade transparency gives more reliable prints after the fact, so small dark pools with weak pre-trade disclosure may still be informative through execution records.

Practitioner Note 3.2.115. *Fragmented Routing Objective*
A practical router ranks venues by expected net cost:
expected cost = spread/impact + fees − rebates + latency risk + information cost.
Patient orders can seek rebates and hidden liquidity; urgent orders pay for immediacy and certainty.

Empirical Finding 3.2.116. *Primary-Market Variance Feedback*
If an alternative venue delivers volatile fills, rational routers reduce future allocation to it.
Lower variance makes routing fractions easier to estimate; higher variance pushes flow back to the primary market and can lower the alternative venue's share.

3.3 Information, Price Formation, and Spreads

3.3.1 Information and Market Efficiency

3.3.2 Price Discovery

Definition 3.3.1. *Price Formation Process (PFP)*
PFP is the route by which order flow, information, and trading rules become transaction prices. Cleaner PFP links prices more closely to fundamentals; noisy PFP lets microstructure frictions dominate short-horizon prices.

Principle 3.3.2. *Microstructure to Macro Prices*
Intraday frictions can affect daily risk when they change the closing price, the auction price, or the path used by benchmark investors. The relevant question is not whether intraday noise exists, but whether it survives into the reference prices used outside the trading day.

Taxonomy 3.3.3. *Volatility Scale Segmentation*

i. *intraday volatility*: uncertainty during trading session, (liquidity, spread, and order-flow shocks);
ii. *overnight volatility*: uncertainty between close and next open (information arrival and auction reopening);
iii. *extraday volatility*: close-to-close risk, mixing intraday and overnight components.

Definition 3.3.4. *Intraday and Overnight Returns*
For day t, let O_t be the open price and C_t the close price.
Then $r_t^{\text{intra}} = \log(C_t) - \log(O_t)$ and $r_t^{\text{overnight}} = \log(O_t) - \log(C_{t-1})$.
Pairwise correlations can then be computed separately on intraday and overnight return series.

Caveat 3.3.5. *Correlation Interpretation*
Intraday correlations reflect co-movement during continuous price formation.
Overnight correlations are closer to auction and news revaluation. Spikes in daily index correlation hence come from different mechanisms at different horizons.

Mechanism 3.3.6. *Fragmentation and Price Formation*
Fragmentation can improve prices when venues remain synchronised by routing, arbitrage, and consolidated information. It can damage price formation when venues lose a common reference, when primary quotes become stale, or when market participants pause routing during stress.

Mechanism 3.3.7. *Fast Liquidity and Price Discovery*
Fast market makers help price discovery when they update quotes and arbitrage stale venues. They hurt it when speed investment mainly earns queue priority or fee rebates without adding durable liquidity. Fee schedules therefore affect the price formation process, not only venue revenue.

Caveat 3.3.8. *Outage Price Discovery*
If alternative venues keep trading while primary data or primary matching is unavailable, prices may temporarily split into parallel formation processes rather than one consolidated market price.

3.3.3 Bid-Ask Spread Decomposition

3.3.4 Inventory Models

3.3.5 Information-Based Models

Model 3.3.9. *Kyle Linear Price Impact*
Kyle's one-period model separates informed flow from noise flow. The informed trader observes terminal value $p_\infty \sim N(P^\star, \sigma_p^2)$; noise traders submit $n_\omega \sim N(0, \sigma_n^2)$. The market maker observes total flow $y = Q(p_\infty) + n_\omega$ and sets a linear price $f_P(y) = \tilde{P} + \lambda y$.
Here Q is informed order size, $\tilde{P}$ is the unconditional price, and λ is price impact per unit of net flow. Noise flow is camouflage: the market maker reacts to total flow because it cannot observe which part is informed.

Model 3.3.10. *Kyle Linear Equilibrium*
In the linear equilibrium, $Q(p_\infty) = \frac{p_\infty - \tilde{P}}{2\lambda}$, $\tilde{P} = P^\star$, $\lambda = \frac{\sigma_p}{2\sigma_n}$. The coefficient λ is a signal-to-noise price: higher information variance σ_p raises impact; higher noise-trader variance σ_n hides informed flow and lowers impact.

Principle 3.3.11. *Adverse Selection Through Flow*
The market maker does not need to identify the informed trader. It protects itself by moving price with observed net flow, because aggressive flow is statistically informative.

3.3.6 Strategic Trading Models

3.3.7 Liquidity and Market Impact

Taxonomy 3.3.12. *Market Impact Lens Segmentation*

i. *theoretical*: strategic splitting and information inference;

ii. *reduced-form*: price response to signed volume, participation, volatility, and time;

iii. *practitioner*: cost of urgency, benchmark choice, and venue access.

Definition 3.3.13. *Temporary and Permanent Impact*
Temporary impact is the price concession needed to obtain immediacy and is expected to decay as liquidity replenishes. Permanent impact is the residual price shift after decay, but it may reflect information in the order rather than mechanical pressure.

Mechanism 3.3.14. *Propagator Intuition*
A trade has an impulse effect on price, and that effect decays through time. The observed price path is the sum of many signed trade impulses with different decay profiles. This separates immediate pressure from later relaxation: a print can move price now without requiring the full move to be permanent.

Caveat 3.3.15. *Permanent Impact Identification*
Calling a long-run price move permanent impact is risky unless market moves, sector moves, autocorrelated flow, and the signal that caused the trade have been separated.

3.3.8 Volatility and Liquidity Interaction

Principle 3.3.16. *Volatility Is a Liquidity Price*
Volatility raises inventory risk, adverse-selection risk, and the probability that displayed liquidity becomes stale. Liquidity providers respond through wider spreads, smaller displayed size, faster cancellations, or withdrawal.

Definition 3.3.17. *Garman-Klass Intraday Volatility*
For a fixed intraday interval, the Garman-Klass estimator uses open, high, low, and close prices.
Let O, H, L, and C denote the interval open, high, low, and close:

$$\sigma_{GK} = \sqrt{\frac{1}{2}\left(\log\frac{H}{L}\right)^2 - (2\log 2 - 1)\left(\log\frac{C}{O}\right)^2}.$$

It is useful for comparing intraday uncertainty across markets when transaction-level volatility is too noisy.

Model 3.3.18. *Spread-Volatility Regression*
A compact empirical check regresses spread on volatility: $\psi_i = a_0 + a_1\sigma_i + \epsilon_i$.
Here ψ_i is spread observation, σ_i is realised volatility, a_0 is baseline spread, a_1 is spread sensitivity to volatility, and ϵ_i is residual noise. $\hat{a}_1 > 0$ is consistent with liquidity providers charging more when uncertainty rises.

Model 3.3.19. *Log-Log Spread-Volatility Model*
If spread and volatility are skewed, use $\log\psi_i = a_0 + a_1\log\sigma_i + \epsilon_i$.
Returning to levels gives the multiplicative relation $\psi_i = A_0\sigma_i^{a_1}e^{\epsilon_i}$ with $A_0 = e^{a_0}$. If $\epsilon_i \sim N(0,\sigma_\epsilon^2)$, the unbiased level predictor is $\hat{\psi} = \hat{A}_0\sigma^{\hat{a}_1}e^{\hat{\sigma}_\epsilon^2/2}$.
Here A_0 is the level intercept, σ_ϵ^2 is residual variance, and hats denote estimates.

Caveat 3.3.20. *Spread-Volatility Identification*
A fitted relation is descriptive unless tick size, time of day, venue, depth, and trade intensity are controlled. In large-tick assets, volatility may move through depth rather than spread.

Mechanism 3.3.21. *Liquidity Spiral*
A large order can consume depth, widen spreads, and raise volatility. Higher volatility makes liquidity providers reduce size or cancel quotes, forcing the next order to trade through less liquidity and amplifying the move. The state variable is confidence in replenishment: once traders doubt the book will refill, everyone demands more price concession at the same time.

Case Study 3.3.22. *Flash Crash Mechanism*
The 2010 Flash Crash illustrates a liquidity spiral: a large automated futures sell program met thin liquidity, high-frequency traders recycled inventory quickly, aggressive flow migrated into cash equities, spreads widened, and some routers or venues paused.
The trigger was an order; the crash mechanism was fragile liquidity under linked-market stress.

Model 3.3.23. *Depth-Driven Flash Crash Toy Model*
On one side of the book, let F_t be the market-order arrival rate, $\bar{F}$ the normal refill rate, $r_t = F_t - \bar{F}$ the overflow, and $I_t = 1/D_t$ inverse depth. A compact depth-impact dynamic is $\frac{dI_t}{I_t} = \sigma_I dW_t + r_t dt$.
Here D_t is market depth, I_t is impact, σ_I is liquidity-shock volatility, and W_t captures random liquidity shocks. Positive overflow r_t makes inverse depth compound upward, so each additional trade has larger price impact than the last.

Mechanism 3.3.24. *Hot-Potato Amplification*
If a fraction ρ of traders follows volume mechanically, initial trading rate r_0 can amplify roughly as $r_t \simeq \frac{1-\rho^t}{1-\rho} r_0$. When ρ is high, overflow consumes depth faster than passive liquidity can refill it. The loop is mechanical: trading creates volume, volume targets force more trading, and depth absorbs the loop until it fails.

Taxonomy 3.3.25. *Flash-Crash Control Segmentation*

i. *rate controls*: cap participation rates or stop volume-following feedback;
ii. *depth controls*: require or incentivise fast refill near the touch;
iii. *price controls*: pause trading after large price moves;
iv. *cross-market controls*: coordinate futures, ETFs, cash equities, and options;
v. *data controls*: detect overflow, quote withdrawal, and broken references in real time.

Caveat 3.3.26. *Systemic Risk Link*
Fragmentation is not systemic risk by itself. It becomes dangerous when market participants react to stress in the same direction, when related venues desynchronise, or when liquidity withdrawal in one instrument propagates to hedges, ETFs, futures, and correlated stocks.

Taxonomy 3.3.27. *Shock Propagation Channels*

i. *cross-asset hedging*: futures, ETFs, options, and cash baskets transmit inventory pressure;
ii. *routing feedback*: routers chase fading liquidity or pause when fills become unreliable;
iii. *quote withdrawal*: market makers widen or cancel when volatility exceeds their risk budget;
iv. *reference-price dependency*: pegged and benchmark orders inherit stale or broken references;
v. *correlated constraints*: margin, risk limits, and kill switches cause simultaneous de-risking.

Empirical Finding 3.3.28. *Trade Count as HFT Proxy*
Rising trades per day can proxy for higher high-frequency participation when trade sizes shrink and message intensity rises. It is not sufficient alone: the same count can reflect genuine liquidity, excessive slicing, or stress-induced churn.

3.4 Order Flow and Empirical Microstructure

3.4.1 Trade Classification and Signed Flow

3.4.2 Order Flow Imbalance

3.4.3 Stylized Facts of High-Frequency Data

Taxonomy 3.4.1. *High-Frequency Stylized Facts*

i. *heavy tails*: short-horizon returns have more extremes than Gaussian returns;

ii. *volatility clustering*: large moves tend to arrive near other large moves;

iii. *long memory*: signs, durations, and volatility can remain persistent;

iv. *intraday seasonality*: volume, spread, volatility, and depth depend strongly on clock time.

These facts are data conditions, not trading signals by themselves.

Model 3.4.2. *Signature Plot*
Let S_t be price, $X_t = \log S_t$ log-price, and Δ the sampling step.
The realised variance estimator is $\hat{V}_R(\Delta) = \sum_{j=1}^{T/\Delta} r_\Delta(j)^2$, where $r_\Delta(j) = X_{j\Delta} - X_{(j-1)\Delta}$. Here T is the horizon and $r_\Delta(j)$ is the jth log return. In frictionless continuous time, $\hat{V}_R(\Delta)$ converges to integrated variance as $\Delta \downarrow 0$; in real data, bid-ask bounce and quote noise make the estimator rise at very fine sampling.

Mechanism 3.4.3. *Microstructure Noise in Volatility*
Observed log-price can be written $X_t = M_t + \epsilon_t$, where M_t is the efficient semimartingale price and ϵ_t is observation noise. At high frequency, returns include both efficient-price changes and noise differences, so realised variance can diverge as sampling becomes too fine.

Definition 3.4.4. *Epps Effect*
The Epps effect is the fall in estimated cross-asset correlation as the sampling interval becomes very short. It is driven by asynchronous trades, lead-lag relations, tick effects, and other microstructure frictions.

Model 3.4.5. *Realised Covariance*
For two log-price processes X^1 and X^2, realised covariance at step Δ is $\hat{C}_R(\Delta) = \sum_{j=1}^{T/\Delta} r^1_\Delta(j) r^2_\Delta(j)$.
Here $r^a_\Delta(j)$ is the jth return of asset a. The estimator needs synchronised observations; refresh-time or previous-tick sampling is usually required before interpreting high-frequency correlations.

Model 3.4.6. *Time-Series Decomposition*
Microstructure time series can be decomposed into trend, seasonality, and residual variation: $X_t = C_t + S_t + \epsilon_t$ or $X_t = C_t S_t + \epsilon_t$. Here X_t is the observed series, C_t is the trend, S_t is the seasonal component, ϵ_t is unexplained noise, and t indexes equally spaced observations.

Definition 3.4.7. *Centred Moving Average*
For odd window length $2k+1$, the centred moving average is $MA_t = \frac{X_{t-k}+\cdots+X_t+\cdots+X_{t+k}}{2k+1}$. For even window length $2k$, use half weight at the endpoints: $A_t = \frac{0.5X_{t-k}+X_{t-k+1}+\cdots+X_{t+k-1}+0.5X_{t+k}}{2k}$.
Here MA_t is the local trend estimate and k controls the smoothing width. The window should match the seasonal period, or an integer multiple of it, when estimating seasonality.

Model 3.4.8. *Additive Deseasonalisation*
For period index i and within-period slot j, write $X_{ij} = C_{ij} + S_j + \epsilon_{ij}$.
If MA_{ij} estimates C_{ij}, then $\hat{S}_j = \frac{1}{n}\sum_{i=1}^{n}(X_{ij} - MA_{ij})$, $S_j = \hat{S}_j - \frac{1}{p}\sum_{k=1}^{p}\hat{S}_k$. Here p is the number of slots per period, n is the number of periods, and S_j is centred so the seasonal effects sum to zero.

Model 3.4.9. *Multiplicative Deseasonalisation*
For proportional seasonality, write $X_{ij} = C_{ij}S_j + \epsilon_{ij}$ and estimate $\hat{S}_j = \frac{1}{n}\sum_{i=1}^{n}\frac{X_{ij}}{MA_{ij}}$, $S_j = \hat{S}_j / \left(\frac{1}{p}\sum_{k=1}^{p}\hat{S}_k\right)$.
Here S_j is normalised to average one. This is natural for volume curves because activity is often a relative multiplier of the day's baseline.

Model 3.4.10. *Volume Seasonality Factorisation*
A compact volume model is $V(y,m,d,n,\omega) = \{f(n)+g(d,n)\}F(m)G(y)v(n,\omega)$, where $f(n)$ is the usual intraday profile, year y, month m, day d, time slot n, and event state ω, $g(d,n)$ captures special-day effects, $F(m)$ is monthly seasonality, $G(y)$ is yearly scale, and $v(n,\omega)$ is residual slot-level variation.

Practitioner Note 3.4.11. *Seasonality Before Signal*
Remove predictable clock-time effects before comparing venues, estimating impact, or training intraday features. Otherwise a noon lull, close auction, expiry day, or macro window can look like alpha.

3.4.4 Price Impact Estimation

Definition 3.4.12. *Execution-Time Impact*
Execution-time impact is the price move observed while the order is being traded, measured from the perspective of the order side and often normalised by spread or volatility.

Empirical Finding 3.4.13. *Impact Has a Hump Shape*
Intraday impact usually rises during execution and partly decays afterwards. The remaining shift is the candidate permanent component; the decayed part is temporary impact and liquidity replenishment.

Taxonomy 3.4.14. *Impact Driver Segmentation*

i. *stock characteristics*: liquidity, volatility, spread, tick size, and typical depth;
ii. *order characteristics*: size, duration, participation rate, side, and aggressiveness;
iii. *timing*: trend, rebalancing pressure, signal arrival, auctions, and market-wide events;
iv. *venue access*: lit depth, dark liquidity, queue priority, and routing constraints.

Model 3.4.15. *Participation-Duration Shape*
A compact empirical specification writes market impact as increasing in duration and participation:

$$\text{Market Impact} \propto \text{Duration}^{\alpha} \times \text{Participation}^{\gamma}.$$

Here α is the duration exponent, γ is the participation exponent. The exponents are calibrated from executions; in many datasets the participation exponent is concave, so extra participation hurts, but less than linearly. The model is not structural; it is a compact way to express the practitioner trade-off between urgency and footprint.

Caveat 3.4.16. *Timing Is Not Noise*
Two orders with identical size and participation can have different impact because one trades with the market and the other trades against it. Impact estimation therefore has to separate execution mechanics from the signal that caused the trade.

3.4.5 Market Impact Models

Taxonomy 3.4.17. *Model Family Segmentation*

i. *Almgren-Chriss*: optimise a schedule by balancing expected impact cost against execution risk;
ii. *linear impact*: use tractable instantaneous or temporary cost proportional to trading rate or signed size;
iii. *propagator*: model each signed trade as a decaying impulse in future prices;
iv. *empirical square-root*: summarise metaorder cost as concave in order size relative to market volume.

Principle 3.4.18. *Permanent Impact or Price Anticipation*
Long-horizon impact mixes metaorder pressure and information revealed by trading. Remove the information effect before calling the residual permanent impact.

Taxonomy 3.4.19. *Permanent-Impact View Segmentation*

i. *mechanical view*: prices move because buying and selling pressure are imbalanced;
ii. *informational view*: prices move because the trade reveals new expectations about value;
iii. *hybrid view*: executions contain both pressure and information, so identification matters more than labels.

Model 3.4.20. *Propagator Midprice Dynamics*
In trade time, a standard propagator writes the midprice as $m_k = m_{-\infty} + \sum_{k'<k} G(k-k')\epsilon_{k'} + \eta_k$.
Here m_k is the midprice after trade k, $\epsilon_k \in \{-1, +1\}$ is trade sign, $G(\cdot)$ is the decaying impact kernel, η_k is fair-price noise, and $m_{-\infty}$ is an initial price level.

Mechanism 3.4.21. *Trade-Sign Memory and Kernel Decay*
Trade signs are positively autocorrelated because large orders are split and similar agents trade together.
A decaying kernel offsets this persistence: impact must decay enough that predictable order flow does not imply predictable price drift.

Model 3.4.22. *Price Response Curve*
For horizon h, define the response curve by $\mathcal{R}(h) = \mathbb{E}[(m_{k+h} - m_k)\epsilon_k]$.
Here $\mathcal{R}(h)$ measures the expected signed price move after a trade. It is estimated with the trade-sign autocorrelation $C(\ell) = \mathbb{E}[\epsilon_{k+\ell}\epsilon_k]$ and the kernel increments $\Delta G(\ell) = G(\ell+1) - G(\ell)$.

Taxonomy 3.4.23. *Propagator Kernel Segmentation*

i. *non-parametric*: estimate discrete kernel increments directly from response and sign autocorrelation;

ii. *exponential*: impose fast decay with a small number of parameters;

iii. *power law*: allow slow decay and long-memory order flow;

iv. *agent-specific*: estimate separate kernels or intensities by participant class.

Process 3.4.24. *Debiasing Long-Horizon Impact*

i. classify trades by whether they are likely information-driven or liquidity-driven;

ii. remove market and sector moves with a benchmark or factor model;

iii. adjust for autocorrelated metaorder flow after the execution day;

iv. measure the remaining post-execution drift over several days.

Model 3.4.25. *CAPM-Like Post-Execution Decomposition*
For event day D, a stock's post-execution move can be split into systematic and idiosyncratic parts:

$$\log(P_d) - \log(P_{d-1}) = \beta\,[\log(I_d) - \log(I_{d-1})] + \Delta W_d.$$

Here P_d is the stock close, I_d the reference index, β the index exposure, and ΔW_d the residual stock-specific move. Cumulative ΔW_d isolates the profile after removing index co-movement. The intuition is attribution: do not charge the execution for a market move that the whole index experienced.

Caveat 3.4.26. *Autocorrelated Flow Bias*
Metaorder flow can remain signed and autocorrelated for days. If later same-direction executions are ignored, their impact may be falsely attributed to the original order.

3.4.6 Trade Duration and Event Time

3.4.7 Statistical Models for Order Flow

Definition 3.4.27. *Point-Process View*
Order-book events can be modelled as point processes: random arrival times for insertions, cancellations, and transactions. The intensity λ is the expected event rate over a short interval.

Definition 3.4.28. *Counting Process and Duration*
For event times $(\tau_i)_{i\geq 1}$, the counting process and duration process are $N_t = \sum_{i\geq 1} \mathbf{1}_{\{\tau_i \leq t\}}$, $\Delta\tau_i = \tau_i - \tau_{i-1}$. Here N_t counts events up to time t, τ_i is the ith event time, and $\Delta\tau_i$ is the waiting time between consecutive events.

Definition 3.4.29. *Conditional Intensity*
For filtration $\mathcal{F}_t$, the event intensity is $\lambda(t \mid \mathcal{F}_t) = \lim_{h\downarrow 0} \frac{1}{h}\mathbb{P}(N_{t+h} - N_t > 0 \mid \mathcal{F}_t)$. Here $\mathcal{F}_t$ is the information available before t. The intensity is the instantaneous event rate conditional on that information.

Taxonomy 3.4.30. *Order-Book Event Segmentation*

i. λ^+: limit-order insertions that add liquidity;

ii. λ^c: cancellations that remove resting liquidity;

iii. λ^x: transactions that consume resting liquidity.

Model 3.4.31. *Homogeneous Poisson Baseline*
The simplest model treats λ^+, λ^c, and λ^x as constant and independent of the book state. It is a useful null model, but it cannot reproduce realistic queue-size distributions because event rates are strongly state-dependent. Its value is diagnostic: any predictive model should beat this memoryless arrival clock.

Definition 3.4.32. *Integrated Intensity*
The integrated intensity over an inter-event interval is $\Lambda(\tau_{i-1}, \tau_i) = \int_{\tau_{i-1}}^{\tau_i} \lambda(s)\,ds$. Here Λ measures event time in intensity units. Under the time-change view, a correctly specified point process becomes a unit-rate Poisson process after this clock transformation.

Model 3.4.33. *Linear Hawkes Process*
A one-dimensional Hawkes intensity is $\lambda(t) = \lambda_0(t) + \int_{-\infty}^{t} \nu(t-s)dN_s = \lambda_0(t) + \sum_{\tau_i<t} \nu(t-\tau_i)$. Here $\lambda_0(t)$ is baseline intensity, $\nu(\cdot)$ is the excitation kernel, and past events raise current intensity. Hawkes models fit clustered liquidity because trades or messages make near-future events more likely.

Model 3.4.34. *Exponential Hawkes Kernel*
With $\nu(u) = \alpha e^{-\beta u}\mathbf{1}_{\{u\geq 0\}}$ and constant λ_0, the intensity is $\lambda(t) = \lambda_0 + \sum_{\tau_i<t} \alpha e^{-\beta(t-\tau_i)}$. Here α is jump size, β is decay speed, and α/β is the branching ratio. Stationarity requires $\alpha/\beta < 1$, and then $\mathbb{E}[\lambda(t)] = \lambda_0/(1-\alpha/\beta)$.

Mechanism 3.4.35. *Conditioning Improves Queue Dynamics*
Queue-reactive models condition event intensities on current and neighbouring queue sizes. Small best queues are easier to consume; large queues can trigger nearby cancellations. The model becomes more realistic because the book state changes trader incentives before the next event arrives.

Taxonomy 3.4.36. *Conditioning Segmentation*

i. *unconditional*: event type only;
ii. *own queue*: event type and current queue size;
iii. *neighbour queues*: own queue plus adjacent bid and ask queues;
iv. *event time*: count events rather than seconds when clock time is less informative.

Taxonomy 3.4.37. *Marked Flow Segmentation*

i. *event type*: insertions, cancellations, market orders, and trades;
ii. *side*: buy-initiated and sell-initiated flow;
iii. *agent class*: broker, HFT, investor, dealer, or internaliser when identifiers exist;
iv. *venue*: lit, dark, auction, and internalised pools.

A marked Hawkes model lets one class of event excite or inhibit another, useful when liquidity arrives in clusters.

3.4.8 Feature Engineering for Microstructure Alpha

3.4.9 Microstructure Data Engineering

Taxonomy 3.4.38. *Market-Data Depth Segmentation*

i. *L1*: best bid, best ask, sizes at the touch, and last trade;
ii. *L2*: visible depth by price level, usually aggregated by side and venue;
iii. *L3*: order-level messages, allowing queue reconstruction and cancellation analysis;
iv. *trades*: prints with price, size, time, venue, and often trade condition codes.

The empirical question should choose the shallowest feed that identifies the event cleanly.

Definition 3.4.39. *Microstructure Proportion*
A microstructure proportion is the share of events satisfying a binary condition: trade at the bid, execution on a venue, hidden-liquidity fill, HFT-tagged message, or cancellation before queue execution.

Model 3.4.40. *Bernoulli Proportion Estimator*
For independent indicators $X_1, \ldots, X_n \in \{0, 1\}$ with success probability p,
$\hat{p} = \frac{1}{n}\sum_{\ell=1}^{n} X_\ell$, $\mathbb{E}[\hat{p}] = p$, $\mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}$. The usual large-sample interval is $\hat{p} \pm z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$.
Here X_ℓ is the indicator for event ℓ, n is the number of observations, p is the true success probability, $\hat{p}$ is the sample proportion, α is the error probability, and $z_{1-\alpha/2}$ is the corresponding standard-normal quantile.
For a 95% interval, $z_{1-\alpha/2} \approx 1.96$.

Caveat 3.4.41. *Normal Approximation Boundary*
The normal interval is fragile when the sample is small or the estimated proportion is close to zero or one. A practical check is $n\hat{p} > 5$ and $n(1-\hat{p}) > 5$; otherwise use an exact or adjusted binomial interval.

Taxonomy 3.4.42. *Sampling Unit Segmentation*

i. *trade count*: estimates the probability that the next print has a property;
ii. *traded value*: estimates where economic turnover is concentrated;
iii. *message count*: estimates quoting, cancellation, or latency behaviour;
iv. *time bucket*: estimates realised activity inside a comparable clock segment;
v. *security bucket*: estimates whether the result is driven by large caps, small caps, sectors, or venues.

Changing the sampling unit changes the estimand, even when the same symbol $\hat{p}$ is used.

Model 3.4.43. *Simple Linear Regression Baseline*
For observations (x_i, y_i), a simple linear regression writes $y_i = a_0 + a_1 x_i + \epsilon_i$, $1 \le i \le n$.
Here y_i is the dependent variable, x_i is the explanatory variable, a_0 is the intercept, a_1 is the slope, ϵ_i is the residual, and n is the sample size.
Ordinary least squares estimates $\hat{a}_1 = \left(\sum_i (x_i - \bar{x})(y_i - \bar{y})\right) / \left(\sum_i (x_i - \bar{x})^2\right)$, $\hat{a}_0 = \bar{y} - \hat{a}_1 \bar{x}$. Here $\bar{x}$ and $\bar{y}$ are sample means, and $\hat{a}_0, \hat{a}_1$ are the estimated intercept and slope.

Definition 3.4.44. *Regression Fit Quality*
For fitted values $\hat{y}_i$ and residuals $\hat{\epsilon}_i = y_i - \hat{y}_i$, $R^2 = 1 - \frac{\sum_i \hat{\epsilon}_i^2}{\sum_i (y_i - \bar{y})^2}$.
Here $\hat{y}_i$ is the fitted value, $\hat{\epsilon}_i$ is the residual, and R^2 is explained variability relative to total variability. It measures fit, not causality or trading usefulness.

Caveat 3.4.45. *Regression Is an Empirical Approximation*
Linearity is often a local summary for spread, volatility, turnover, and impact relations. Check residuals, non-stationarity, heteroskedasticity, and whether a log or event-time transformation gives a more stable relation.

Caveat 3.4.46. *Averaging Effect*
An average intraday profile is not a typical realised path. Averaging volatility, volume, or spread across days can hide event days, close auctions, and isolated liquidity shocks.

Model 3.4.47. *Average of Regressions vs Regression of Averages*
For an index $I_t = \sum_{k=1}^N w_k S_t^k$, spreads aggregate linearly as $\psi^I = \sum_{k=1}^N w_k \psi^k$, but volatility generally does not satisfy $\sigma^I = \sum_{k=1}^N w_k \sigma^k$. Here w_k is the weight of stock k, S_t^k its price, ψ^k its spread, and σ^k its volatility.
Regression coefficients estimated on index quantities therefore need not equal weighted averages of single-stock regression coefficients.

3.5 Market Making and Liquidity Provision

3.5.1 Economics of Market Making

3.5.2 Avellaneda-Stoikov Framework

3.5.3 Adverse Selection and Toxic Flow

3.5.4 Optimal Quoting and Inventory Control

3.5.5 Latency and Queue Competition

Principle 3.5.1. *Latency as Queue Option*
Speed buys queue priority. It is most valuable when queues are thin, stale, or information-sensitive.

Model 3.5.2. *HFMM and Investor Race Model*
Split participants into high-frequency market makers and longer-horizon investors.
A market maker earns spread surplus $s\Delta$ net of maker fee c_M; an investor earns value surplus Γ net of spread and taker fee c_T: $\pi_M = s\Delta - c_M$, $\pi_T = \Gamma - s\Delta - c_T$, $\bar{c} = c_M + c_T$.
Here s is spread in ticks, Δ is tick size, Γ is investor gross value gain, c_M, c_T are maker/taker fees, π_M, π_T are net per-trade gains, and $\bar{c}$ is venue revenue per trade. The model asks who should be subsidised to invest in speed, not whether speed is socially free.

Mechanism 3.5.3. *Race Intensity*
Let μ_m be market maker m's arrival intensity and τ_n investor n's arrival intensity, with averages $\bar{\mu}$ and $\bar{\tau}$.
If technology costs are quadratic, $\Pi_M(m) = \frac{\mu_m}{\bar{\mu}}\pi_M\mathcal{R} - \frac{1}{2}\beta\mu_m^2$, and $\Pi_T(n) = \frac{\tau_n}{\bar{\tau}}\pi_T\mathcal{R} - \frac{1}{2}\gamma\tau_n^2$.
Here $\mathcal{R}$ is trade rate, Π_M, Π_T are total expected gains, and β, γ are speed-cost coefficients. The ratios $\mu_m/\bar{\mu}$ and $\tau_n/\bar{\tau}$ act like race shares; lower cost raises optimal speed because winning one more race becomes cheaper.

Taxonomy 3.5.4. *Latency-Cost Segmentation*

i. *market-maker cost*: co-location, feed handling, quoting engines, and hedge automation;
ii. *investor cost*: broker connectivity, routing latency, algo responsiveness, and monitoring;
iii. *venue fee split*: maker fee or rebate versus taker fee;
iv. *surplus split*: spread capture for makers versus value capture for investors;
v. *competition level*: number of fast makers relative to investor flow.

Definition 3.5.5. *Efficient Maker Fee*
In the stylised fee-choice model, the efficient maker fee satisfies $c_M^\star = s\Delta - \frac{\Gamma - \bar{c}}{1 + \left(\frac{M}{N}\frac{\gamma}{\beta}\right)^{1/3}}$, $\bar{c} = c_T^\star + c_M^\star$.
Here M, N are the numbers of market makers and investors.
Negative $c_M^\star$ is a maker rebate; negative $c_T^\star$ is a taker rebate.

Caveat 3.5.6. *Fee Rebates Are Comparative*
Maker rebates fit scarce or costly market-making; taker rebates fit crowded makers and costly investor access.

Practitioner Note 3.5.7. *Queue Competition Diagnostic*
Do not classify a fee schedule as good or bad in isolation. Segment by spread size, maker count, investor urgency, queue stability, latency cost, and whether rebates improve displayed liquidity or merely pay for fleeting quotes.

3.5.6 Cross-Asset and Cross-Venue Market Making

3.5.7 Risk Management for Market Makers

3.6 Execution and Algorithmic Trading

3.6.1 Optimal Execution Problem

Principle 3.6.1. *Impact-Risk Trade-Off*
Optimal execution balances two forces: market impact rewards slower trading, while price risk rewards faster completion. A schedule is a compromise between footprint and exposure.

Model 3.6.2. *Discrete Execution Schedule*
Plan a buy order of size $v^\star$ over N slices. During slice n, trade v_n shares, expected market volume is V_n, volatility is σ_n, and the fair price evolves as $S_{n+1} = S_n + \sigma_n \xi_{n+1}$, $\sum_{n=1}^{N} v_n = v^\star$.
Here ξ is a standard Gaussian shock. The schedule controls timing; the market supplies uncertainty.

Model 3.6.3. *Temporary Impact Cost*
A simple execution price for slice n is $\widetilde{S}_n(v_n) = S_n + \kappa\sigma_n \left(\frac{v_n}{V_n}\right)^\gamma$, $W = \sum_{n=1}^{N} v_n \widetilde{S}_n(v_n)$.
Here W is total execution notional, and κ and γ are stock-dependent impact parameters.
Participation v_n/V_n is the key lever: trading more of the available volume costs more.

Definition 3.6.4. *Expectation-Minimising Schedule*
Ignoring risk, the minimiser allocates more to high-volume and low-volatility slices: $v_n^{\exp} = \frac{V_n/\sigma_n^{1/\gamma}}{\sum_{\ell=1}^{N} V_\ell/\sigma_\ell^{1/\gamma}} v^\star$.
If V_n and σ_n are constant, the schedule is uniform: $v_n = v^\star/N$.

Mechanism 3.6.5. *Risk-Only Schedule*
If impact is ignored and only price-risk variance matters, the optimal action is to trade immediately. Waiting leaves remaining inventory exposed to random price moves.

Model 3.6.6. *Mean-Variance Execution Objective*
Let x_n be the remaining quantity after slice n, with $x_0 = v^\star$ and $x_N = 0$.
A compact objective is $\mathcal{C} = v^\star S_0 + \kappa \sum_{n=1}^{N} \sigma_n \frac{v_n^{\gamma+1}}{V_n^\gamma} + \lambda \sum_{n=1}^{N} x_n^2 \sigma_n^2$.
The middle term is impact cost; the last term is inventory risk. Higher risk aversion λ front-loads the schedule, while higher impact sensitivity back-loads it.

Taxonomy 3.6.7. *Execution Schedule Segmentation*

i. *risk-neutral*: follows volume and volatility to reduce expected impact;
ii. *risk-only*: finishes immediately to remove exposure;
iii. *mean-variance*: trades along a convex remaining-quantity path;
iv. *tactical overlay*: lets dark pools, SORs, alpha, and limits modify child orders inside the schedule.

3.6.2 Execution Benchmarks

Principle 3.6.8. *Benchmark Matches Trading Need*
Each execution benchmark expresses a different risk trade-off: avoid detection, follow market volume, capture price opportunity, or avoid missing liquidity. Benchmark choice should depend on order size, market depth, region, and investment style.

Taxonomy 3.6.9. *Regional Benchmark Preference*

i. *US*: liquidity seeking is natural when HFT is pervasive and stealth matters;
ii. *Asia*: VWAP and PoV help compare local market efficiency and fragmented market conventions;
iii. *Europe*: IS and routing discipline matter because orderbooks and venue interaction are fragmented.

Caveat 3.6.10. *Benchmark Is Not an Objective Alone*
The benchmark justifies the comparison, not the whole execution policy. Market conditions, order size, liquidity, and control freedom still determine the usable algorithm.

Benchmark	Region	Order characteristics	Market context	Risk hedged
PoV	Asia	Large order size; more than 10% of ADV, using average daily consolidated volume	Possible negative news	Rapid news repricing
VWAP/TWAP	Asia and Europe	Medium size; 5–15% of ADV	Unusual volume is negligible	Slow information drift
Implementation Shortfall (IS)	Europe and US	Small size; 0–6% of ADV	Possible price opportunities	Vanishing price opportunity
Liquidity Seeker	US	Any size	Stock expected to oscillate around fair value	Liquidity bursts; relative-price dislocations

Table 1. Typical uses of benchmarks for trading algorithms.

3.6.3 Execution Algorithms

Principle 3.6.11. *Algo as Layered Decision System* (Lehalle and Laruelle, 2018)
An execution algorithm is not only a schedule. It is a layered system that turns a parent order into child orders, adapts to market state, and records enough context to explain the realised cost afterwards.

Taxonomy 3.6.12. *Algorithm Layer Segmentation*

i. *execution parameters*: side, size, benchmark, timing, limits, participation, urgency, and venues;
ii. *risk-control layer*: sets the curve from impact, timing risk, alpha, and context;
iii. *tactical layer*: uses SORs, liquidity seeking, pegging, and dark probes to realise it.

Risk control sets the envelope; tactics work inside it.

Principle 3.6.13. *Benchmark Customisation*
Execution styles can be combined: a VWAP can carry minimum participation, limit-price, or would-level constraints; a PoV can be capped by urgency; an IS schedule can still use an SOR and dark-pool probes. The benchmark defines the reference, while constraints define acceptable behaviour around it.

Caveat 3.6.14. *Hard Mixtures Are Fragile*
Binary benchmark switches are brittle in noisy markets. A trading envelope gives the schedule room to adapt without abandoning risk control.

Definition 3.6.15. *Trading Envelope*
A trading envelope is a band of acceptable execution trajectories around the target schedule. Being too fast implies pausing or posting passively; being too slow implies crossing the spread or increasing participation.

Taxonomy 3.6.16. *Envelope Construction Segmentation*

i. *benchmark path*: VWAP, TWAP, PoV, IS, or minimum-impact curve;
ii. *risk width*: tolerance for timing risk, price drift, and completion uncertainty;
iii. *market inputs*: volume curve, volatility, spread, depth, and impact estimates;
iv. *constraints*: limits, participation caps, venue restrictions, and common portfolio end times;
v. *tactics*: liquidity capture, dark probes, pegging, and routing choices used inside the band.

Process 3.6.17. *Customisation Workflow*

i. choose the desired risk profile and inject hard constraints into the envelope;
ii. choose liquidity tactics consistent with investment style and observed market state;
iii. monitor whether the realised path remains inside the envelope.

Process 3.6.18. *Execution Decision Loop*

i. initialise the parent order and benchmark;
ii. compute a target trajectory from historical profiles and model parameters;
iii. observe quotes, trades, fills, rejects, cancellations, and venue responses;
iv. revise child-order size, price, urgency, and venue split;
v. publish monitoring and performance indicators for the trader.

Taxonomy 3.6.19. *Pre-Trade Segmentation*

i. *expected cost*: spread, impact, volatility, and typical quantity on the book;
ii. *profile*: expected volume curve, duration, and participation bounds;

Benchmark	Stock type	Trade type	Main feature
PoV	Medium to large market depth	Long-duration position trade	Flow-following; reactive within participation bounds
VWAP/TWAP	Any market depth	Hedging order; long-duration position; tracking-error unwind	Volume-curve schedule; passive when possible
IS (Implementation Shortfall)	Medium liquidity depth	Alpha extraction; non-linear hedge, typically gamma; inventory-driven trade	Fast on opportunity; cuts losses on adverse moves
Liquidity seeker	Poor fragmented market depth	Alpha extraction; opportunistic position mounting; already split or scheduled order	Pool search; dispersed liquidity; low signalling

Table 2. Typical features of trading algorithms.

iii. *context*: news, volatility regime, sector, country, currency, and event risk;

iv. *portfolio coupling*: common end times, exposure constraints, and cross-order synchronisation.

Practitioner Note 3.6.20. *What-If Controls*
Useful tools support live counterfactuals: finish now, slow down, switch benchmark, seek dark liquidity, or cap participation. The aim is disciplined comparison before changing trajectory.

3.6.4 Transaction Cost Analysis (TCA)

Definition 3.6.21. *TCA*
Transaction cost analysis compares realised execution prices with relevant benchmarks and explains the gap by market context, order design, broker behaviour, and trading style.

Principle 3.6.22. *Pre-, Live-, and Post-Trade Consistency*
The same indicator should have a pre-trade expectation, a live monitored value, and a post-trade assessment. A bid-ask-spread indicator, for example, can move from expected average, to updated live likelihood, to realised influence on performance.

Taxonomy 3.6.23. *Performance Attribution Segmentation*

i. *immediate price*: the reference available when the order decision is made;

ii. *planned price*: the price expected from the risk-control trajectory;

iii. *unexpected market move*: exogenous price variation during execution;

iv. *trading efficiency*: quality of liquidity-seeking tactics relative to the plan;

v. *market impact*: price movement caused by the order itself.

This segmentation is useful only when the benchmark, horizon, and side are explicit.

Caveat 3.6.24. *TCA Uses Wider Samples Than Post-Trade Review*
Post-trade review explains one execution in detail. TCA usually aggregates months of comparable trades to identify contexts where an algorithm, broker, or parameter set is systematically strong or weak.

Taxonomy 3.6.25. *TCA Context Segmentation*

i. *liquidity*: spread, queue size, turnover, trade size, free float, and trading rate;

ii. *risk*: volatility, jumps, intraday trend, and index or sector co-movement;

iii. *order design*: size, duration, participation rate, benchmark, and limit constraints;

iv. *market state*: news, auctions, rebalances, corporate events, and venue outages.

Practitioner Note 3.6.26. *Broker and Algo Selection*
The practical output of TCA is a routing table: for a given market context and trading objective, send the next order to the broker or algorithm whose historical performance is best after adjusting for risk and liquidity.

3.6.5 Smart Order Routing

Definition 3.6.27. *Execution-Side SOR*
In execution, a smart order router chooses where each child order should go after the parent algorithm has chosen when and how much to trade. It is the venue-optimisation layer inside the execution stack.

Definition 3.6.28. *Marketable Price Across Venues*
With several books, marketability is venue-specific. A buy order can be marketable on one venue because its limit reaches the local ask, while not marketable elsewhere if the same instrument is offered at a higher price.

Taxonomy 3.6.29. *Routing Logic Segmentation*

i. *observable liquidity*: quotes, depth, queue position, hidden-liquidity signals, and auction state;

ii. *fill probability*: venue hit rate, cancellation speed, adverse selection, and reject history;

iii. *net economics*: explicit fees, maker-taker rebates, clearing, and broker-specific schedules;

iv. *latency*: feed delay, gateway delay, matching-engine delay, and stale-quote risk;

v. *information cost*: signalling from sweeping, probing, or repeatedly visiting the same venue.

Process 3.6.30. *Child-Order Routing Loop*

i. receive the child quantity, side, limit, and urgency from the execution algorithm;

ii. score venues by expected net cost and completion probability;

iii. split or sequence messages across venues;

iv. update venue scores from fills, partial fills, rejects, and missed liquidity.

Principle 3.6.31. *Best Execution Differs by Regime*
US routing leans on protected consolidated quotes. European brokers choose the consolidation layer, so routing quality includes venue access, data quality, and auditability.

Caveat 3.6.32. *Displayed Best Price Is Not Enough*
A venue with the best quote can still be a poor destination if the queue is unstable, the quote is stale, fees are high, or fills are followed by adverse price moves.

Mechanism 3.6.33. *Active Sweep Segmentation*

i. *local best price*: cheapest visible venue first;

ii. *consolidated price*: split across venues to minimise average price;

iii. *policy coverage*: restrict destinations to venues covered by the broker mandate;

iv. *residual choice*: when venues tie at a level, choose by fee, latency, and fill reliability.

Caveat 3.6.34. *Fee-Adjusted Price Priority*
Let P be the price on V, δ the relative price increase on V', Q the size, f the fee on V, and ϕ the fee saving on V'. For a buy order, V' is cheaper when $P(1+\delta)(1+f-\phi)Q < P(1+f)Q$.
Roughly, the price penalty must be below the fee saving.

Caveat 3.6.35. *Latency Makes Liquidity Probabilistic*
Displayed depth is probabilistic: stale or duplicated quotes can vanish before the router arrives.

Principle 3.6.36. *Liquidity Seeking Objective*
A liquidity seeker scores the quantity likely to be available on arrival, using visible depth, hidden-liquidity signals, latency, duplicate quotes, fees, and fill rates.

Taxonomy 3.6.37. *Passive Posting Segmentation*

i. *queue rank*: post more where fewer shares stand ahead;

ii. *distance to touch*: prefer venues closer to becoming marketable;

iii. *trading rate*: prefer venues where opposing flow consumes the level faster;

iv. *resting-time target*: split to equalise expected completion time.

Definition 3.6.38. *Minimum Resting-Time Split*
Let D be total passive child size, Q_n queue ahead, r_n opposing-flow rate, f_n posted fraction, and $\tau^\star$ common completion time. For posted size x on venue n, $\tau_n(x) = \frac{x+Q_n}{r_n}$ is the expected completion time, so equal-time posting gives $\tau^\star = \frac{D+\sum_n Q_n}{\sum_n r_n}$ and $f_n = \frac{\tau^\star r_n - Q_n}{D}$. If $f_n < 0$, post nothing on venue n.

Criterion	1st venue	2nd venue	3rd venue
Rank in queue	better queue	next queue	deepest queue
Distance to best	nearest touch	next touch	farthest touch
Trading rate	fastest flow	middle flow	slowest flow
Minimum resting time	highest adjusted rate	next adjusted rate	residual venue

Table 3. Passive routing criteria can rank the same venues differently.

Process 3.6.39. *Liquidity-Seeker Build*

i. estimate hidden liquidity, mirrored orders, latency, fees, and adverse selection;

ii. remove liquidity when the probability-adjusted net price is attractive;

iii. post passively where queue, distance, and trading-rate criteria support fast completion;

iv. keep the parent order inside the execution envelope set by the benchmark layer.

3.6.6 High-Frequency Trading Strategies

3.6.7 Reinforcement Learning and ML in Execution

Model 3.6.40. *Stochastic-Control Execution View*
Stochastic-control execution chooses a trading rate v_t from state variables such as price S_t, remaining inventory Q_t, cash X_t, and forecast market flow μ_t. A compact sell-side value function is

$$H_t(v) = \mathbb{E}_t\left[X_T^v + Q_T^v(S_T - AQ_T^v) - \phi\int_t^T (Q_s^v)^2 ds\right]$$

Here $H_t(v)$ is the time-t value of control v, $\mathbb{E}_t$ is conditional expectation at time t, T is the terminal time, and the superscript v indicates the state path generated by that control. A penalises terminal inventory and ϕ penalises carrying inventory through time.

Model 3.6.41. *Cartea-Jaimungal State Dynamics*
One useful specification lets exogenous market flow μ_t interact with the trader's own rate v_t:

$$dS_t = b_t(\mu_t - v_t)dt + \sigma dW_t, \qquad \widehat{S}_t = S_t - \left(\frac{\psi}{2} + \kappa v_t\right),$$
$$dQ_t^v = -v_t dt, \qquad dX_t^v = \widehat{S}_t v_t dt.$$

Here b_t is permanent-impact sensitivity, σ is price-diffusion volatility, W_t is a Brownian shock, $\widehat{S}_t$ is the execution price, ψ is the bid-ask spread, and κ is temporary-impact slope.

Principle 3.6.42. *Value Function Intuition*
The control trades off three terms: realised cash, leftover inventory value, and inventory-risk penalty. A larger ϕ or A pushes faster liquidation; stronger temporary impact κ pushes smoother trading.

Taxonomy 3.6.43. *Control vs Learning*

i. *stochastic control*: specify dynamics and solve for an optimal feedback policy;

ii. *reinforcement learning*: learn the policy or value function from simulated or historical interaction;

iii. *hybrid execution*: use control structure for constraints and ML for forecasts, fills, and regime detection.

3.6.8 Production Considerations

Principle 3.6.44. *Trading Architecture*
Automated trading needs a production architecture that synchronises market data, order gateways, historical databases, models, monitoring, and TCA. The hard part is not one model; it is keeping the decision chain coherent under fragmented venues and live latency.

Taxonomy 3.6.45. *Trading Tool Input Segmentation*

i. *market data*: trades, quotes, depth, timestamps, and book updates, preferably consolidated across venues;

ii. *venue connectivity*: private order-entry messages and public market-data feeds;

iii. *historical data*: raw events, real-time analytics, order messages, fills, and market context;

iv. *models*: impact, risk, alpha, volume, and context models with calibrated parameters;

v. *calendar*: holidays, auctions, expiries, index events, and expected halt days.

Mechanism 3.6.46. *Analytics Close to the Feed*
Spread, effective spread, volatility, book imbalance, and queue-size measures should be computed as soon as raw data arrive. Fast analytics become state variables for execution, monitoring, and later post-trade explanation.

Caveat 3.6.47. *Connections Are Two-Way*
An execution engine sends orders, cancellations, and modifications, then receives acknowledgements, rejects, fills, and public book updates. In dark venues, the private responses can also act as probes of hidden liquidity.

Taxonomy 3.6.48. *Model Component Segmentation*

i. *formula*: the functional relation between state variables and decisions;
ii. *estimation*: the procedure that calibrates parameters from historical data;
iii. *accuracy check*: the live or post-trade test that decides whether the model is still usable.

Process 3.6.49. *Indicator Publication*

i. models and algorithms publish their state as indicators;
ii. traders monitor exceptions, concentration, performance, and context shifts;
iii. post-trade systems store the same indicators for diagnosis and TCA.

This makes trading algorithms stateful agents in the production stack rather than black boxes.

3.7 Asset-Class-Specific Microstructure

3.7.1 Equity Market Microstructure

Principle 3.7.1. *Equities as Microstructure Laboratory*
Equities are the cleanest empirical laboratory because trades, quotes, venue identifiers, and regulatory changes are comparatively well recorded. Lessons from equities transfer to other markets only after adjusting for contract standardisation, post-trade process, leverage, and dealer intermediation.

Caveat 3.7.2. *Asset-Class Transfer*
An equity-style CLOB is not the default architecture for every asset.
Futures, options, FX, fixed income, and crypto differ in tick size, margin, clearing, settlement, inventory risk, and whether liquidity is naturally multilateral or dealer-mediated.

3.7.2 Futures Market Microstructure

Principle 3.7.3. *Centralised Futures Trading*
Liquid futures are highly electronic and usually trade on centralised limit order books. Fragmentation is limited because margining, clearing, delivery rules, and contract fungibility make post-trade duplication costly.

Caveat 3.7.4. *No Call Auction Norm*
Many futures markets do not rely on equity-style opening and closing call auctions. Instead, participants agree on liquid trading hours during which the contract can support reliable price discovery and hedging.

Mechanism 3.7.5. *Lead-Lag With Cash Markets*
Index futures and their cash baskets form prices jointly.
Futures may lead when they are cheaper to trade, more leveraged, or open during information arrival; cash markets may lead when constituent-level information dominates.

Principle 3.7.6. *Large Tick Futures*
Futures often have economically large ticks. A binding tick creates bid-ask bounce and price reversals, but it also supports large displayed quantities at the best limits relative to comparable equities.

Taxonomy 3.7.7. *Futures Architecture Drivers*

i. *standardisation*: contracts are fungible by expiry and specification;
ii. *central clearing*: margin and default management are built into the market;
iii. *leverage*: small price moves create large economic exposure;
iv. *hedging link*: futures connect to cash baskets, ETFs, options, and swaps.

3.7.3 FX Market Microstructure

3.7.4 Options Market Microstructure

Principle 3.7.8. *Options Are Surface Markets*
An option quote is linked to the underlying price, volatility, rates, dividends, strike, and maturity.
Liquidity therefore lives on a surface, not in one instrument, and market makers hedge across the underlying, futures, and related options.

Caveat 3.7.9. *Post-Trade Complexity*
Options fragmentation is constrained by clearing, exercise, assignment, margin, and position-risk management. The trading interface may look electronic, but the economics are dominated by cross-contract risk.

Mechanism 3.7.10. *Market-Maker-Centric Liquidity*
Options markets rely heavily on professional market makers posting two-sided liquidity. Buy-side traders typically consume liquidity because maintaining coherent quotes across many strikes and maturities requires specialised models and hedging infrastructure.

Taxonomy 3.7.11. *Options Liquidity Segmentation*

i. *moneyness*: at-the-money contracts are usually deepest;
ii. *expiry*: near expiries trade differently from long-dated optionality;
iii. *underlying liquidity*: hedging cost transfers into option spreads;
iv. *volatility regime*: high uncertainty widens quotes and reduces size;
v. *complex orders*: spreads and multi-leg trades may be more relevant than outrights.

3.7.5 Fixed Income Microstructure

Taxonomy 3.7.12. *Fixed-Income Instrument Range*

i. *government bonds*: most standardised and electronically tradable;

ii. *corporate bonds*: fragmented by issuer, maturity, covenant, and liquidity;

iii. *credit derivatives*: standard indices trade more centrally than bespoke risk;

iv. *swaps and structured products*: often remain dealer-mediated because contracts differ.

Principle 3.7.13. *RFQ Dominance*
Many fixed-income products trade through RFQ because each bond or swap line is less standardised and less continuously active than an equity. Dealers compete by quote, balance sheet, inventory, and relationship rather than only queue position.

Mechanism 3.7.14. *Dealer Balance Sheet as Venue*
For less liquid bonds, the dealer's inventory and willingness to warehouse risk are part of the market structure. Liquidity is therefore partly bilateral and may disappear when balance-sheet cost rises.

Rule 3.7.15. *TRACE and Post-Trade Transparency*
TRACE reports U.S. corporate bond trades after execution, improving post-trade transparency even when pre-trade negotiation remains bilateral. Similar transparency pushes in Europe aim to make fixed-income liquidity easier to measure without forcing all products into a CLOB.

Caveat 3.7.16. *Pre-Trade Transparency Limits*
Publishing too much pre-trade interest in illiquid bonds can reveal inventory needs and increase dealer risk. Fixed-income transparency policy must balance market monitoring against the cost of exposing fragile liquidity.

Practitioner Note 3.7.17. *Data Changes the Market*
Richer order, quote, and trade records let asset managers measure transaction costs, compare dealers, and link investment decisions to liquidity. Electronification changes execution quality and governance of trading desks.

3.7.6 Securities Lending and Synthetic Shorts

3.7.7 Crypto Market Microstructure

4 Research

4.1 Alpha Research

4.2 Research in AI

4.2.1 Reading AI Papers

Remark 4.2.1. *Reading Wide*
Navigate through literature reading small amounts of individual research papers, build and improve mental model of research topic.
On *Papers with Code*, check out top benchmark models, read abstract and take note of key information only. Next step for each model, look into the datasets used by the papers and make notes.
On *Google Scholar*, check if there is any survey papers (reviews and describe state of problem space with challenges and opportunities) to get up to speed, read each paper, and make notes.
Next find related works (recently published) to understand how researchers in the field traditionally approached the problems, and what the emerging trends are. Related works will populate reading list.

Remark 4.2.2. *Reading Deep*
First pass will not understand more than 10% of research paper, and may require reading of another more fundamental paper. Then read subsequent passes until $70\% \sim 80\%$ understanding.
In introduction, highlight problems and challenges, solutions to challenges, main contributions of the work. Should be able to extract a problem-solution chain from the introduction and proposed solution.
In methods, maintain list of concepts not yet understood. If there is link to paper references, keep track.
In experiments, highlight data setup, main table of results. Keep track of unfamiliar evaluation metrics.

Remark 4.2.3. *Practical AI Research Tools*
Experiment tracking tools include 'Weights & Biases', 'Tensorboard', 'Neptune'.
If using 'Weights & Biases', model artefacts can be stored directly on a system of choice.
To train deep learning pipelines by using a config which an modify depending on which dataset, model, or configuration is used, Python's Hydra package may be used.

Remark 4.2.4. *Identifying Gaps in Research Paper*

i. Identify gaps in research question by comparing with the research hypotheses of compiled papers
ii. Identify gaps in experimental gaps, such as shortcomings in evaluation of methods, the way the comparisons were chosen or implemented, and whether the experimental setup tests the research hypothesis decisively.
iii. Identify gaps through expressed limitations, implicit and explicit.

4.2.2 Writing AI Papers

Remark 4.2.5. *Generating Ideas for Building on Research Paper*

i. Change task of interest. Can the main ideas be applied to a different modality, a different data type? Can the method or learned model be applied to a different task? Can the outcome of interest be changed?

ii. Change the evaluation strategy. Can it be evaluated on a different dataset or different metric? Explore why something works well/breaks. Make different comparisons.

iii. Change the proposed method. Can training dataset or data elements be changed? Can the pre-training/training strategy be changed? Can the deep learning architecture or problem fomulation be changed?

Remark 4.2.6. *Iterating on Research Ideas*

i. Search for whether idea has been tried. Construct titles for new paper ideas, see if there is Google result.

ii. Read important related works and follow up works.

iii. Geed feedback from domain experts on drafted ideas in written form. Email to authors of the work being built on, share idea and plan, ask their opinion.

Remark 4.2.7. *Global Structure of ML Papers*
ML Papers follow the following pattern (in 6 to 7 sections):

i. Abstract (answers 5 to 6 canonical questions in $100 \sim 250$ words):

   1. What is the background and gap?
   2. What is the key desideratum?
   3. What is the proposed solution?
   4. What are its main components?
   5. What are its strengths?
   6. What are the notable results (with tasks, numbers)?

ii. Introduction (start and end position are nearly identical across papers):

   1. Context, success of prior approaches
   2. Weaknesses or gaps of prior methods
   3. Desiderata for an improved solution
   4. Proposed method: key components, contributions
   5. High-level experimental overview, positive results

iii. Related Work/Background (typically 2 to 3 subsections), each paragraph conveys:

   1. High-level mapping of approach categories
   2. Evolution of methods over time
   3. How the proposed method compares to each category
   4. What gaps persist

iv. Methods (content always includes):

   1. Overall approach description
   2. Architecture, input/output flow
   3. Loss functions, training objectives
   4. Implementation details
   5. (Sometimes) dataset descriptions or task-specific usage

v. Experiments (how results are conveyed. Always structured as such):

   1. Overall evaluation setup
   2. Dataset description (if not earlier)
   3. Implementation details
   4. Results per task type (tables/figures)
   5. Ablations at the end
   6. References to earlier figures and comparisons to prior models

vi. Conclusion, Broader Impacts (mirrors abstract but expands):

1. Solution, components
2. Strengths, notable results
3. Future directions
4. Limitations and societal considerations
5. Motivations for follow-up work

Remark 4.2.8. *Figure Progression of ML Papers*
ML Papers follow the following pattern for figures:

i. Method overview diagram (always first)
ii. Lower-level architecture or objective illustration
iii. Comparisons vs. prior models across multiple tasks
iv. Ablation studies
v. (Optional) qualitative examples, predictions, dataset samples

5 Appendix

Miscenllaneous supplementary material that does not fit elsewhere.

5.1 Financial Calculator Guide (TI BA Pro)

Method 5.1.1. Settings calibration

i. *Reset calculator*: $\boxed{2ND}$ $\boxed{+|-}$

ii. *Increase to 9 decimal*: $\boxed{2ND}$ $\boxed{.}$ (FORMAT) $\boxed{9}$ $\boxed{\text{ENTER}}$

iii. *Set period to 1 year*: $\boxed{2ND}$ $\boxed{\text{I/Y}}$ (P/Y) $\boxed{1}$ $\boxed{\text{ENTER}}$

iv. *Set as AOS mode*: $\boxed{2ND}$ $\boxed{.}$ (FORMAT) $\boxed{\uparrow}$ $\boxed{2ND}$ $\boxed{\text{ENTER}}$

Method 5.1.2. Calculator Reset

i. *Backspace button*: $\boxed{\rightarrow}$, i.e., pressing $\boxed{2}$ $\boxed{\times}$ $\boxed{3}$ $\boxed{\rightarrow}$ $\boxed{2}$ $\boxed{=}$ will give 4.

ii. *Clear previous entry*: $\boxed{CE|C}$

iii. *Clear everything*: $\boxed{CE|C}$ $\boxed{CE|C}$

iv. *Clear TVM worksheet*: $\boxed{2ND}$ $\boxed{FV}$ (CLR TVR)

Method 5.1.3. Memory management

i. *Store in memory*: $\boxed{STO}$ ($\boxed{0}$ to $\boxed{9}$)

ii. *Recall from memory*: $\boxed{RCL}$ ($\boxed{0}$ to $\boxed{9}$)

iii. *Recall last answer*: $\boxed{2ND}$ $\boxed{=}$

iv. *Clear all memory and store values*: $\boxed{2ND}$ $\boxed{0}$ $\boxed{2ND}$ $\boxed{CE|C}$

Method 5.1.4. Computation Shortcuts

i. *Set up calculator for single variable statistics*: $\boxed{2ND}$ $\boxed{8}$, then $\boxed{2ND}$ $\boxed{\text{ENTER}}$ until 1-V, then $\boxed{CE|C}$.
Enter data setting and clear the data: $\boxed{2ND}$ $\boxed{7}$ $\boxed{2ND}$ $\boxed{CE|C}$.
Enter single-var data: [VALUE] $\boxed{\text{ENTER}}$ $\boxed{\downarrow}$ $\boxed{\downarrow}$, enter value in X (data), and leave Y as 1 (frequency).
Enter stats function and toggle $\boxed{\downarrow}$ to see mean, sample s.d., population s.d.
For weighted returns, use X as the return, and Y as the weights.

ii. *Covariance and correlation*: $\boxed{2ND}$ $\boxed{8}$, then $\boxed{2ND}$ $\boxed{\text{ENTER}}$ until [LIN], then $\boxed{CE|C}$.
Enter data setting and clear the data: $\boxed{2ND}$ $\boxed{7}$ $\boxed{2ND}$ $\boxed{CE|C}$.
Enter data: [VALUE] $\boxed{\text{ENTER}}$ $\boxed{\downarrow}$ $\boxed{\downarrow}$, enter value in X and Y.
Enter stats function and toggle $\boxed{\downarrow}$ to see r, Sx and Sy, then compute covariance as $Sx \times S_y$.
Correlation is simply the value r computed earlier.

iii. *Time value of money*: Input values into all except one of these: $\boxed{\text{N}}$ $\boxed{I/Y}$ (%), $\boxed{\text{PV}}$, $\boxed{\text{PMT}}$, $\boxed{\text{FV}}$. Then use $\boxed{\text{CPT}}$ on the target variable to solve for the results.

iv. *Interest rate conversion*, i.e., convert nominal 10%, $m = 12$ payments per year into effective rate.
$\boxed{2ND}$ $\boxed{2}$ (ICONV) $\boxed{\uparrow}$ $\boxed{12}$ $\boxed{\text{ENTER}}$, $\boxed{\downarrow}$ $\boxed{10}$ $\boxed{\text{ENTER}}$, $\boxed{\downarrow}$ $\boxed{\text{CPT}}$ to get effective rate.

v. *Cash flow computation*: clear memory with $\boxed{\text{CF}}$ $\boxed{2ND}$ $\boxed{CE|C}$, then input [VALUE] $\boxed{\text{ENTER}}$ $\boxed{\downarrow}$.
Enter interest rate with $\boxed{\text{NPV}}$ [VALUE] $\boxed{\text{ENTER}}$ $\boxed{\downarrow}$, then $\boxed{\text{CPT}}$ to get present value, PV.

vi. *Amortisation schedule*: i.e., \$1000 on 3-year loan, interest rate of 10%.
Check payment per year, make sure it is 1 (with $\boxed{2ND}$ $\boxed{I/Y}$).
Input information with $\boxed{3}$ $\boxed{\text{N}}$ $\boxed{10}$ $\boxed{I/Y}$ $\boxed{1000}$ $\boxed{\text{PV}}$ $\boxed{\text{CPT}}$ $\boxed{\text{PMT}}$.
Before using amortisation worksheet, clear memory with $\boxed{2ND}$ $\boxed{\text{PV}}$ (AMORT) $\boxed{2ND}$ $\boxed{CE|C}$.
To see interest and principal repayment at each time period, set $P1$ as $\boxed{t}$ for year t, then use $\boxed{\text{CPT}}$ $\boxed{\downarrow}$ to see the values at each time period.

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